



Traffic Congestion Estimation and Control: A Comprehensive Review of the Applied Computational Intelligence Models

Younus Hasan Taher¹ · Jit Singh Mandeep¹ · Haydar Abdulameer Marhoon^{2,3} · Hamdi A. Al-Jamimi^{4,5} · Hamzah Luqman^{4,6} · Farag Azzedin⁴ · Muhammad Abdullah^{7,8} · Hassan M. Al-Ahmadi^{7,8} · Zaher Mundher Yaseen⁷

Received: 4 November 2024 / Accepted: 21 June 2025

© The Author(s) under exclusive licence to International Center for Numerical Methods in Engineering (CIMNE) 2025

Abstract

Congestion control is a demanding functionality for intelligent transportation systems. It involves various functionalities and tasks, namely, traffic estimation and forecasting, sensing and communication, and traffic signal control. The literature on traffic congestion estimation and control has emerged significantly, and different methodologies and approaches have evolved. In this survey, an exploration of the existing systems in traffic congestion estimation and control was conducted. Furthermore, this review provides the primary taxonomy of the current methodologies and approaches; namely, for congestion estimation: machine learning, fusion methods, and for congestion control: machine learning and data-driven models, optimization techniques, technology-driven approaches, fuzzy logic and systems, game theory and decision making, and hybrid traffic management. In addition, this survey is considered the first to address traffic congestion handling in intelligent transportation systems from sensing, estimation, detection, recognition, and control perspectives. Finally, the survey provides an overview of the existing issues and challenges for traffic congestion in intelligent transportation systems, as well as future research directions.

✉ Zaher Mundher Yaseen
z.yaseen@kfupm.edu.sa

Younus Hasan Taher
You1993nis@gmail.com

Jit Singh Mandeep
mandeep@ukm.edu.my

Haydar Abdulameer Marhoon
haydar@alayan.edu.iq

Hamdi A. Al-Jamimi
aljamimi@kfupm.edu.sa

Hamzah Luqman
hluqman@kfupm.edu.sa

Farag Azzedin
fazzedin@kfupm.edu.sa

Muhammad Abdullah
muhammad.abdullah@kfupm.edu.sa

Hassan M. Al-Ahmadi
ahmadi@kfupm.edu.sa

² Information and Communication Technology Research Group, Scientific Research Center, Al-Ayen University, Thi-Qar, Iraq

³ College of Computer Sciences and Information Technology, University of Kerbala, Karbala, Iraq

⁴ Information & Computer Science Department, King Fahd University of Petroleum and Minerals, 31261 Dhahran, Saudi Arabia

⁵ Research Excellence, King Fahd University of Petroleum and Minerals, 31261 Dhahran, Saudi Arabia

⁶ SDAIA-KFUPM Joint Research Center for Artificial Intelligence, King Fahd University of Petroleum and Minerals, 31261 Dhahran, Saudi Arabia

⁷ Civil and Environmental Engineering Department, King Fahd University of Petroleum and Minerals, 31261 Dhahran, Saudi Arabia

⁸ Interdisciplinary Research Center for Smart Mobility and Logistics, King Fahd University of Petroleum and Minerals, 31261 Dhahran, Saudi Arabia

¹ Department of Electrical, Electronic and System Engineering, Faculty of Engineering and Built Environment (FKAB), Universiti Kebangsaan Malaysia (UKM), 43600 Bangi, Selangor, Malaysia

1 Introduction

The creation of a well-ordered road transportation system is critical in urban areas and major cities. An intelligent transportation system (ITS) increases traffic safety and efficiency by allowing cars to produce and transmit messages. In both industrialized and developing countries, ITS is the backbone of smart cities [1]; they benefit the economy, save time, and make travel safer. ITS relies on technologies, such as the Internet of Things (IoT) [2], deep learning modes, cloud computing, Vehicular Ad Hoc Networks (VANETs), machine learning (ML), fog computing [3, 4], mathematical/trust models [5, 6], service-oriented architectures, and artificial intelligence (AI) algorithms. Collectively, these technologies enable object and lane detection, route planning, accident avoidance, and traffic management. Furthermore, ITS is crucial in mitigating environmental effects produced by CO₂ emission [7].

In the context of ITS, traffic represents the flow of different types of vehicles on the road despite the nature of the road. Traffic flow control refers to enabling traffic signals or other mechanisms to control and optimize the flow of traffic to prevent bottlenecks, long queues, delays, etc., [8]. On the other hand, traffic control requires another

functionality—traffic flow prediction, to avoid sudden traffic jams. Figure 1 is presented as the conceptual diagram of a traffic management system that includes intelligent traffic lights, an actuator, sensors for problem detection, cloud management, and control applications.

The handling of traffic control and prediction using artificial intelligence approaches is a new trend with different directions and perspectives [9]. Some researchers approach the problem from the perspective of sensing infrastructures, while others emphasize the role of communication systems. Meanwhile, a third group concentrates on models and algorithms for control and prediction, aiming to conduct both qualitative and quantitative comparisons in tackling traffic control and prediction challenges. This article presents traffic flow control and estimation/prediction from the main perspectives of sensors, communication, and management.

1.1 The Existing Studies

The literature includes various recent surveys of traffic scheduling in intelligent transportation (ITS). For instance, the work by [10] investigated the traffic police scheduling architecture and its components. They examined the various traffic police scheduling techniques for reallocating resources depending on current traffic densities. In addition,

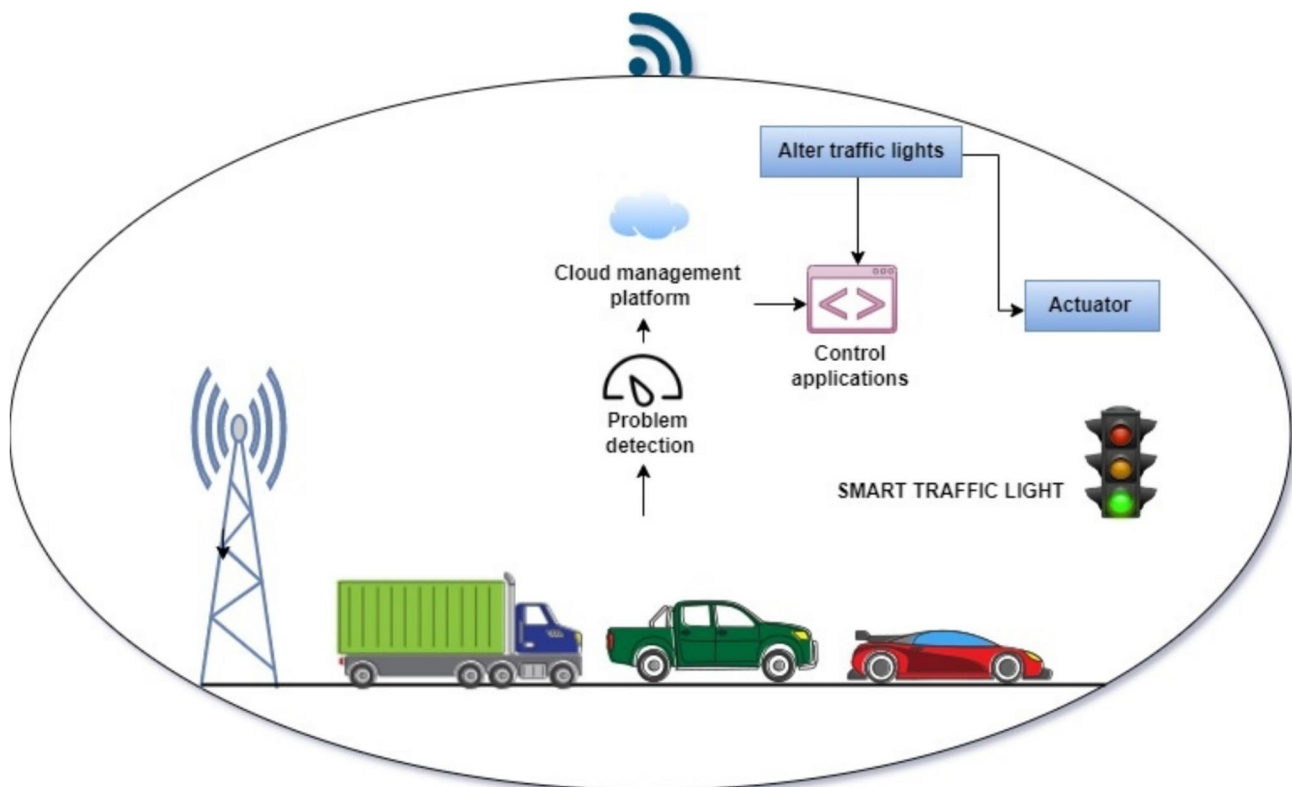


Fig. 1 A conceptual diagram of the traffic management system

they described the traffic police schedule taxonomy in three parts. To start, the data gathering process covered numerous cutting-edge sensing and communication technologies, including high-definition cameras, Wi-Fi, HetNet 5G, and sensors for traffic data collection. A selection of ML methods was provided in the data analysis area for use in data processing and traffic analysis. Also, the study discussed various scheduling algorithms for traffic scheduling and their objectives. Lastly, the problems and difficulties with traffic planning and the necessity for technical developments in traffic management were discussed. In another survey [11], a method for congestion prediction was proposed following a comprehensive summary of the existing research conducted using various AI methodologies, most notably different ML models regarding traffic prediction. In the work of [12], the study focused on the state of deep learning applications at the moment in terms of tasks involving congestion detection, prediction, and relief. The two types of congestion—regular congestion and irregular congestion—were also discussed.

Few studies have also focused on the big data component of ITS applications. For instance, the work of [13] provides insight into the use of Big Data methods in ITS, highlighting many applications and integrating models and applications. In the work of [14], a survey of traffic congestion from three perspectives, namely, sensing, communication, and computation was conducted. Other types of surveys have considered certain aspects of traffic congestion. Authors in [15] presented a survey of traffic congestion prediction with a focus on the use of graph neural networks. Table 1 presents an overview of some of the existing surveys on congestion handling in ITS.

Overall, none of the existing surveys have provided a review of traffic congestion sensing, detection, prediction, and control with a focus on the relation between these elements. The work of [12] only addressed these elements with a focus only on deep learning approaches. Deep learning-based approaches are useful for recognition and detection; however, they are not sufficient for handling all the mentioned traffic elements. Hence, this survey aims at covering ITS from the perspective of traffic congestion sensing, estimation, detection, and control. Furthermore,

a general taxonomy that covers the different solutions to these problems, as well as the comparison between the partial solutions were developed in this study.

1.2 Contribution

This survey presents several contributions that can be stated as follows:

- i. To our knowledge, this is the first survey that addresses traffic congestion handling in ITS from the aspect of sensing, estimation, detection, recognition, and control. The goal is to offer a comprehensive understanding of the existing literature on traffic congestion, as well as the solutions that have been proposed to address it.
- ii. A general taxonomy of the existing algorithms and their relationships was developed to facilitate comparison. This will enable researchers to identify the various methods of traffic congestion and the relationship between them.
- iii. This study addresses the existing issues and challenges in traffic congestion and provides future research directions.

1.3 Taxonomy

The main topic discussed in this article is traffic estimation and control; this application has two levels to be discussed, namely, sensing and communication in the first level and traffic control or traffic estimation in the second level. Traffic estimation was separated from traffic control because each one has its methods and techniques. For traffic estimation, there are four main methods used—sensor fusion, Kalman filtering, neural networks, and hybrid methods. On the other hand, traffic control has four methods used, namely, reinforcement learning, neural networks, game theory, fuzzy logic, and hybrid.

Table 1 Overview of the existing surveys on congestion handling in ITS

Survey	Taxonomy	Sensing	Communication	Computation	Congestion estimation/prediction	Traffic control
[10]	✓	✓	✓	✓	×	✓
[11]	✓	✓	×	×	✓	×
[12]	×	✓	×	×	✓	✓
[14]	×	✓	✓	✓	✓	✓
[13]	×	✓	×	×	✓	×
This study	✓	✓	✓	✓	✓	✓

Note ✓ element was considered, × element wasn't considered

2 Congestion Estimation Models

This section presents the congestion estimation methods documented in existing literature. These methods are categorized into two subsections, namely: ML methods and fusion methods. In a more understanding manner, Fig. 2 presented the whole divisions of the congestion estimation models.

2.1 Machine Learning Models

2.1.1 Supervised Learning

Various supervised learning techniques have been suggested for traffic congestion estimation, including traditional, Bayesian, evolutionary, neural network, deep learning, and hybrid methods. These studies encompass a range of approaches from conventional methods to advanced neural network and deep learning techniques.

2.1.1.1 Traditional Methods The researchers conducted studies aimed at developing a methodology to quantify the extent of traffic congestion in urban settings. Liu et al. employed a multiple linear regression model to forecast the urban traffic congestion index, taking into account diverse factors such as peak periods, weekdays versus weekends, weather conditions [16], and road conditions. The primary methodology encompassed model development, data collection, and subsequent application of the model to estimate the congestion index.

The execution of models using distinctive denoising procedures was tried by [17] using activity volume information collected from three circle locators set on a thruway close to Minneapolis. Five denoising techniques—Experimental Mode Deterioration, Outfit Observational Mode Decay, Moving Normal, Butterworth channel, and Wavelet were considered for the comparison. Four types of wavelets—coiflet, Daubechies, haar, and Symlet were also investigated. The Support Vector Machine (SVM) is used to predict how traffic flow patterns change based on denoised traffic data. This helps improve the accuracy of traffic flow predictions in the study.

Gokasar and Timurogullari employed computer vision to gather vehicle location, lane, and time data from the D100 Highway [18]. They manually constructed event matrices to capture occurrences such as shoulder violations, police presence, and weather conditions. Utilizing Logistic Regression, they assigned weight coefficients to amalgamate diverse event dynamics into an "Event Score." Subsequently, they merged traffic density and event datasets to forecast future traffic frames. They compared results

to determine the most accurate prediction algorithm and interval. Among the algorithms tested, SVM and Recurrent Neural Networks (RNN) demonstrated the highest accuracy in predicting traffic density at 1, 3, and 5-min intervals. Computer vision algorithms and manual event matrices facilitated the analysis of vehicle behavior, while data mining tools and ML algorithms were instrumental in forecasting traffic density across various time frames.

Sujatha et al. tackled the traffic congestion challenges prevalent in South Africa's Gauteng province by devising traffic flow prediction models [19]. The overarching objective was to achieve accurate prediction and identification of road congestion levels, ultimately aiming to mitigate the adverse socioeconomic impacts of congestion on the province's economy and daily life. Their proposed approach involved the classification of congestion levels (light, heavy, and jam) using a decision tree learning model (J48), with ratings and traffic rates being utilized. In addition, they applied a moving pattern technique to capture consecutive moving average velocities, and road users' judgments and related information are processed through the J48 classification algorithm.

Cheng et al. proposed a Random Forest–based model to estimate the travel time of traffic flow [20]. The model incorporated seven variables, including the mean and median travel time of floating cars, traffic state parameters, vehicle density, and speed. Data were collected from 133 simulations conducted using VISSIM software, which were then partitioned into training and test sets. The results were obtained compared against two other models: a quadratic polynomial regression model and a BP neural network model.

Moreover, Naive Bayes (NB) has been employed for predicting congestion and incidents in urban road networks [21]. The NB model operates under the assumption of probabilistic independence among predictor variables and classifies the state of congestion or incident occurrence. These predictor variables encompass network environment factors like time of day, day of the week, and weather, as well as traffic condition variables such as speed on bottlenecks. Wang and Kim employed a data-driven approach to construct and validate NB models, utilizing actual traffic, incident, and weather data from Brisbane, Australia, in 2014. The findings demonstrated the model's potential for accurate prediction of congestion and incidents, emphasizing its adaptability across diverse regions and potential integration into real-time traffic and incident management systems.

To foresee a short-term activity stream, a sample-rebalanced and outlier-rejected k-nearest neighbor (K-NN) relapse show was utilized by [22]. To fathom the awkwardness issue, they utilized a degree for evolutionary traffic flow patterns and reconstructed balanced training sets via relative transformation. At that point, the analysts made a

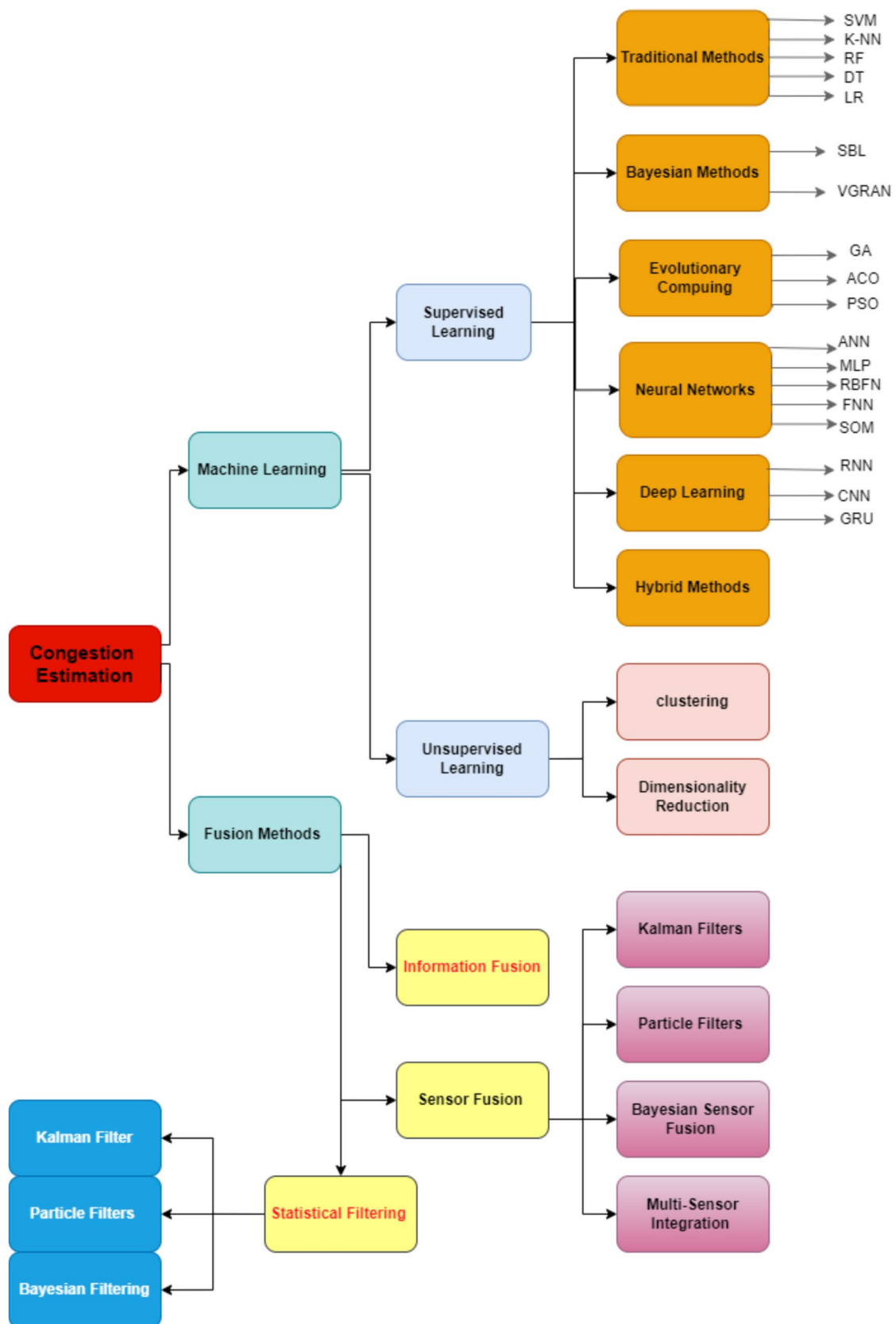


Fig. 2 The main adopted methodologies for the congestion estimation

cross-breed show that incorporates both neighborhood and worldwide information to solve the problem of preparing tests. A distributed KNN-based model for short-term prediction of traffic flow was presented by [23] which accounts for both the temporal and spatial traffic flow distribution. The study implemented the proposed model on the Apache Spark framework as a MapReduce-based algorithm. In addition, Nasution et al. employed the Advanced Traffic Management System (ATMS) to address traffic congestion [24]. The proposed system represents an enhancement of the Intelligent Transportation System and utilizes a query classification strategy, prioritizing information most relevant to the specific traffic issues under consideration.

Classical traffic-flow theory often employs fundamental diagrams (FDs) to model speed–flow–density relationships and to establish early indicators of congestion. For instance, Chen et al. [25] present a slope-change-based FD approach to urban expressway congestion forewarning. By tracking where the slope in the speed–flow curve begins to change significantly, their method identifies a “forewarning point” to alert operators before severe congestion develops. Applied to an expressway in Beijing, this slope-change technique proved effective at quickly flagging imminent congestion while maintaining interpretability, as it relies on well-understood FD principles.

2.1.1.2 Bayesian Methods Various studies have explored the application of Bayesian methods in traffic management and forecasting, showcasing their potential in handling complex traffic data and enhancing prediction accuracy. The estimation of traffic situations via the creation of sparse representations using Block SBL (BSBL) and Sparse Bayesian Learning (SBL) techniques was proposed by Sure et al. [26]. The approach considered handling under-sampled data, irrespective of how the data was acquired. They utilized historical spatiotemporal correlations in traffic data to determine appropriate kernel matrices and estimate unavailable traffic conditions. Yan et al. focused on predicting the duration of traffic accidents using a combination of a weighted K-nearest Neighbor model and a cost-sensitive Bayesian network [27]. Their approach aimed at reducing traffic congestion and facilitating road repairs, though it didn't fully address sample imbalance, characteristic interplay, and cost-sensitive issues.

Bayesian Networks have been also employed for classification tasks in traffic management. a brand-new Bayesian framework called Variational Graph Recurrent Attention Neural Networks (VGRAN) was proposed by [28]. The system utilizes dynamic graph convolution operations to gather time-varying road sensor readings and identify latent variables related to sensor representation and traffic patterns. Their proposed probabilistic technique provides a flexible generative model, effectively capturing the sensor's

stochastic nature and temporal traffic correlations through variational inference. Similarly, the use of a probabilistic congestion estimation method based on Bayesian Networks was proposed by Afrin and Yodo [29]. They incorporated both speed and volume-related indicators to provide a comprehensive assessment of congestion situations. The study created two different BN models and implemented them on real-time datasets for both non-recurrent and recurring traffic congestion.

2.1.1.3 Evolutionary Methods Evolutionary methods have been employed to address various challenges in traffic management and modeling. Using genetic optimization, Guo et al. proposed a model for user-equilibrium traffic-driven area-wide traffic signal time optimization [30]. The model was developed as a multi-dimensional search problem that aims at minimizing the product of the variance of journey time per unit distance and the overall trip time connected to the urban street network. The genetic algorithm is used to develop a model solution. The area-wide traffic signal control system's logic framework and functional module were developed using an area-wide micro-simulation control protocol integrated into the PARAMICS software tool. The proposed model was validated through a computational experiment conducted in the Central Business District of Nanjing, China. In addition, a ground-breaking rule-based method was proposed for forecasting the severity of traffic accidents and to achieve a balance among conflicting objectives [31]. The proposed approach utilized decision trees (DTs) for congestion estimates and a non-dominated sorting genetic algorithm (NSGA-II) to enhance and recognize rules that satisfied identified criteria.

Ant colony optimization (ACO) and complex graph theory were utilized to examine alterations in road network connectivity post-CO policy enforcement [32]. The study focused on assessing the impact of implementing the Community Opening (CO) policy on traffic congestion in major Chinese cities like Beijing, Shanghai, Guangzhou, and Shenzhen. The ACO algorithm evaluates changes in travel times and distances between road nodes, while the complex graph theory calculates attraction indices for roads, assisting in analyzing potential traffic congestion areas. The study aimed to understand changes in road network structure, resident commuting efficiency, and traffic congestion areas post-implementation of the CO policy in these urban settings.

2.1.1.4 Neural Networks Zhu et al. developed a traffic prediction model based on GPS-derived information data from vehicles [33]. The A-Dijkstra weighed shortest path algorithm was proposed based on the V-Clique clustering algorithm to evaluate historical vehicle GPS data and the artificial neural network (ANN) to build the predictive model. Another ANN-based approach was proposed to cat-

egorize traffic congestion of various routes in a city using traffic data gathered by in-road stationary sensors [34]. In addition, Fahs et al. introduced a novel approach to alleviate traffic congestion issues through the utilization of four distinct ML techniques: Feed Forward Neural Networks (FFNN), Radial Basis Function Neural Networks (RBFNN), simple linear regression, and polynomial linear regression [35]. The proposed framework encompasses the collection of an extensive dataset encompassing diverse traffic congestion causes, the proposition of prediction techniques employing real-time traffic data and ML algorithms, and a comprehensive comparative analysis of these techniques to identify the most suitable algorithm for the specific area.

A multi-layered perceptron model has been employed to analyze crashes on six rural two-lane roads [36]. The crash data was systematically collected spanning the years 2013–2016. The study acknowledges the severe consequences of traffic accidents and seeks to enhance modeling accuracy by including the parameter of drop-offs to account for elevation variations among roads. In addition, Gu et al. proposed an enhanced hybridization of the Bayesian Combination model with deep learning (IBCM-DL) [37]. The study determined the relevance of the traffic flow during the current moment to the traffic flow during the past time using Correlation analysis. The IBCM architecture incorporates three sub-predictors to benefit from each technique: the gated recurrent unit neural network (GRUNN), Radial basis function neural network (RBFNN), and autoregressive integrated moving average (ARIMA). They extended the BCM framework put forward by [38] by proposing a progressed Bayesian combination approach. To coordinate the non-linear reliance and the long-term reliance of the complex transient flow of the activity stream, the IBCM-DL combines a parametric approach, a non-parametric strategy, and a profound learning strategy.

Self-Organizing Maps (SOM) offers a powerful tool for understanding and predicting urban traffic patterns which facilitate urban planning and management. For instance, Guo et al. proposed a traffic modeling pipeline using SOM to recognize dominant traffic patterns in cities and predict common traffic occurrences or probable pattern locations [39]. The research utilized traffic data collected from online map vendors for 32 Swiss cities during two months. The utilization of SOM assists in finding the traffic flow patterns and determine the similarity of road segments within and across different cities. In addition, Gu et al. proposed a data-based approach to assess traffic congestion between bus stops [40]. The proposed methodology involved extracting three traffic indicators from the smart card and bus trajectory data, utilizing a SOM to identify traffic patterns in these segments. It was validated using real-world data. A congestion index for ranking SOM clusters was developed to determine congested bus stops. The study also employed a

genetic algorithm to explore public transport network (PTN) optimization based on congested bus stops. The SOM successfully reflected traffic characteristics and estimated congestion, offering insights into PTN optimization strategies, which can benefit urban transport system management and road traffic status estimation.

2.1.1.5 Deep Learning Different studies explored Deep Learning (DL)-based solutions for addressing the Traffic congestion prediction problem. This includes various techniques such as Convolutional Neural Networks (CNN), RNN, and Gated Recurrent Unit (GRU). The CNN was employed to design a system that detects road congestion using a multidimensional feature traffic congestion [41]. The proposed approach consists of three stages. At the primary stage, the density of foreground items was computed based on gray-level co-occurrence matrix. Then, the speed of moving objects was calculated using the Lucas-Kanade optical flow with pyramid implementation. Finally, Gaussian mixture model was used to explain the background, and CNN is used to precisely differentiate the final foreground from any prospective foreground. In addition, Guo et al. proposed a progressive chart convolution repetitive neural organization for activity expectation [42]. The geographic data of the street is taken as a chart. The proposed strategy utilizes a data-driven approach to learn an optimal chart during the preparation stage. It was validated through three real-world case studies, demonstrating superior performance compared to existing activity prediction methods. CNN model was employed to determine the precise and accurate degree of congestion [43]. They used a traffic parameter layer to develop an image-based traffic congestion estimation framework that can shorten processing time. In addition, Chakraborty et al. proposed another approach for traffic congestion prediction from camera photos [44]. The proposed approach used two different deep learning techniques (you only look once (YOLO) and deep convolution neural network (DCNN)). The dataset's congested photos were labeled, and the models were trained using occupancy data from surrounding radar sensors. CNN suits image-based tasks, ideal for detecting traffic congestion in CCTV footage using specific deep learning techniques like YOLO and DCNN.

Li et al. proposed the MF-TCPV system based on ML and fluffy comprehensive evaluation for the perception and expectation of activity clog [45]. The proposed system integrated multi-source heterogeneous activity information about clog prediction using DataX and DataV. For the long-term expectation of activity clog characteristics, a deep prediction model called LSTM-SPRVM was developed. This model is based on deep learning algorithms, machine learning methodologies, and Spark parallelization technology. It aims to enable accurate future prediction of traffic

congestion characteristics. Based on Fluffy Comprehensive Assessment and activity blockage characteristics counting, the normal speed, street inhabitation rate, and activity stream thickness, MF-TCPV isolates activity blockage into six stages. The MF-TCPV was validated based on genuine information from White Mud Drive in Canada. In addition, Ganapathy et al. utilized the Sequence Convolution-based Auto-Encoder Long Short-Term Memory (SCAE-LSTM) for vehicle speed estimation [46]. They aimed to estimate traffic flow and improve arterial junctions using spatial–temporal traffic sequences. The Spatial-Temporal AI Reconnect (STAR) algorithm was proposed to assess traffic flow on highways and reduce travel delays. The study compared the STAR algorithm with baseline deep learners in experiments conducted on Chennai state highways.

A gated recurrent unit neural network was employed to train on a substantial digital tachograph dataset containing vehicle velocity and location records [47]. The authors compared their proposed approach with other models such as ARIMA and LSTM. The GRU-based approach demonstrated superior prediction accuracy. Furthermore, the GRU-based approach offers a computational advantage, resulting in lower costs.

2.1.1.6 Hybrid Methods Different hybrid models have been proposed to address the complexities of traffic flow prediction and analysis. Zhang et al. proposed a hybrid approach to predict changes in traffic flow using a Quantum Genetic Algorithm—Learning Vector Quantization (QGA-LVQ) neural network algorithm [48]. The approach aimed to utilize the global optimization ability of OGA to overcome some shortcomings of LVQ neural network. The experimental simulation results demonstrate that the QGA-LVQ neural network achieves outstanding prediction accuracy and convergence speed. Luo et al. used GA optimization to combine an enhanced seasonal autoregressive integrated moving average (ISARIMA) model with a multi-input autoregressive (AR) model [49]. GA was used to select stations that have a strong link with the forecast station, while the ISARIMA model was used to forecast the traffic flow within the test station. A multi-input AR model, incorporating traffic flow data from strategically chosen stations, was employed to forecast traffic flow at the test station.

Another study introduced a deep learning-based method to automatically extract built-in characteristics from traffic flow data using hybrid and multiple-layer architectures [50]. They built an attention-based Conv-LSTM module to extract spatial and short-term temporal information. The proposed hybrid algorithm was able to recognize the significance of traffic patterns at different time intervals by naturally assigning diverse weights using the consideration instrument. It also proposed a bidirectional LSTM (Bi-LSTM) module designed to capture the fluctuating inclination of the activity

stream. This module utilizes separate daily and weekly special highlights to model changes from both forward and backward directions.

Traffic Flow Dependency and Dynamics-based Deep Learning (TD2-DL) approach for predicting high-resolution traffic speed propagation across an entire network was proposed by [51]. The proposed approach integrated temporal-spatial flow dependency, traffic flow dynamics, and deep learning techniques. It initially used a graph theory-based method to identify local temporal-spatial traffic dependencies, adaptively considering the prediction horizon and traffic delays. Then, the v-CTM model was employed to mathematically describe the traffic speed propagation on each road. The LSTM model was used to predict boundary conditions, factoring in traffic's temporal-spatial dependencies and historical data predicted by the v-CTM model. The approach integrated physical models with deep learning, the Extended Kalman Filter (EKF) assimilated predicted traffic speeds, and a Feedforward Neural Network (FNN) imputed the missing data. The conducted experiments demonstrated the effectiveness of TD2-DL in network-wide traffic speed prediction.

In addition, hybrid ANN with Particle Swarm Optimization (PSO) was developed to model traffic flow [52]. The authors investigated traffic flow for long and short trucks on South Africa's N1 freeway, known for its significant traffic volume and frequent congestion caused by trucks. The traffic data were collected via loop detectors and video cameras. The study aimed to evaluate truck traffic flow, emphasizing real-world traffic scenarios from a developing country, employing PSO algorithms in traffic flow analysis.

Modern hybrid approaches integrate model-driven insights (e.g., FD, CTM) with powerful machine-learning frameworks to improve both interpretability and accuracy in congestion estimation. One example is Pan et al. [53], who developed a Fundamental Diagram + Markov + LSTM pipeline. Their model first employs the FD to detect congested vs. uncongested states, then uses a Markov model to account for probabilistic transitions between states. Finally, any unexplained dynamics are captured via an LSTM network. Experiments on real-world data (Beijing, Los Angeles) show that this combined strategy significantly outperforms standalone machine-learning and classical statistical methods.

Another representative hybrid method is proposed by Li et al. [54], who introduce a Cell Transmission Model (CTM) + Transformer framework for urban signalized intersection traffic state prediction. They construct a spatial–temporal graph of lane segments, factoring in dynamic signal phases, and then apply a Transformer network to capture global temporal dependencies. The CTM provides physically grounded constraints on traffic flow, while the Transformer excels at sequence learning. Validated using the large-scale pNEUMA dataset, their approach demonstrates how

blending model-based and data-driven features can yield robust intersection-level congestion forecasts.

2.1.2 Unsupervised Learning

The availability of large unannotated data motivated researchers to utilize unsupervised-based learning techniques for traffic congestion prediction. These techniques can benefit from the huge publicly available data without the need for data annotation, which is expensive and time-consuming. Several unsupervised-based learning techniques have been proposed in the literature for estimating traffic congestion. These approaches can be categorized into clustering, dimensionality reduction, and fuzzy-based methods.

2.1.2.1 Clustering Benjamin et al. proposed an approach to predict the Top of Descent (TOD) location in air traffic to manage air traffic congestion [55]. The approach involved unsupervised learning to identify common TOD clusters that represent the macroscopic perspective of air traffic controllers. A DBSCAN-K-means algorithm was used for TOD clustering. This algorithm is also used by Radhika et al. to address urban transportation challenges [56]. The authors proposed an adaptive traffic control system based on ML and the IoT. IoT sensors installed in vehicles and smart devices collect and transmit traffic data, while A ML-driven DBSCAN clustering algorithm is integrated to identify anomalies. This approach allows the adjustment of traffic light based on traffic flow and neighboring intersection predictions. Another attempt to estimate the TOD was by [57]. A decision tree model was trained using operational flight factors like proximity to predict TOD, current flight level, fleet, and aircraft type. The proposed system was evaluated in a Singapore flight information region case study and a maximum prediction error range of +31NM was reported. These results highlight the feasibility of ML in predicting TOD based on prior cruise characteristics and pilot operational considerations.

Hidden Markov Model (HMM) was employed to predict a congested road's impact on neighboring road traffic conditions by assessing spatial and temporal relationships [58]. The proposed approach employed traffic state clustering through Grey Level Co-occurrence Matrix (GLCM) and spectral clustering techniques. These methods were used to identify similar congested road segments based on time intervals. These congested road segments are used by HMM to predict the traffic state. By analyzing spatial and temporal features of roads without direct connectedness, the study identified road relationships based on congestion occurrence. The reported results indicate the effectiveness of the proposed HMM model in estimating traffic state impacts in related areas, contingent upon the congested road's proximity to neighboring roads.

2.1.2.2 Dimensionality Reduction A novel unsupervised model for traffic forecasting was proposed by [59], through conceptualizing the traffic data as a latent variable model. Leveraging the compressed latent space, the study applied prediction models to enhance forecasting accuracy by addressing challenges in traffic prediction using unsupervised Variational Autoencoder (VAE) model. This model was used to learn the generation and inference processes of traffic data from a lower-dimensional latent space. In addition, an online unsupervised imputation method was introduced to handle missing traffic data in three real-world traffic datasets that were used to evaluate the proposed approach.

2.1.2.3 Fuzzy-Based Methods (Including DST) In a smart environment, multi-sensor data fusion is widely utilized to combine data from different sensors. Sensors, on the other hand, produce noisy and imprecise data, posing a significant problem for researchers. Data uncertainty is a major stumbling block for decision-making and data fusion systems. One of the good tools for the modeling and combination of ambiguous data is the evidence theory of Dempster-Shafer. Ullah et al. combined data using the Dempster-Shafer evidence theory, and the obtained information was assessed for uncertainty using the modified belief entropy of the basic probability assignments (BPAs) [60]. Dempster's rule of combination was applied to integrate all available evidence and generate decision-making outputs, even in the presence of contradictory data. This was done after applying the proposed belief entropy to the data received from the sensors. The reported simulation results show a good performance of the proposed approach and its potential in a smart environment. Dempster's rule of combination was applied to integrate all available evidence and generate decision-making outputs, even in the presence of contradictory data. This was done after applying the proposed belief entropy to the data received from the sensors.

Amini et al. used fuzzy inference model to predict traffic congestion levels within intricate and diverse Hungarian freeway networks [61]. A total of 75 rules were developed from different datasets, percentile distributions, and expert judgments. Matlab fuzzy logic toolbox was used to design the model and develop analyzing steps. The main goal of this work was to combat challenges in interpreting traffic complexity using conventional quantitative methods. Using real traffic data, traffic flow, and segment capacity as inputs, the model employed an output variable representing the congestion level. The results of this study reveal that there is a correlation between input variables and their predictive capability regarding congestion levels. Another fuzzy rule-based system was proposed by Septyanto et al. for traffic congestion monitoring via radio frequency identification (RFID) technology [62]. RFID is used to detect vehicles on roads and retrieve their information. The fuzzification is

then used to calculate the road density by combining average vehicle speed and road occupancy rates. Then, rule-based expert systems using Decision Support System (DSS) are applied to rapidly provide traffic jam information. Fuzzy logic methods, combined with rule-based systems, offer the ability to model intricate, nonlinear functions, allowing linguistic rules to define the relationship between input parameters and the system's output.

Toan et al. focused on the third stage of a multi-stage fuzzy logic controller (MS-FLC) for expressway traffic congestion management [63]. The authors introduced a fuzzy-based decision-making procedure (FDMP) to recommend control strategies and actions based on current and anticipated traffic data. The FDMP operates at strategic and operational levels, achieving a balanced congestion management approach between mainline and ramp traffic. It allows system-wide improvements and addresses both recurring and non-recurring congestion. Despite its comprehensive approach, the FDMP lacks systematic stability analysis and demands extensive network data, posing challenges in precise parameter definitions within the fuzzy logic system. Fuzzy logic approach was also used to develop a methodology for quantifying traffic congestion on expressways using primary traffic variables like density and speed [63]. The fuzzy inference system (FIS) was applied using data from Interstate 880 in California. It generated congestion indices that closely align with the Levels of Service (LOS) defined by the Highway Capacity Manual. The research highlights the insufficiency of single-variable congestion representation and advocates for a combined approach to accurately reflect expressway traffic conditions, especially under heavy congestion. It also recommends incorporating temporal and spatial aspects of congestion into the model for more comprehensive traffic control strategies.

A cloud-based intelligent traffic congestion prediction model empowered by a hybrid Neuro-Fuzzy approach was proposed by Gollapalli et al. [64]. This system aims to alleviate traffic delays at city junctions by leveraging data from embedded IoT sensors. It integrates cloud-based preprocessing with a Neuro-Fuzzy engine to enhance decision-making with reduced errors. The proposed model employed neural networks with fuzzy logic to optimize fuzzy membership functions through neural network training to minimize miss-rates and enhance traffic congestion prediction accuracy. According to the reported results, a superior performance of this hybrid neuro-fuzzy model compared to state-of-the-art algorithms was obtained, emphasizing its potential in real-time environments. Adaptive Neuro-Fuzzy Inference System (ANFIS) with artificial neural networks were used by Alawad et al. for real-time risk assessment by developing an intelligent framework to manage risk in railway stations [65]. The study introduced a crowd management system in station platforms, focusing on single-platform overcrowding,

essential for safety and service quality. The proposed model handles overcrowding risk by analyzing transfer efficiency and retention rates. Real data has been collected from several stations and utilized in this study to predict overcrowding in various locations. ANFIS model was used in to address traffic congestion problems at signalized road intersections in the South Africa transportation system [66].

The traffic density, vehicular speed, and traffic volume variables were used as decision variables by the proposed model, while triangular and Gaussian membership functions structured the algorithm. The ANFIS model achieved training and testing performances of 0.8722 and 0.9370, respectively. This study emphasizes the importance of updated traffic data including speed, density, and volume to understand traffic patterns.

A reinforcement learning approach using Genetic Fuzzy Systems (GFS) was used in [67] to facilitate smooth traffic flow. The paper proposed a cooperative automated driving system using GFS models and local information that can contribute to next-generation transportation system management. The Decentralized Cooperative Automated Vehicles (CoAVs) were trained to make individual decisions using GFS to merge cars into high-density traffic based on local information. The GFS model guides lane changes and acceleration, considering various CoAV behavioral parameters under different market penetration rates (MPRs) and congestion levels. The outcomes demonstrate the effectiveness of increasing the CoAV MPR and its potential for strategic cooperation with human-driven vehicles to enhance traffic performance.

To aid traffic operator decision-making during expressway incidents, a multi-stage Fuzzy Logic Controller (MS-FLC) was developed by [68]. The fuzzy-logic-based procedure focused on proactive ramp control during incidents to predict the traffic conditions. The MS-FLC involved two primary stages: evaluating traffic conditions at incident onset and predicting traffic conditions during incidents. The approach provided a systematic decision-support method that accommodates both proactive and reactive control strategies. The fuzzy-based methodologies help effectively managing uncertain and imprecise data which improved expressway incident management. The controller's design facilitated a comprehensive grasp of traffic conditions, offering operators easily interpretable linguistic expressions to assist in decision-making.

2.2 Fusion Methods

2.2.1 Sensor Fusion Methods

Fusing sensor data plays an important role in improving the accuracy of traffic management systems including traffic congestion and flow control [69]. It also has been utilized

with vehicles tracking and positioning [70–72]. To reduce the uncertainty of a single sensor, several researchers fused data from different sensors or sources. Yahia et al. proposed an approach for estimating and detecting traffic congestion based on unmanned aerial vehicles (UAVs) [73]. The proposed technique integrates speed-density sensor data with UAV observations using Kalman filter. The proposed method relied on the covariance matrices provided by a two-state ensemble Kalman filter (EnKF) and the A-optimal information measure (mean uncertainty) for the navigation task. Trinh et al. proposed an incremental Unscented Kalman Filter (UKF) for handling data from different sources for a real-time prediction of traffic problems [74]. The UKF has been compared to incremental Extended Kalman Filter (EKF) in terms of prediction accuracy and the reported results show similar performance of both algorithms. However, the proposed incremental UKF was more reliable due to its lower variance estimations, especially during free-flow times. The proposed framework was also able to anticipate speed and flow when there is a lack of data.

Kalman filter is combined with the K-nearest neighbor (KNN) for predicting passenger flow in public transport [75]. A dynamic weight coefficient adjustment was proposed for anticipating values produced by Kalman filtering and the KNN technique. Different patterns in data variance can be handled by both the KNN technique and the Kalman filtering. By giving more weight to the correct expected technique, the dynamic weight coefficient can more accurately forecast the outcome. A case study using real data showed that the proposed fusion technique outperformed Kalman filtering, the KNN approach, and the support vector machines in anticipating flow values of departing passengers and boarding passengers.

Various particle filtering (PF) methods have been utilized to address the state estimation problem for heterogeneous traffic [76]. This problem is related to vehicles with distinct driving behaviors. Three variations of PF have been introduced in this work by employing deterministic partial differential equations and additive state-independent process noise. The variations include parameter-adaptive PF, spatially-correlated noise PF, and a combination of the two. Numerical experiments involving complex traffic scenarios and real-world traffic data have been used to evaluate these filters. The reported results show that the enhanced PF variations achieved 80% and 46% accuracies with synthetic and real traffic data, respectively. The study marks a preliminary step for further research in heterogeneous traffic state estimation, highlighting potential improvements in accuracy through enhanced PF techniques, opening avenues for complex model observability analyses, and emphasizing the importance of realistic deployment settings and complex traffic data assessment. A column-based matrix decomposition

method was proposed to select crucial segment boundary measurements for the PF in traffic state estimation [77]. This method aims to reduce the measurement space without compromising estimation accuracy. Employing this method offers a trade-off between computational efficiency and estimation accuracy. This approach has been evaluated on a simulated road segment and the reported results show a 9.06% improvement when selecting half the original number of measurements while maintaining accurate traffic state estimations, offering potential benefits for larger urban road networks.

Mil et al. proposed a framework for travel time estimation by fusing various sensor-derived data through a modified Bayesian data fusion approach and Gaussian mixture models [78]. The study incorporates extra features to enhance the accuracy of the proposed model, such as traffic condition differences and individual sensor bias estimation. The proposed method performs traffic classification using Gaussian mixture models and travel time estimation via the modified Bayesian fusion approach. The model exhibits advantages over the basic Bayesian method, enhancing robustness against noisy data, reducing individual estimation biases, and providing more precise travel time estimates. Evaluation utilizes real-world and simulated datasets across distinct traffic conditions, including free flow, transitional flow, and congested flow regimes.

Bachmann et al. evaluated several multi-sensor data fusion techniques for freeway traffic speed estimation, comparing loop detectors and probe vehicle data via micro-simulation modeling [69]. The research aims to identify effective fusion methods, considering the impact of probe vehicle quantity and varying traffic conditions. The reported results show that most fusion techniques outperformed single sensor approaches. To reduce the uncertainty of a single sensor for traffic management, Gitahi et al. proposed a multi-sensor data fusion approach to improve travel time and average segment speed estimation [79]. Floating Car Data (FCD) from HERE, INRIX, and TomTom service providers have been fused. The proposed method utilizes virtual 250m sub-segments and assesses speed differences across datasets. Performance analysis against ground truth data showed the reliability of FCD for real-time congestion tracking with some limitations at intersections. The study emphasizes the necessity of long-term evaluations and defines acceptable error rates for different traffic scenarios. Multi-sensor fusion methods show potential in improving the robustness of traffic estimation and expanding temporal coverage. However, the selection of appropriate fusion algorithms should align with specific traffic management needs, taking into account computational resources and data quality. Table 2 presented the reviewed research on the implementation of AI models for congestion estimation models.

Table 2 Congestion estimation related literature studies adopted using the application of AI models

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[42]	Dataset	Convolution neural network	Proposes an OGCRNN that outperforms baseline methods on real-world datasets (e.g., 30-minute-ahead MAE < 2.6 mph). The authors plan to design an attention-based approach for building a fully dynamic Laplace matrix, enabling the model to adapt spatiotemporal dependencies in real time for more accurate traffic prediction	OGCRNN reduced MAE by over 1.5% vs. GCGRU and GRU; improved MSE by 2.3% vs. ARIMA; computational efficiency robustly verified across D.C., Philadelphia, PeMSD4 datasets for 15 and 30-min predictions
[43]	Not mentioned	Convolution neural network	Proposes an image-based CNN framework that directly computes a new “congestion level” (0–1 scale) from traffic camera images, outperforming traditional background-subtraction and SSD-based methods. Achieves up to ~99% precision and F1 in both sunny and rainy conditions, with fewer missed detections and reduced processing complexity for real-time congestion estimation	The proposed CNN-based method achieved precision of 98.74%, recall of 99.09%, and F-measure of 98.91%, outperforming traditional methods (MOG, Multilayer, PBAS) and SSD deep learning in traffic congestion estimation under varied weather and congestion conditions
[41]	Surveillance video	Fusion of Visual Features and Convolutional Neural Network	In a controlled dataset test, their method achieved 94% accuracy with MSE 0.03, surpassing Method A by 4%, though with a 0.2 s/inference runtime—highlighting a superior accuracy–speed balance for real-time applications	The third-order asymptotic method significantly improved accuracy, reducing the Kolmogorov distance error to 0.004 at $N = 10$ compared to 0.069 for Gaussian approximation, providing robust estimations of processing capacities for distributed physical data processing systems using queuing networks
[44]	Camera	CNN for detecting traffic congestion in CCTV	For the deep models, it was discovered that AUCs were higher than 0.9. This demonstrates that the models are also effective in difficult circumstances	YOLO achieved the highest accuracy (91.4%) followed by DCNN (90.2%) and SVM (85.7%) in traffic congestion detection from CCTV images. Deep models consistently outperformed SVM, demonstrating robustness across diverse camera configurations and weather conditions (day, night, poor resolution)."
[45]	Different data sources	RNN(LTSM)	Proposes MF-TCPV, combining LSTM-SPRVM and Fuzzy Comprehensive Evaluation, achieving a traffic congestion prediction accuracy of 92.36% on real-world Whitemud Drive data. It outperforms existing methods, clearly visualizes congestion levels, and highlights the need for integrating adjacent-road features via graph neural networks (GNN)	"MF-TCPV framework achieved 92.36% accuracy in traffic congestion prediction using LSTM-SPRVM, outperforming LSTM, CNN-LSTM, CNN-GRU, and LSTM-SVR models. It significantly reduced MSE and improved computational efficiency, validated on Whitemud Drive, Canada

Table 2 (continued)

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[46]	Not specified	SCAE-LSTM and STAR	STAR algorithm efficiently estimates traffic flow during peak hours with reduced computational complexity	"SCAE-LSTM with STAR algorithm predicted traffic flow up to 3 × faster than GRNN and 2.4 × faster than LSTM. It reduced path computation time for 200 OD pairs during peak hours, with stable performance across large-scale arterial networks
[47]	DTG (Dynamic Tachometer Graph)	GRU (Gated Recurrent Unit) neural network	The GRU-based model outperforms ARIMA and LSTM, with lower computational cost, for highway speed prediction	The GRU model achieved superior highway speed prediction, with MAE of 3.42 for bus lanes and 7.36 for regular lanes, outperforming LSTM (4.06, 8.37) and ARIMA (4.24, 10.61), while significantly reducing computational cost compared to LSTM
[19]	GPS devices on cars	Decision tree learning model (J48)	The decision tree model achieved 91% accuracy in detecting and classifying traffic congestion levels	The Decision Tree (J48) algorithm achieved an overall accuracy of 91% with a precision of 94% in classifying traffic congestion levels (light, heavy, jam), demonstrating effective performance for congestion detection using mobile sensor velocity data
[16]	GPS	Multivariate Linear Regression	Develops a hybrid approach combining support vector machines (SVM) with data denoising schemes (EEMD, EMD, MA, BW, Wavelets), significantly improving traffic flow prediction accuracy. The SVM + EEMD achieves the lowest prediction error, with MAE reduced to 2.59 and MAPE to 15.28%	The multi-logarithmic linear regression model achieved high accuracy ($R^2 = 0.878$) in predicting urban traffic congestion duration and average delay time across New York City's five districts, demonstrating reliability for urban main roads and highways
[25]	RTMS Sensors	Slope-based model using Van Aerde Fundamental Diagram	Uses speed-flow slope changes to forewarn congestion using sparse data with physical interpretability	Achieved 93.75% forewarning success rate over 16 weekdays; model thresholds: 35.5 km/h speed and 1,160 veh/h/ln flow; Van Aerde fit $R^2 = 0.95$
[17]	Inductor loops, remote microwave, Bluetooth, video, float cars with GPS navigation	Denoising schemes and SVM	The model performed better predictions than those without a denoising approach	The hybrid model (SVM-EEMD) significantly improved traffic flow prediction accuracy, achieving MAE reductions (e.g., from 4.19 to 3.38 vehicles/2 min) compared to SVM alone, validated on Minneapolis highway datasets, outperforming EMD, Wavelet, Butterworth, and Moving Average denoising methods
[54]	HD Map + Vehicle Data	Cell Transmission Model + Transformer (CeT)	Fuses cell-based modeling with Transformer to predict traffic states at signalized intersections using vehicle-type and signal-phase embeddings	CeT outperformed GWN and AGCRN with 13.36% MAE and 12.93% RMSE reduction, and 4.78% accuracy gain in 4-phase signals (MAE = 0.272, ACC = 84.36%)

Table 2 (continued)

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[18]	Cameras	SVM, RNN	Successfully predicted traffic density using SVM and RNN based on computer vision data	SVM predicted lane-wise traffic density with R^2 up to 0.63 (right lane, 1 min ahead). RNNs performed similarly but were less stable. Event-weighted predictions outperformed baseline models, demonstrating effective integration of visual data and incident records
[80]	Microscopic traffic simulation software by PTV Group for transportation analysis	Random Forests	Random Forests model accurately estimates travel time, outperforming other models with more variables	Random Forests significantly outperformed BP Neural Networks and polynomial regression, reducing MAD from 5.51 to 0.80 s for freeway travel time estimation, accurately capturing traffic states (free-flow, transition, congestion) with robust computational efficiency using VISSIM simulations
[21]	Not mentioned	Naive Bayes classifier for traffic prediction	The Naive Bayes models successfully predict congestion and incidents with desired accuracy using data-driven approaches based on ML techniques	Naive Bayes model achieved average precision of 0.56, recall of 0.85, and F1-score of 0.66 for congestion prediction across 35 bottleneck links in Brisbane. Incident prediction was less accurate ($F1 = 0.04$), varying by incident type
[53]	Loop Detectors	Hybrid FD-Markov-LSTM (Statistical + Deep Learning)	Combines FD for state classification, Markov for transition prediction, and LSTM for residual modeling	Achieved 39% lower MAE, 35% lower RMSE, and 7.4% lower MAPE than ARIMA, RF, KNN, and pure LSTM across 3 datasets (Los Angeles & Beijing)
[24]	Collected manually using data from Google Maps traffic flow	K-nearest neighbor	The accuracy value is 0.795 for $k=1$, meaning that the accuracy of the congestion data tested is 81%	The k-NN-based traffic flow prediction model achieved 79.4% accuracy, precision of 0.795, F1-score of 0.794, and AUC of 0.853 on data from the Mohammad Toha junction in Bandung, validated using MAPE and RMSE error metrics
[22]	MONICA sensors	k-Nearest Neighbor	For $k=1$, the accuracy result is 0.795, indicating that 81% of the congestion data examined is accurate	The SrOrkNNr model reduced RMSE by up to 5.79% compared to baseline kNN across four Amsterdam highways, achieving $R^2 > 0.95$ and outperforming SVR, ARIMA, ANN, Kalman Filter, Decision Trees, and LSTM in short-term traffic flow forecasting

Table 2 (continued)

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[23]	Not mentioned	K-nearest neighbor	Their model is suitable for real-time prediction due to its high prediction accuracy and swift execution	The distributed spatial-temporal kNN model achieved lowest MAE (0.1378) and MAPE (2.93%) compared to TDUD-KNN (MAE 0.1443, MAPE 3.02%) and SARIMA (MAE 0.1459, MAPE 3.01%), while ensuring real-time scalability using Apache Spark
[26]	Traffic sensing systems	Sparse Bayesian Learning	Provides an effective trade-off between computational time and NMAE	The SBL + Kalman model achieved NMAE < 3.5% at SI = 0.5, outperforming SVD (9%) and MLR-based methods. BSBL matched SBL accuracy at SI = 0.26. SBL + K offered the best trade-off between accuracy and computation, enabling real-time estimation on PeMS freeway data
[27]	Various vehicle sensors, like coils, ultrasonic sensors, video detectors, and piezoelectric sensors	A two-step model using a cost-sensitive Bayesian network and weighted K-nearest neighbor (KNN)	Proposes a hybrid Xception-SVM (XPSVM) model for short-term traffic congestion prediction, achieving 97.16% accuracy, precision of 98.21%, recall of 98.8%, and F1-score of 98.5%. It significantly outperforms models like Xception, InceptionV3, ResNet101, VGG16, and Mobile Net in error metrics	The cost-sensitive Bayesian + weighted KNN model reduced MAPE to 0.234 for long-duration accidents and improved classification accuracy by 3.8% over standard Bayesian networks, lowering misclassification of long events from 26.48% to 12.65% on Xiamen traffic data
[28]	Loop detectors, meteorological sensors, and cameras	A novel Bayesian	Outperforms cutting-edge methods while capturing the inherent uncertainty of the expected outcomes	VGRAN achieved lowest RMSE (2.32), MAE (1.53), and MAPE (3.7%) on METR; and RMSE (2.88), MAE (1.76), MAPE (4.2%) on PEMS, outperforming DCRNN, GraphWave, OGCRNN, APTN, and other state-of-the-art traffic prediction models
[29]	Traffic tracker dataset	Bayesian Network	Helpful in giving a complete picture of the speeds and vehicle counts, together with a probabilistic assessment of the congestion status	A probabilistic estimation of traffic congestion using Bayesian network
[33]	GPS data	Artificial neural networks (ANN)	Proposes an Optimized Graph Convolution RNN (OGCRNN), outperforming baseline methods with 30-min ahead prediction errors (MAE: ~2.12 mph, MAPE: ~6.31%) on real-world datasets. Plans future integration of an attention mechanism for fully dynamic Laplace matrices	The ANN-based A-Dijkstra model achieved lower MAPE than SVR and average-based models on Beijing taxi GPS data, with prediction error generally below 5%, enabling more accurate and time-saving urban path planning under real-time traffic conditions

Table 2 (continued)

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[34]	Road stationary sensors	Neural network	Provides a methodology for intelligently classifying the condition of traffic congestion using ANN	The ANN model accurately classified congestion into high, medium, and free flow using density and speed, matching field observations. It outperformed velocity-threshold models by avoiding misclassification under low-speed conditions unrelated to congestion, achieving 100% agreement on test cases
[35]	Vehicle sensors like coils, ultrasonic sensors, video detectors, and piezoelectric sensors	ML techniques, including Feed Forward Neural Networks (FFNN) and Radial Basis Function Neural Networks (RBFNN)	Finding: FFNN achieves 97.6% accuracy in predicting traffic congestion, outperforming other models in urban traffic analysis	The FFNN model achieved 97.6% accuracy with lowest RMSECV (0.01) and REP (1.00), outperforming RBFNN ($R^2 = 0.74$), polynomial regression ($R^2 = 0.78$), and simple regression ($R^2 = 0.08$) in predicting traffic congestion using real data from Hamra Street, Beirut
[36]	Not specified	Multilayered Perceptron (MLP) and Artificial Neural Networks (ANN)	predicts accidents using a neural network with an R2 accuracy measure (indicator of model fit) of 0.83 and an MSE parameter (mean squared error) of 0.59	The MLP model achieved R^2 of 0.83 (overall), 0.94 (Qazvin), and 0.98 (Hamadan-Malayer), with MSE ranging from 0.33 to 0.61. It accurately predicted crash counts across six Iranian roads using geometric and traffic flow features
[37]	The actual volume of traffic measured by microwave sensors installed on the expressways	Bayesian combination and deep learning	Ignoring additional sensors like those for weather, traffic accidents, speed, and occupancy	SOM-based ANN model identified 161 congested road segments with mean ATS-bus = 21.8 km/h and ATS_taxi = 26.0 km/h, highly correlated ($R^2 = 0.94$). Peak congestion occurred at 7–10 AM and 5–6 PM, guiding GA-based bus stop optimization
[39]	Not specified	Self-Organizing Map (SOM)	Proposes a SOM-based traffic model using online map data from 32 Swiss cities, accurately identifying common urban traffic patterns. Results quantify lockdown effects, with highway congestion decreasing ~ 30%, while traffic near city centers and borders increased by ~ 15%	The SOM-based model accurately recognized urban traffic patterns across 32 Swiss cities, achieving R^2 up to 0.85 and high categorical accuracy. It predicted traffic congestion levels and spatial patterns under varying conditions, including lockdowns, using real-time map data
[40]	Not specified	Self-Organizing Map (SOM)	Proposed methodology estimates congestion in road segments between bus stops, aiding public transport optimization, and validated with real-world data for sustainable improvements	SOM-based ANN identified 161 congested segments with ATSbus $\approx$ 21.8 km/h (validated by taxi data, ATS $\approx$ 26.0 km/h). High correlation ($R^2 = 0.94$) confirmed model reliability. PTN optimization via GA reduced total passenger travel time using real trip demand data

Table 2 (continued)

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[55]	ADS-B data	DBSCAN-K-means, Decision Tree	Predicts TOD with ML, error within 31 nautical miles	The decision tree model predicted Top of Descent within ± 31 NM (≈ 4 min) for A380 flights in Singapore FIR, using ADS-B cruise parameters. TODDIFF (micro) showed higher consistency than macro clustering, enabling reliable trajectory forecasting for CDA support The study did not report specific quantitative performance metrics
[56]	IoT-based ITM sensors	ML, DBSCAN clustering	The study demonstrates an IoT and ML-based traffic management system, significantly improving traffic flow and reducing congestion and accidents	
[59]	IoT-based ITM sensors	ML, DBSCAN clustering	The study demonstrates an IoT and ML-based traffic management system, significantly improving traffic flow and reducing congestion and accidents	VAE reduced RMSE by up to 69.6% (NMAR data) and 54.9% (MCAR-40) over PCA and AE, and improved forecast RMSE (7.49 vs. 7.51 raw) while compressing inputs by 44.6 $\times$, outperforming other imputation and feature extraction methods
[60]	IoT environment	Dempster-Shafer Theory	Proposes a modified Dempster-Shafer fusion method using improved Deng entropy, significantly reducing uncertainty (by $\sim 25\%$) and increasing data fusion accuracy ($\sim 90\%$), compared to traditional DS methods, by effectively handling redundant information in multi-sensor IoT environments	The DFUDS scheme improved fusion accuracy by $\sim 25\%$ over Dempster-Shafer theory and $\sim 20\%$ over Deng entropy, achieving higher belief confidence (up to 0.75) for target detection, while reducing redundant sensor data and energy consumption in smart environments
[61]	IoT-based ITM sensors	ML, DBSCAN clustering	Proposes a Mamdani fuzzy inference-based model for congestion prediction, utilizing freeway data from 2446 segments in Hungary. Results show rapid, reliable predictions (over 90% accuracy), effectively modeling complex, uncertain relationships between traffic flow, segment length, lanes, and congestion severity	"The Mamdani FIS model accurately predicted congestion levels using flow, length, and lane data from 2446 freeway segments in Hungary, achieving real-time estimation (e.g., LOC = 281 for flow = 253, 2 lanes, 5.16 km) with minimal computational cost

Table 2 (continued)

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[81]	Inductive loop detectors	Fuzzy logic-based system	The study introduced a fuzzy logic-based methodology for quantifying traffic congestion on expressways using speed and density data from inductive loop detectors. This approach allowed for a more nuanced and accurate representation of congestion levels, combining both speed and density data, and providing a clearer picture of traffic states compared to traditional methods. The methodology was validated using real-world data from Interstate 880 in California	The FIS model mapped speed and density into congestion index (0–1 scale) using 19 fuzzy rules. Compared to LOS measures, it provided smoother, more accurate transitions. $CI(v^*k)$ grew stably with LOS, outperforming single-variable indices $CI(v)$ and $CI(k)$
[62]	RFID technology	Fuzzy rule-based system	This study implements a fuzzy rule-based system for monitoring traffic congestion using RFID technology, enhancing accuracy and efficiency in traffic management	The system identified 54 vehicles with average speed 8.27 m/s and road occupancy 59.04%, classifying congestion as 'Light-Heavy.' Fuzzy membership values: speed (low = 0.173, middle = 0.769), occupancy (middle = 0.699), demonstrating accurate congestion monitoring using RFID and fuzzy rule-based logic
[63]	Traffic surveillance data	Multi-stage fuzzy logic controller	Enhances expressway congestion management through a fuzzy-based decision-making procedure, improving traffic control efficiency	The fuzzy-based decision-making procedure (FDMP) within MS-FLC outperformed ALINEA/Q in incident simulations, reducing ramp queues and improving total travel time, time spent, and mean density under high demand and severe capacity reduction scenarios on expressway networks
[64]	IoT-enabled sensors	Neuro-fuzzy approach	Proposes a hybrid neuro-fuzzy prediction model for smart-city traffic congestion management, achieving 99.21% accuracy during training and 98.72% during validation, outperforming existing models (91%–98%). It significantly reduces prediction errors (miss rate ~ 1.27%) using IoT-based data	The proposed hybrid Neuro-Fuzzy model achieved 99.21% accuracy (training) and 98.73% (validation), outperforming prior methods by up to 8% in smart city congestion prediction, with minimal miss rate (0.79%) and high sensitivity (99.59%) and specificity (97.72%)
[67]	Vehicle onboard sensors	Genetic fuzzy systems	This study improves freeway merge efficiency in mixed-traffic conditions using decentralized cooperative driving automation with genetic fuzzy systems	The GFS-based decentralized CoAV model improved throughput by up to 16.6% and reduced average delay by over 35% as MPR increased, enabling safe and efficient freeway merges using only local information under both aggressive and conservative lane change modes

Table 2 (continued)

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[65]	RFID and GPS	Fuzzy logic and neural networks	The study demonstrates improved traffic management and congestion prediction using RFID and GPS data, employing fuzzy logic and neural networks for efficient control	The ANFIS model achieved near-perfect accuracy ($R^2 = 1.0$, $MSE = 0.0000419$, $MAE = 0.0055$) using GPR regression for predicting overcrowding risk. It effectively assessed risk levels using real-time transfer efficiency and platform retention rate data from Beijing metro
[82]	Inductive loop detectors, video cameras	Adaptive neuro-fuzzy inference system	The study demonstrates effective traffic flow management at signalized intersections using ANFIS, significantly improving accuracy in predicting traffic volume based on vehicle speed and density	ANFIS model achieved R^2 of 0.8722 (training) and 0.9370 (testing) using vehicle speed and traffic density as inputs and volume as output, demonstrating strong predictive accuracy for optimizing traffic flow at signalized intersections in South Africa
[58]	IoT traffic sensors	Hidden Markov Model	The study applies a Hidden Markov Model to predict traffic flow impacts on neighboring roads using data from IoT sensors, achieving 89% prediction accuracy	The proposed HMM model with traffic state clustering achieved up to 92.5% accuracy (avg. 89.3%) in predicting neighboring road traffic states, outperforming HMM-Avg (80.5%), using IoT sensor data from Aarhus, Denmark without connected road information
[73]	UAVs + speed density sensory data	Dual state ensemble Kalman filter	When speed-density statistics are useless due to traffic congestion, UAV observations help find occurrences	UAV-EnKF reduced average density estimation error (δ) and achieved 100% incident detection under all traffic loads, outperforming EnKF and California algorithm. It accurately estimated incident parameters (u_s , p_s) even in congested conditions, improving robustness in non-recurrent congestion
[75]	Traffic data from public transport	Kalman Filtering Combined K-Nearest Neighbor	new method for short-term passenger flow prediction in urban public transport, combining Kalman filtering with K-nearest neighbor to enhance forecasting accuracy	The KK fusion method reduced MRE to ~3.3–4.7% and RMSE to 1.10–12.79 pax, outperforming KNN, KF, and SVM (MRE up to 15.5%). It achieved 92.1% hit rate within 10% error across five transit stations in Changchun
[76]	Vehicle class-specific sensors	Enhanced particle filters	Proposes enhanced particle filters (PF) for heterogeneous traffic estimation, achieving up to 80% error reduction on simulated data and up to 46% accuracy improvement on real-world data compared to standard particle filters, especially in complex overtaking scenarios	"Enhanced particle filters (PAPF + SCNM) reduced MAE by up to 56.6% (p_1) and 48.8% (p_2) over standard PF in overtaking scenarios, and up to 46% in real heterogeneous traffic, significantly improving estimation accuracy under complex vehicle interactions

Table 2 (continued)

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[78]	GPS, loop detectors, mobile data	Bayesian data fusion model	The study develops a modified Bayesian data fusion model to estimate travel time more accurately, addressing issues of spurious data and varying traffic conditions. It enhances travel time prediction by effectively combining data from multiple sensor types and adapting to different traffic scenarios	The proposed model reduced MAPE by 16.3% over baseline, achieving 3.44% overall MAPE and lower variance across simulated and real datasets. It outperformed median, mean, and Choi & Chung methods under free-flow, transitional, and congested traffic conditions
[79]	Radar, Bluetooth/WiFi/BLE	FCD data fusion techniques	"Proposes a multi-sensor data fusion approach combining INRIX, TomTom, HERE, radar, and Bluetooth/WiFi sensors, reducing congestion detection errors (MAPE ~ 8.4–20%) and significantly improving speed estimation reliability and accuracy compared to single-source methods, particularly during peak traffic periods"	"Multi-sensor fusion reduced FCD speed estimation errors to 8–20% (MAPE) on A8 Highway and 28–35% on Ajaccio urban roads. INRIX showed highest accuracy; fusion improved granularity, reliability, and congestion detection using 250m segments and real-time confidence weighting"
[69]	Probe vehicles, loop detectors	Algorithm Type: Various data fusion techniques	The research compares multiple data fusion methods for freeway traffic speed estimation using microsimulation modeling. It evaluates the effectiveness of fusing data from probe vehicles and loop detectors, demonstrating improvements in accuracy over single-sensor approaches under varying traffic conditions	Measurement Fusion Kalman Filter achieved lowest RMSE (≤ 1.70 km/h) across all traffic conditions with 40% probe penetration, outperforming loop detectors and probe data alone. Bar-Shalom/Campo and Simple Convex combinations also showed consistent accuracy improvements in free-flow and mixed states."
[70]	GPS, RADAR, cameras	Deep learning-based DBIF	The study introduces an effective smart vehicle tracking and positioning system using a deep learning-based discrete behavior information fusion (DBIF) model, showing significant improvements in accuracy, error reduction, and complexity management. This advanced approach utilizes a multi-sensor fusion technique, enhancing the capability of smart transportation systems in various real-world applications	"DBIF achieved 92.08% accuracy and 0.9741 fusion rate, reducing error to 0.0662, complexity to 0.0717, and fusion time to 0.9938s—outperforming EKF + SVM, ICP-DA, and TAAWUN in smart vehicle tracking and positioning across varied conditions"
[71]	Smartphone for data collection	Multi-sensor information fusion	Improves the accuracy ratio to 96.22% during the determination of the ideal journey time and data transmission to autonomous guided vehicles	IoT-AGV achieved 96.22% classification accuracy and lowest delay rate among five methods (e.g., 0.67s vs. 1.89s for CP-PNN), enhancing travel time estimation and decision accuracy in autonomous guided vehicles using DNN and real-time multisensor data fusion

Table 2 (continued)

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[72]	Taxi orders between Jan 1st, 2013 to Oct 1st, 2016	Multiple Spatio-Temporal Information Fusion Networks (MSTIF-Net)	The knowledge of urban ride-hailing demand is made more foreseeable by the combination of multi-graph information with pixel-level scenario information	"MSTIF-Net achieved lowest RMSE (17.23), MAPE (20.98%), and JS divergence (0.0231) on Haikou dataset, outperforming DCRNN, T-GCN, Conv-LSTM, and traditional methods by up to 7.45% RMSE and 7.94% MAPE in two large-scale urban datasets
[74]	Loop detectors (LD), GPS	Incremental Unscented Kalman Filter	This study presents a framework using the incremental Unscented Kalman Filter (UKF) and GPS data to estimate traffic parameters, offering accuracy and system complexity management	Incremental UKF achieved RMSE of 103.47 veh/h/lane (flow) and 5.36 km/h (speed), with MAPE of 5.41% and 6.82%, outperforming EKF in reliability during free-flow conditions while maintaining real-time performance (12 ms per step) on simulated LA motorway data
[77]	Induction loops, radar, GPS	Particle filtering with matrix decomposition	Reduced sensor data improves efficiency, maintains accuracy in traffic state estimation, beneficial for large-scale urban networks	Using only half of the measurements, the particle filter reduced computation time by 9.06% (from 23.83 to 21.67 min), while maintaining acceptable accuracy (RMSE for speed increased slightly: 12.10 $\rightarrow$ 13.05 km/h) on a 4km simulated road segment
[48]	Building sensors for real-time data transmission	Quantum Genetic Algorithm—Learning Vector Quantization (QGA-LVQ)	The study proposes a QGA-LVQ neural network algorithm for short-term traffic flow prediction, overcoming the limitations of LVQ neural network. It outperforms GA-BP and wavelet neural networks in accuracy and real-time performance, aiding in real-time traffic congestion prediction for smart city management	The Random Forest (RF) model achieved 87.5% accuracy in classifying traffic congestion states (smooth, congestion, blockage) using 1124 samples. It demonstrated fast computation and strong generalization with minimal parameter tuning and effective attribute importance analysis
[83]	Not specified	RF classification algorithm	Model predicts traffic with 87.5% accuracy, fast calculation speed, and strong applicability	The ANN-PSO model achieved R^2 of 0.9991 (training) and 0.9930 (testing), with RMSE as low as 3.196. It outperformed alternative configurations in modeling truck traffic flow on South Africa's N1 freeway using real-world data
[84]	Previously trained data sets	Machine Learning (ML) and Genetic Algorithms (GA)	When employed under incident settings, the new BGA-ML is substantially quicker than the old GA algorithm and can cut the overall journey time in half	The genetic algorithm-based signal optimization model reduced peak period average delay by 10.73%, travel time by 5.33%, and variance by 23.72%; during non-peak periods, delay dropped 4.47%, travel time 0.79%, and variance by 40.36% in Nanjing CBD

Table 2 (continued)

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[30]	Radio frequency identification (RFID)	Genetic algorithm	mobility improvements are achieved after applying the proposed model along	The genetic algorithm-based signal optimization model reduced peak period average delay by 10.73%, travel time by 5.33%, and variance by 23.72%; during non-peak periods, delay dropped 4.47%, travel time 0.79%, and variance by 40.36% in Nanjing CBD
[31]	Database of Tehran Police Department	A novel multi-objective genetic algorithm	According to the classification metrics, it performed better than ANN, SVM, and traditional DTs in classification tasks	The proposed NSGA-II rule-based method achieved 88.2% accuracy, outperforming ANN (85.4%), SVM (81.2%), CART (61.3%), and ID3 (44.6%). It also yielded high rule quality: support 0.79, confidence 0.74, and comprehensibility 0.53
[32]	GPS floating car data (FCD)	ACO	The study demonstrates that the "Community Opening" policy improves road network connectivity and commuting efficiency, reducing traffic congestion in major Chinese cities	Opening gated communities improved road connectivity by 9.43–29.80% and reduced short-trip commuting time by 1.51–3.11% and distance by 2.57–4.50% across Beijing, Shanghai, Guangzhou, and Shenzhen, effectively alleviating urban congestion and redistributing traffic flow
[52]	Not specified	Particle swarm optimization	The study's details on findings are not available in the snippet provided. More information is needed to summarize the findings accurately	The ANN-PSO model achieved R^2 of 0.9991 (training) and 0.9930 (testing), with RMSE as low as 3.196. It effectively modeled long and short truck traffic flow on South Africa's N1 freeway, outperforming other configurations via sensitivity analysis
[68]	Not specified	Fuzzy logic control	The study's specific findings are not available in the provided document snippet. Further details are needed for an accurate summary	
[49]	Transportation Research Data Lab (TDRL)	Improved seasonal autoregressive integrated moving average (ISARIMA) & genetic algorithm (GA)	Compared to other approaches, it performs predictions well since weather, incidents, and other factors can alter traffic flow statistics	The ISARIMA-GA model achieved RMSE of 16.18 and 87.21% accuracy, improving 14.7% over SARIMA and 6.7% over SVR, while reducing operation time to 12.4 min—matching SARIMA speed with ANNs-level accuracy in 15-min traffic flow prediction
[50]	Sensors for flow data	Deep Learning Model with Attention-Based Conv-LSTM	A class of attention-based graph neural networks, such as GMAN [47], offers an effective alternative to the conventional CNN-LSTM hybrid networks for traffic prediction on complex and extended road networks	The hybrid Conv-LSTM + BiLSTM model with attention achieved lowest MAE (4.09), RMSE (6.91), and MAPE (8.74%) on urban data, outperforming SVR, SAE, LSTM, DNN-BTF, and DCRNN in both urban and freeway scenarios with increasing accuracy at longer horizons

Table 2 (continued)

Research	Sensing	Algorithm type	Finding	Quantitative Performance Summary
[42]	Road segments, traffic measurements	Graph convolution RNN	The study introduces an optimized Graph Convolution Recurrent Neural Network (OGCRNN) for traffic prediction, which effectively captures spatial-temporal traffic data characteristics and outperforms existing methods in real-world scenarios	OGCRNN achieved the lowest errors: MAE (2.63), MSE (11.24), MAPE (4.84%) on D.C.; MAE (2.85), MSE (13.06), MAPE (5.41%) on Philadelphia; and MAE (17.13), MSE (825.1) on PeMSD4, outperforming GRU, GCGRU, SVR, RF, and ARIMA

3 Congestion Control

This section presents the congestion control methods documented in the existing literature. These methods are categorized into two subsections, namely: ML methods and fusion methods. In more understanding manner, Fig. 3 presented the whole divisions of the congestion control models.

3.1 Machine Learning and Data-Driven Approaches

3.1.1 Traditional Machine Learning Methods

Supervised ML algorithms are frequently employed for traffic congestion mitigation. For example, one approach to traffic congestion monitoring utilizes an improved k-nearest neighbors (kNN) strategy, as presented in the work of [85]. It integrates a piecewise switched linear traffic (PWSL) model with a Kalman filter for traffic state estimation, acting as a virtual sensor. The system employs kNN-based Shewhart and exponential smoothing schemes for congestion detection. The kNN algorithm assesses traffic state deviations from normal to identify congestion. Kernel density estimation is also used for setting nonparametric control limits. Another approach uses Xception Support Vector Machine (XPSVM), a hybrid classifier model to forecast Short-Term Traffic Congestion (STTC) at 5-min intervals over an hour [86]. The model combines SVM and Xception classifiers to provide accurate predictions by utilizing weight regularization and a precisely calibrated binary hyperplane mechanism. The photos used in this study were taken from Google Maps in Bangalore, Karnataka, India, and were retrieved using the Selenium automation tool. The results highlight how crucial precise short-term traffic congestion prediction is for traffic management. An approach aimed at training a ML algorithm i.e., logistic regression to predict whether a road is congested or not [87]. The training data is collected through two beacons installed at each corner of the road and uploaded to the cloud platform and helps in determining travel time and road density. The logistic regression result, which determines whether or not the route is crowded, can be utilized to provide drivers with real-time information via navigation systems in order to alleviate congestion.

3.1.2 Deep Learning Methods:

Deep learning models have several applications in traffic monitoring and control. A proposed smart road traffic congestion control system employs ML techniques, specifically Artificial Back Propagation Neural Networks (ABPNN) to manage and minimize traffic congestion [88]. The system uses ABPNN to predict traffic flow and congestion points.

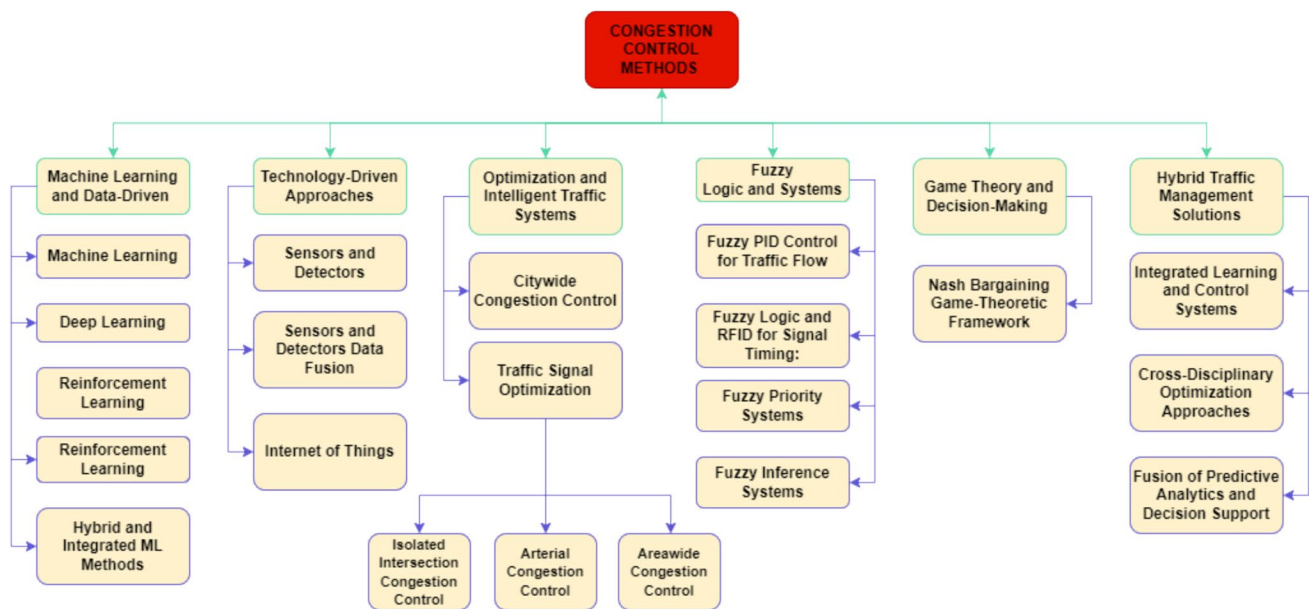


Fig. 3 The main adopted methodologies for congestion control

It integrates sensory data, processes it through the neural network, and identifies congestion points. The approach includes using RFID sensors for vehicle communication and Google Maps for alternate routing, focusing on enhancing traffic flow efficiency and reducing congestion. Another traffic management framework based on backpropagation neural networks aims at reducing urban congestion. It does so by recommending routes to users that contribute to overall congestion mitigation in the city [89]. The neural network is trained using the images extracted from Google Maps API. An application allows users to enter their beginning point and destination. The algorithm then divides the route into road segments and uses a trained neural network to predict traffic congestion. Subsequently, it proposes an alternative path to users in order to ease city congestion. ML models can be employed on the IoT based frameworks to mitigate congestion. An IoT based framework, FITCCS-VN, collects and exchanges data between junctions using IoV-enabled devices, which communicate via Vehicular Network (VN) [90]. Sensory data from wireless communication vehicles is preprocessed, trained, performance evaluated, and validated, with edge computing used for efficient processing. The training and validation phases use a dataset gathered from traffic control sensors on multiple VNs, with preprocessing techniques used to improve data quality. Traffic congestion predictions are made using ML approaches such as Artificial Neural Networks (ANN) and Support Vector Machines (SVM). The simulation results indicate that the proposed framework is better than the earlier proposed approaches.

Other neural network architectures besides feedforward neural networks have also been used to mitigate

congestion. For instance, LSTM neural networks can be used to predict and control congestion at the neighborhood level as proposed by [91]. They proposed an LSTM model which required four inputs to forecast a congestion score for a neighborhood. Their results indicated that congestion score can indeed be predicted using the four inputs. The authors further explained the usefulness of the model for traffic control schemes at neighborhood level. Similarly, another deep learning approach for traffic control aimed at developing a RNN-based model for predicting short-term traffic flow [92]. The study involved collecting real-time traffic data, preprocessing it for use in the model, and then training the RNN to forecast future traffic conditions. The model's effectiveness was assessed by comparing its predictions against actual traffic flow data, focusing on its accuracy in short-term forecasts. In addition, a novel framework has been proposed that uses deep learning to forecast traffic parameters such as speed and traffic flow etc., with the goal of improving traffic management in smart cities [93]. This framework makes use of both historical and real-time data from a variety of sources, including social media feeds, weather updates, and road status reports. Traffic data is evaluated and identified as congested or non-congested using Bidirectional Recurrent Neural Networks (BRNNs) with soft Gated Recurrent Units (GRU). This proposed methodology shows promise in terms of improving traffic prediction accuracy and efficiency, with possible applications including route optimization, traffic control, and congestion alleviation.

3.1.3 Image Recognition Methods

Image recognition has also been used to monitor and control traffic congestion. You Look Only Once (YOLO) is an object detection algorithm that has been employed for various purposes in the transportation sector. A deep learning-based framework that utilizes YOLO algorithm was proposed by [94]. The video cameras capture photos and videos, which are then processed using deep learning techniques to determine traffic intensity. This information is then used to adjust traffic signal timings, with the goal of effectively reducing delays and congestion. Another method aimed at optimizing traffic signal settings in real-time using YOLO to detect vehicle counts through video cameras covering each approach of isolated intersections [95]. The vehicle counts were then used to determine whether to adjust the signal settings depending on certain pre-defined thresholds. An approach to enhance traffic light control in smart cities using image and video processing techniques was proposed by [96]. In this approach, traffic data is collected via cameras, processed using Raspberry Pi and OpenCV, and analyzed to determine vehicle density. Traffic lights are then scheduled dynamically based on this data. Further, a novel framework attempted to reduce congestion by utilizing image processing and clustering technologies [97]. Images were collected using properly positioned cameras, and feature extraction was carried out using the KAZE features algorithm. Congestion was then classified as light, medium, or heavy using a clustering method. The captured image can be matched to these categories to assess the level of congestion. Such information can help authorities control congestion.

3.1.4 Reinforcement Learning Methods

Reinforcement learning includes agents learning to make decisions by interacting with their surroundings in order to maximize rewards. Reinforcement learning allows for the creation of traffic management systems that continuously improve decision-making, hence reducing congestion in urban areas. Since signalized intersections are the main bottlenecks in any city, reinforcement learning frameworks often mitigate congestion by optimizing signal settings at intersections. A reinforcement learning framework which leverages a deep Q-learning neural network for optimal action selection in traffic signal control aims to improve road network efficiency [98].

The approach includes a dynamic discount factor in the Bellman equation to address time step variability in reinforcement learning. The neural network is trained to produce optimal actions for traffic management, with preliminary results showing a 20% reduction in system total delay. The algorithm uniquely integrates actions into the input layer, focusing on a value-based reinforcement learning

model for efficient traffic signal control. Because single occupancy vehicles tend to increase road congestion, high occupancy vehicles are sometimes given priority. Human-Light is a decentralized adaptive traffic signal control algorithm that leverages reinforcement learning techniques. Its primary goal is to optimize people throughput at intersections by rewarding high-occupancy vehicle (HOV) commuters with travel time savings [99]. A proposed distributed multi-agent Q-learning framework optimizes traffic lights for both motorized and non-motorized traffic, considering real-time data from multiple intersections [100]. The system integrates real-world constraints and simulates performance using real-world maps and traffic data, demonstrating superior performance over existing solutions in terms of traffic flow and efficiency. These studies used distributed reinforcement learning approaches where the agents (i.e., intersections) work interdependently to learn their individual policies. Since the actions of an agent (i.e., an intersection) affect other intersections, a joint-action reinforcement learning framework was proposed where agents not only learn an individual policy, but they also learn in groups [101].

The comparison of the joint-action approach with the scenario when agents learn individually yielded that joint learning leads to lesser waiting times and thus less congestion. Similarly, a framework to address adaptive traffic signal control in multi-intersection systems using a cooperative deep Q-network with Q-value transfer (QT-CDQN) was proposed by [102]. This approach models a multi-intersection traffic network as a multi-agent system, where each agent (intersection) employs a deep Q-network to control its traffic signals. The innovation lies in integrating Q-value transfer among agents, enhancing cooperation and optimizing traffic flow across the network. The algorithm is tested under various traffic conditions, demonstrating its efficiency and adaptability to dynamic traffic environments.

A Q-learning-based traffic signal control (TSC) framework was presented for managing traffic in smart cities by maximizing vehicle throughput at intersections and adjusting signal timings [103]. This system, adaptable to changes in intersection structures, employs throughput and standard deviation of queue lengths as key parameters. Unlike model based TSCs, the proposed system dynamically adapts to real-time traffic conditions, offering a flexible solution suitable for various intersection types. Another reinforcement learning-based approach for traffic signal control in smart cities was proposed by [104]. It introduces a model that is adaptable to different traffic network topologies without requiring frequent retraining. The model uses reinforcement learning, specifically Deep Q-learning, and incorporates a novel state design that is invariant to road length. This approach allows the model to efficiently adapt to varied traffic conditions and network configurations. The goal is to optimize traffic signal control for enhanced traffic

flow, reducing congestion and improving sustainability in urban environments. The research emphasizes adaptability, environmental efficiency, and the use of advanced ML techniques in traffic management.

Traditionally, the traffic signal optimization problem is formulated as a single-objective problem with the objective of minimizing delay or maximizing efficiency. However, the problem of traffic signal optimization was extended by formulating it as a multi-objective problem to simultaneously optimize safety, efficiency and decarbonization [105]. A deep reinforcement learning based adaptive traffic signal control system was proposed that integrates the Dueling Double Deep Q Network (D3QN) framework. The application of the framework on a simulated intersection showed that the system outperforms the traditional adaptive traffic signal control approaches.

3.1.5 Hybrid and Integrated Machine Learning Methods

Traditional ML techniques are sometimes integrated with optimization approaches to provide effective congestion control frameworks. A novel traffic control optimization methodology integrates ML algorithms with Genetic Algorithms (GA) for traffic signal timing optimization at urban intersections, particularly for non-recurrent traffic incidents [84]. The approach starts with a typical GA, optimizing phase durations to minimize total travel time. Then, regression models predict travel times, with the best model further refined. The result is a new algorithm, BGA-ML, which blends GA with the selected ML model (Extreme-Gradient Decision-Tree) for faster, more effective traffic control under incident conditions, demonstrating considerable time reduction in simulations.

Similar to the integration of traditional ML methods with optimization approaches, deep learning and reinforcement learning are often used in conjunction with other techniques including optimization and game theory to improve traffic flow and reduce congestion. One such framework uses a reinforcement learning approach by adopting a two-mode multiagent architecture to evaluate both independent and cooperative procedures for controlling congestion [106]. For each intersection, Q-learning was employed for independent mode and game theory was employed for cooperative mode. Experimental results indicated that cooperation between the agents in a multiagent system is essential to control congestion.

Reinforcement learning algorithms differ from analytical approaches for predicting traffic and reducing road congestion because they adapt dynamically to changing settings and learn optimal tactics through interaction. Unlike analytical approaches that rely on established models and assumptions, reinforcement learning agents learn via experience, modifying their actions in response to feedback from

the environment. However, the capabilities of analytical approaches can be combined with reinforcement learning to enhance traffic management. One such approach called ‘analytically guided reinforcement learning’ aimed at combining the capabilities of an analytical technique with deep reinforcement learning to optimize traffic signals, which outperformed both ML and analytical approaches in reducing congestion [107].

An integrated congestion control approach for vehicle traffic networks using a pattern-learning controller with memory and prediction (PLMP) was proposed by [108]. It employs deep learning to recognize and exploit spatial and temporal patterns in traffic, focusing on metropolitan grid networks. The method involves storing recurring traffic patterns and predicting future patterns to optimize traffic light sequences, thereby reducing congestion. The study utilizes two synthetic datasets to evaluate the performance of the proposed model against various metrics and compares it with other baseline methods.

3.2 Optimization Techniques for Congestion Control

3.2.1 Optimization Techniques for Citywide Congestion Control

Optimization techniques play an important role in urban traffic management. The problem of reducing congestion can be formulated as optimization problems such as maximizing throughput at intersections, minimizing delays for vehicles, and optimizing route selection etc. For instance, an optimization-based framework was developed to control congestion [109]. The fitness function consisted of average waiting time for all the vehicles in the road network and was minimized using the genetic algorithm. In addition, ACO has been frequently used for traffic congestion control. ACO involves artificial ants exploring trails through a network. They leave a "pheromone" trace, which indicates path quality. Shorter trails accrue more pheromones, which attract subsequent ants. This feedback loop directs the colony to the best course for activities such as traffic light regulation. A framework consisting of a multi-agent ACO algorithm to create an effective vehicular traffic management system was proposed by [110]. It involved simulating vehicles as agents, using ACO for optimal path finding in traffic networks, and applying dynamic pheromone updates for real-time traffic management. In another study, an approach based on IoV and ACO was proposed to implement an effective traffic control system [111]. It involved segmenting street maps, applying ACO for route optimization, and using fuzzy logic for traffic intensity modeling. The method is aimed at optimizing route selection and traffic flow leading to reduced congestion.

A model for traffic congestion control using a Concurrent Switching Dynamics (CSD) approach was presented by [112]. It focuses on optimizing traffic signal timing in a network of interconnected roads by modeling traffic evolution as a system with cyclic and deterministic transitions. The method involves solving a constrained receding horizon optimization problem with quadratic cost and linear constraints, using traffic density and boundary inflow data. This approach aims to enhance traffic flow efficiency and reduce congestion in urban areas.

3.2.2 Isolated Intersection Optimization for Congestion Control

Signalized intersections are the main bottlenecks in urban areas. Poorly designed traffic signals at these intersections cause delays to vehicles and other road users. In the worst-case situations, long queues may reach the upstream intersections causing congestion to propagate at the network level. Therefore, the problem of optimizing traffic signal settings at intersections has long been a topic of interest for researchers and practitioners. For instance, an optimization problem was formulated to optimize signal timings and signal phasing by minimizing average delay for three-leg intersections [113]. It was solved using a population-based metaheuristic, differential evolution. The simulation results indicated that the proposed optimization problem outperformed fixed-time signal control and actuated signal control for sample scenarios. ACO algorithm to determine the next phase of the traffic signal [114]. The ACO algorithm was applied to the data collected through the wireless sensor network. The simulation results showed that the system outperformed the pretimed traffic controllers. Several hybrid approaches using ACO algorithm have been proposed to optimize signal timings at signalized intersections. For example, dynamic ACO algorithm with the look-ahead mechanism (DALM) is one such real-time traffic signal control approaches whose data requirements are rather small compared to other approaches [115]. Using the lookahead mechanism along with the current vehicle queue, the framework is able to predict vehicle queue lengths and consequently the optimal phasing scheme. The simulation results indicate that the framework can greatly improve the efficiency of signalized intersections. Aside from ACO and genetic algorithms, several other optimization algorithms have been used for traffic signal control. A study formulated two optimization problems to optimize signal settings in order to reduce delay and congestion at signalized intersections [116]. One optimization problem aimed at optimizing the phase plan, sequence and signal timing, whereas the other aimed at optimizing the cycle length for a fixed-time signal control. The former was solved using flower pollination algorithm (FPA) and the latter using differential evolution algorithm. The comparison with the base

model indicated that the flower pollination algorithm effectively reduces delay and congestion at isolated signalized intersections.

Traditionally traffic signal settings are optimized by minimizing delays for vehicles. More recently, researchers and practitioners have started focusing on minimizing delays for people. For instance, a mixed-integer nonlinear optimization problem was proposed to minimize person-delay at an isolated intersection while prioritizing transit vehicles in conflicting directions [117]. The application of the system at an isolated intersection in Athens, Greece showed that the system effectively minimizes person-delays while giving priority to transit vehicles in conflicting directions.

3.2.3 Arterial Intersection Optimization for Congestion Control

While optimizing signal timings at isolated intersections is important to minimize congestion, designing traffic signals independently along an arterial may result in unnecessary delays for certain vehicles. In fact, intersections along a corridor can be optimized to create a green wave, allowing vehicles to encounter green signals at consecutive upcoming intersections. For example, an optimization problem was formulated to improve the flow along an arterial by optimizing the traffic signal control parameters [118]. The decision variables included signal cycle, green light time, and phase offset. The formulated optimization problem was solved using the artificial bee colony algorithm resulting in an improved traffic control along the arterial. Many traffic signal optimization approaches require traffic detectors which are often expensive to install and operate. A large-scale traffic signal optimization approach for fixed-time traffic signals was developed which uses a small percentage of vehicle trajectories and does not require detector data [119].

The optimization programs were developed for both isolated and coordinated intersections and were solved using gradient descent and genetic algorithms, respectively. A practical test of the system was carried out in Birmingham, Michigan, revealing decreases in both delay and the number of stops at signalized intersections by up to 20% and 30%, respectively. Traffic signal coordination along an arterial is often formulated as a bandwidth maximization problem. The term bandwidth refers to the maximum amount of green time that vehicles can pass uninterrupted over a coordinated arterial. One such optimization problem aimed at maximizing the sum of weighting bi-directional progression bandwidths for double-cycle signal control [120]. The simulation results indicated that the proposed model is effective at reducing average delay compared to an earlier proposed revised MULTIBAND model.

Real-time signal coordination along corridors is frequently required to adaptively adjust signal settings at

coordinated traffic signals in response to fluctuating traffic conditions. For this purpose, a framework was proposed for real-time arterial coordination control based on dynamic intersection turning fraction estimation [121]. The framework consisted of three sub-models: the first optimization model aimed at estimating the dynamic intersection turning fractions, the second optimization model aimed at minimizing the delay to optimize the time-varying public cycle for the arterial road based on the estimated turning flows from the first model; and the third optimization model aimed at achieving the optimum real-time offset and split for the arterial road. The first and third models were solved using two revised genetic algorithms, whereas the first model was solved using an ordinary algorithm. The simulation results indicated that the proposed framework outperforms the existing methods.

As with the emphasis on reducing person-delay rather than vehicle delay at isolated crossings, reducing person-delay continues to be a topic of attention for optimizing signal control along corridors. For example, a mixed integer linear program for minimizing person-delay at signalized intersections along a corridor was formulated by [122]. Their approach not only accounted for vehicle occupancy but transit vehicles when traveling in conflicting directions. Further, their approach aimed at prioritizing transit vehicles as well as auto platoons. An application of the proposed approach at a four-intersection segment of San Pablo Avenue arterial, Berkeley, California showed that the system outperformed the traditional pretimed traffic signal.

3.2.4 Areawide Intersection Optimization for Congestion Control

Several proposed frameworks focus on either isolated intersections or intersections along a corridor, however, optimizing signal settings at the regional level is also a matter of concern for practitioners and researchers. For this purpose, a graph convolutional network framework was presented for adaptive signal light timing at the regional level [123]. Initially, a multi-intersection traffic flow prediction model, namely, A-GCN⁺ is used to extract spatial and temporal features of the traffic flow. Then, a dynamic regional traffic signal coordination optimization control method utilizes the output and combines it with an improved whale optimization algorithm (IWOA) to obtain the optimal solution for the regional average vehicle delay model. Finally, a multi-strategy fusion graphical method is used to obtain the dynamic main line bidirectional green wave. The experimental results show that the proposed framework effectively increases the regional network traffic efficiency. Another study proposed a model for optimizing traffic signal timing across multiple intersections, aiming to minimize the combined product of total travel time and variance in travel time per unit distance

in an urban network [30]. The approach is based on user equilibrium traffic conditions. A genetic algorithm is developed to solve this multi-dimensional optimization problem, leveraging PARAMICS simulation software for area-wide microsimulation. The model was validated in the Central Business District of Nanjing, China.

The optimization of traffic signal settings involves multiple, often conflicting objectives, such as minimizing delays and maximizing capacity for vehicles. Consequently, numerous research studies have attempted to formulate the traffic signal control problem as a multi-objective optimization problem. For instance, a multi-objective optimization approach was proposed to minimize delay at signalized intersections in road networks [124]. The objective functions consisted of signal control delay and traffic capacity. The conflict delay caused by the conflicts of right-turning vehicles crossing the straight-going non-motor vehicles was also considered in the model. The formulated optimization problem was solved using NSGAIII-DAE algorithm. The results of the numerical experiments, based on the actual traffic data from intersections in Jinzhou, China, indicated that the signal control delay and conflict delay are significantly reduced by the proposed approach compared with the existing algorithm.

3.3 Technology-Driven Approaches for Congestion Control

3.3.1 Sensors and Detectors for Congestion Control

There are various intrusive and non-intrusive sensors and detectors that are used for traffic monitoring and congestion control. These may include inductive loop detectors, WIFI devices, LiDAR, RADAR, RFID, magnetic sensors, and cameras etc. [7]. The effectiveness of traffic detectors along with the published messages through variable message signs (VMS) in reducing congestion has been evaluated by [125]. By collecting the published messages data and the traffic data from the detectors, they evaluated the effectiveness of two types of published messages, namely, guidance messages and notice messages. The results indicated that guidance messages were always more effective in reducing congestion, particularly under severe congestion. An Adaptive Traffic Light Control System (ATLCS) that uses magnetometer sensors to detect vehicles at intersections was proposed to reduce urban road traffic congestion in smart cities [126]. The system's core concept involves synchronizing multiple traffic lights across consecutive junctions. This synchronization is dynamically adjusted based on the number of vehicles at each junction, aiming to minimize stop-and-go driving, thereby reducing overall travel time. ATLCS introduces intentional offsets between green lights

at consecutive junctions in a specific direction, responding to real-time vehicular density at each junction.

Similarly, the use Wi-Fi Direct-enabled smartphones as embedded devices in vehicles to improve traffic flow has been evaluated by [127]. They proposed an intelligent vehicular network framework for smart cities, focusing on efficient route selection in traffic management. The study leverages a vehicular ad hoc network to gather and process traffic data. The system dynamically selects routes based on real-time data from neighboring vehicles, aiming to reduce travel times and manage traffic congestion. The framework is evaluated using different fixed and dynamic route-selection algorithms, assessing resource usage, transmission delay, packet loss, and overall travel time. The results demonstrate reduced travel times, highlighting the framework's potential for improving traffic management in urban areas.

RFID technology has been employed in a variety of transportation applications. For instance, an intelligent traffic management expert system was proposed that uses RFID technology to collect and compute average speed and flow statistics for each road in a city district [128]. It then sends congested road information to the district center server using a communication tool. Using a flooding algorithm, servers share and update congestion information between district centers, allowing dynamic navigation systems to discover the best routes around congested roadways.

3.3.2 Sensors and Detectors Data Fusion

In order to provide a holistic picture of traffic conditions, sensor data fusion for road traffic congestion control combines data from several sensors installed in road networks. Sensor data fusion for traffic control unifies information from multiple sources, including GPS, radar, video cameras, loop detectors, and vehicle-to-infrastructure (V2I) communication systems. For example, an adaptive traffic congestion control system has been proposed that utilizes a combination of RFID technology and image processing to detect emergency vehicles and adjust traffic signals accordingly to facilitate their movement. The approach aims to reduce response times for emergency services while managing overall traffic flow more effectively. Another approach was developed a road traffic congestion elimination framework based on the fusion of millimeter-wave radar sensor and traffic state discrimination algorithms i.e., combined spectral clustering and neural network algorithm [129]. The real-world application indicated that the system can effectively reduce congestion by 20% compared to the traditional pretimed traffic signals.

While most of the adaptive signal design approaches require detectors data, a study aimed at optimizing traffic signal settings by using sample travel time information collected from independent data sources such as GPS/Bluetooth/Wi-Fi sensors [130]. The proposed framework's aims

to minimize the total control delay at an isolated intersection using the concept of shockwaves to estimate the optimal-phase durations based on the previous phase durations. The results showed that the proposed framework reduces delay effectively and requires sample data from only four probe vehicles per phase.

3.3.3 Internet of Things for Congestion Control

IoT uses interconnected devices and sensors to gather and share data in order to improve traffic flow. IoT facilitates real-time analysis and decision-making to reduce traffic congestion by continuously monitoring vehicle movement, speed, congestion, environmental conditions, and infrastructure status. A V2I (vehicle-to-infrastructure) communication technology-based framework was developed to mitigate traffic congestion [131]. Under this proposed framework, the infrastructure such as traffic signal controller provides signal timing information to the vehicles and the vehicles, in turn, adjust their speeds and other actions to cross the intersections with minimal delays. The simulation results indicated that the proposed framework results in significant reduction in average travel times.

IoT-based traffic solutions are frequently used on a regional scale to realize their full potential for congestion reduction. An area-based traffic management and congestion control strategy leveraging fog computing and IoT was developed by [132]. It divides urban regions into traffic management areas, using real-time and static data to manage traffic flow and signal lights, aiming to alleviate congestion. The approach involves dynamic traffic tightness and social features analysis to guide traffic effectively. Similarly, an Intelligent Traffic Management System (ITMS) based on the Internet of Vehicles (IoV) has been proposed [133]. It focuses on improving traffic flow and reducing congestion at intersections in smart cities. The key advancement is an adaptive algorithm that dynamically adjusts traffic signals based on real-time vehicle data. This system prioritizes traffic flow optimization, fairness, and reduced commute times, giving special attention to emergency vehicles. The ITMS operates by processing data from vehicles' communication with smart traffic signals, thus enabling efficient and flexible traffic management. An edge cloud centric IoT framework for traffic management was proposed and evaluated by [134]. According to this framework, loop detectors collect vehicle counts, whereas RSUs collect real-time mobility data from cars. This traffic information is then relayed to the cloud. Then, the time required for each vehicle to reach the signal point along the lane and the set of shortest paths to vehicle destinations are determined. Furthermore, the data is then analyzed to predict waiting queue lengths and wait times in order to estimate how long the green phase will last. All this information is then conveyed to the vehicle through a

navigation system. The predicted green phase duration is communicated independently to the appropriate signal controller, which modifies the green phase timing at the relevant signal point accordingly.

IoT-based traffic management solutions can be combined with data analytics to improve traffic flow and reduce congestion as proposed by [135]. The suggested architecture consists of four layers. The first layer, known as the sensing layer, integrates data from numerous sources to ensure precise and reliable measurements of street traffic conditions before transmitting it to the next layer. The second layer, known as the fog layer, uses this data to make informed decisions before passing it on to the third layer. The third layer, the cloud layer, securely stores this data for analysis and insights. Finally, the fourth tier, the services layer, provides support for traffic control. The suggested framework divides congestion into four levels. Traffic signal settings are adjusted at the outset to relieve congestion. At the second level, traffic is diverted toward less crowded streets to relieve congestion. Moving on to the third level, congested street data is archived to help city planners make decisions. Finally, at the fourth level, assistant applications use this information to relieve congestion by facilitating various tasks.

3.3.4 Advanced Integrated Approaches for Congestion Control

To mitigate traffic congestion, a variety of technical advancements, such as the IoT and cloud-based services, are integrated with various strategies, such as game theory and data-driven techniques. For example, a smart road traffic congestion control system that uses a supervised ML algorithm along with IoT was proposed in [88]. It utilizes IoT-enabled devices for data collection and support vector machine (SVM) for analyzing traffic congestion in a smart city. The system dynamically adjusts traffic signal timings based on traffic density and aims to control or minimize congestion effectively. Similarly, another framework that uses image processing with IoT was proposed aimed at mitigating traffic congestion caused by illegal parking [136]. The system uses an image processing technique to identify vehicles, and IoT for communication. If a vehicle remains illegally parked, details are sent to the police for action. The approach aims to mitigate traffic congestion by addressing the root cause of illegal parking. The Traffic Density-Based Congestion Control Algorithm (TDCCA) for VANETs has been presented in [137]. It utilizes vehicle identification-based methods to manage congestion by modifying beacon signal frequencies according to traffic density. The approach involves calculating the traffic congestion density and adjusting the communication frequency to prevent network overload. This method is aimed at improving the

efficiency of message transmission in high-density vehicular environments.

A framework was presented to optimize traffic flow at multiple intersections by using cooperative game-theoretic approach along with IoT [138]. By leveraging IoT technologies, intersection controllers are viewed as smart agents capable of communication and coordination. A distributed merger and split algorithm for coalition formation among these agents is developed to dynamically control traffic lights. This method involves real-time data analysis and decision-making to minimize vehicle waiting times. The study's approach entails modeling connected intersections, developing a traffic simulation framework, and implementing the cooperative game-theoretic algorithm to evaluate its effectiveness in various traffic scenarios.

Several route guidance systems have also been proposed to regulate traffic flow. For example, a route guidance system called Flow-based Capacity-aware Rerouting (FCR) was proposed in [139]. It computes detour paths in units of flow rather than in units of individual vehicles while considering the capacity of detour routes. Similarly, another advanced route guidance system called ReFOCUS + has been simulated and evaluated for congestion mitigation purposes [140]. It is a dynamic semi-distributed, multi-layer, and Fog-Cloud based advance route guidance system that calculated different traffic parameters using roadside units. This information is then used to reroute vehicles to mitigate traffic congestion. The simulation results indicated that the proposed system outperforms existing solutions in terms of travel time savings.

3.4 Fuzzy Logic and Systems

3.4.1 Fuzzy PID Control for Traffic Flow

Since fuzzy logic shows potential to increase the capacity of roads via automated highway system, some researchers attempt to integrate fuzzy methods with other algorithms. Many research groups proposed adaptive control architectures that seamlessly combine fuzzy logic with standard PID control technology. Adaptive control architectures underscore the controller's inherent capacity to dynamically adapt and respond to evolving conditions, thereby ensuring platoon's resilience. Studies have been conducted on fuzzy controllers to regulate the inter-vehicle distance control in a vehicle platoon. For example, traffic flow control strategy is presented for managing ramp metering at remote downstream congestion points, utilizing a combination of fuzzy logic and PID control to accurately regulate traffic flow [141].

. Ramp meters are traffic signals placed to alleviate congestion by controlling traffic flow entering the freeway and breaking up clusters of vehicles. In this research, FSAPID control strategy is introduced for ramp metering at distant

downstream bottlenecks, integrating fuzzy logic and PID control for precise traffic flow regulation. The effectiveness of the strategy is evaluated through simulation of three downstream bottleneck scenarios using a cell transmission model. Additionally, a road network boundary control method combined with FR-PID is presented in [142]. This method selects the actual road network for verification, which proves FR-PID method has better control effect compared with the traditional control methods.

Other solutions to the overload and congestion problems are presented in [143] 144. These solutions utilize fuzzy adaptive PID control mechanism based on network size. Simulation results are analyzed and show that the mechanisms proposed perform well in terms of convergence speed, stability, and robustness. Fuzzy PID schemes are also utilized for their effectiveness on vehicle dynamics-based applications of fuzzy control [145–147]. There researchers designed PID controllers and compared them with fuzzy-PID controllers. The characteristics and effects of control strategy were further validated and analyzed to verify the feasibility of the schemes in real vehicle tests. An autonomous vehicle dynamic model is developed to verify its accuracy and rationality through simulation experiments [148]. The experiment results show that the model improves the transient response in consideration of the resistance, the vehicle's own transmission efficiency, transmission ratio, engine inertia, and other parameters. Comparative study for an automobile cruise control system [149] is conducted to correlate the effectiveness of PID controllers, fuzzy logic controllers, and PID controllers optimized by GA. The comparison was for the transient performance in addition to the steady state performance i.e. steady state error. The results indicate that the performance of fuzzy controllers have better response regarding the overshoot and the settling time while the PID controllers tuned by using GA gives the shortest rising time.

3.4.2 Fuzzy Logic and RFID for Signal Timing:

Since traditional traffic control systems are not capable of controlling dynamic environments, radio frequency identification (RFID) technology is integrated with fuzzy logic to identify vehicles and hence evaluate their average speed, queue length and waiting time. This integration improves the field of traffic light control system using fuzzy logic. For example, fuzzy logic-based traffic congestion control system was proposed and developed [150]. It utilizes RFID and IR sensors for traffic data collection and employs fuzzy logic for dynamic traffic signal timing adjustment based on real-time traffic conditions. In harmony with other established research [151, 152], the finding was to argue that traffic signals can be the root cause for bottlenecks due to an unfair timing balance. Solving this problem, adaptive traffic

signal controllers are proposed to calculate the phase durations based on the conditions monitored by on-road sensors.

The problem of urban traffic congestion resulting in long waits at intersections is explored in [153]. This paper surveys intelligent traffic signal systems as a solution to help alleviate this problem. This solution incorporates technologies such as fuzzy logic to adapt to real-time traffic conditions, making better use of existing road infrastructure. Intelligent traffic systems are investigated to efficiently manage on-road traffic in real-time, reducing delay at intersections and maintaining road user safety. Despite this, emergency vehicles often struggle to meet response times. Intelligent traffic systems are also susceptible to various attacks, including cyberattacks. To address these challenges, three techniques to enhance the reliability and quality of intelligent traffic systems namely, innovative emergency vehicle priority system, an intrusion detection system, and trust model to assess the trust level of self-driving cars. The performance of the proposed systems is evaluated under various realistic scenarios and the simulation results demonstrate the potential effectiveness of the proposed techniques across all scenarios [154–156]. Intelligent and adaptive systems to promote efficient trade and industry development have been proposed. The studies concluded that (a) traffic congestion is a significant issue leading to wasted time, health risks, and environmental pollution and (b) effective management of road capacity, utilization of traffic control strategies, and implementation of intelligent transport systems are crucial. These articles delve into the importance of real-time traffic signal control technology in addressing congestion at road junctions in smart cities. This highlights the feasibility of implementing interconnected traffic signals to enhance coordination and reduce congestion within a specific area and shows that the benefits of synchronizing traffic signals on key routes enhance the overall smooth flow of traffic at intersections [157].

3.4.3 Fuzzy Priority Systems

Priority-based systems leveraging fuzzy logic and routing algorithms have been utilized to efficiently manage and route data in sensor networks and transportation systems [158–160]. Fuzzy logic is used to handle data uncertainty and make intelligent decisions based on imprecise or incomplete information while priority-based routing algorithms prioritize certain types of data or nodes over others to ensure timely and reliable delivery of critical information in the network. This combination allows fuzzy priority systems to adapt to changing conditions and make optimal routing decisions to improve overall performance and energy efficiency [158–160]. Systems are also developed to control traffic congestion and manage emergency vehicles using fuzzy priority-based strategies. Such systems incorporate

congestion-aware routing algorithm to enhance the transmission of traffic data and utilize fuzzy logic to prioritize traffic phases for congestion management and emergency vehicle coordination [27].

Real-time monitoring with high density of vehicles is an issue investigated in [28]. To tackle this issue, authors propose a protocol with the preemptive-resume priority scheme. The protocol gives priority to packets with higher priority levels, reducing delays of data transmission. Additionally, queuing theory is applied to calculate the total delay for vehicles to a drone receiver, optimizing the transmission interval and minimizing delay to ensure safe driving conditions are met. The effectiveness and benefits are demonstrated through simulation experiments. For vehicles to move smoothly on the road, they rely on traffic signals. However, the fixed length of signal timings can lead to traffic issues, especially when emergency vehicles are not given priority. To address this issue, researchers in [161] propose a new dynamic traffic emergency scheduling algorithm for a four-lane intersection. The algorithm uses an object detection model to differentiate between emergency and normal vehicles on the road. By considering factors such as vehicle density and waiting times, the algorithm prioritizes emergency vehicles, leading to a reduction in average waiting times and leads to optimized signal timings.

Further studies are conducted and focused on investigating the utilization of priority-based constrained routing for road vehicles in Intelligent Transportation Systems (ITS) using ML techniques and dynamic routing algorithms. The objective of these studies is to improve traffic management and efficiency by addressing the limitations of conventional routing methods that do not consider vehicle priorities. By incorporating machine learning models into a priority-based routing system, real-time dynamic priorities can be accurately assigned to vehicles, leading to more effective traffic management decisions [162, 163]. These innovative methods aim to alleviate congestion and reduce travel times for high-priority vehicles, providing a fresh approach to optimizing traffic control strategies. The findings have important implications for practical implementation in ITS systems, as they have the potential to alleviate traffic congestion, enhance traffic flow, increase efficiency, and regulate pollution levels in specific areas.

3.4.4 Fuzzy Inference Systems

Fuzzy inference systems play a key role in fuzzy logic, particularly in practical applications like control. These systems typically operate by accurately mapping non-linear relationships through fuzzy rules that capture expert or common-sense knowledge related to the specific problem being addressed. To enhance traffic management in smart cities using the Mamdani Fuzzy Inference System, fuzzy

logic is used to assess traffic data in order to determine traffic conditions and implement strategies to regulate traffic flow and reduce traffic congestion [48].

A novel control strategy method for self-driving cars, utilizing a learning approach is introduced in [42]. The method involves training an explicit Takagi–Sugeno (TS) controller by feeding input and output data from an existing controller into an Adaptive Neuro-Fuzzy Inference System (ANFIS) algorithm. The method is characterized in the TS format for stability evaluation through Lyapunov theory and Linear Matrix Inequalities. By applying this method to learn the control law from a Model Predictive Controller (MPC), the need for online optimization is eliminated, resulting in reduced computational requirements, and enabling real-time implementation.

The effectiveness of this approach is verified through simulation tests using a small-scale autonomous racing vehicle. ANFIS is also proposed to hinder the effectiveness of intrusion detection systems in defending against zero-day attacks [164]. These attacks can modify data and can have severe consequences such as creating congestion, decreasing traffic safety, and negating the benefits of intelligent transportation systems. To address this issue, hybrid network intrusion detection systems utilizing transformer and adaptive-network-based fuzzy inference systems were proposed to address the issue of imbalanced data samples by first removing noise and then generating new samples [165]. In this research, a feature engineering component was created to incorporate non-linear feature interactions in complex combinations. Finally, the self-attention mechanism of Transformer and the adaptive learning approach of ANFIS were integrated to improve feature representation and decrease reliance on sample data.

3.5 Game Theory and Decision-Making Frameworks:

3.5.1 Nash Bargaining Game-Theoretic Framework:

A decentralized traffic signal control algorithm using Nash Bargaining game-theoretic framework is proposed in [166]. This algorithm dynamically adapts traffic signal timings at intersections, treating each phase as a player in a cooperative game. The focus is on optimizing signal timings to accommodate fluctuating traffic demands efficiently. The algorithm was tested in simulation software on sample networks, demonstrating significant improvements in traffic flow, reduced vehicle delays, and emissions compared to other control methods. Game theory [167] is a branch of mathematics that is used to study decision-making in strategic situations, where the outcome of an individual's choice depends on the choices of others. Furthermore, new multi-objective optimization controller, inspired by the Nash bargaining game,

was proposed to enhance energy efficiency, traffic safety, and driving comfort simultaneously in hybrid electric vehicles [168, 169]. In this approach, the longitudinal dynamic control and energy management strategy are represented as independent players, with the Nash equilibrium serving as the focal point and determined through a linear quadratic game method. The Nash bargaining solution is then calculated using the alternating direction method of multipliers. Simulation results indicate that this innovative controller surpasses both hierarchical and centralized optimization controllers, thereby improving fuel efficiency, optimal performance and robustness in control.

Resource allocation has emerged as a key task in Vehicular ad hoc Network (VANET). Hierarchical resource allocation approaches utilizing Nash bargaining game are proposed [170, 171]. Initially, the distribution of spectrum resources among various Road Side Units (RSUs) is examined, as they receive resources from a central cloud. Subsequently, considering the diversity of vehicular users (VUEs), a matching degree index is established between VUEs and RSUs. The focus then shifts to the distribution of spectrum resources between VUEs and RSUs. To streamline the process, the original problem is divided into two sub-problems namely, power allocation and slot allocation-based on time division multiplexing. Simulation outcomes demonstrate that the proposed strategy is capable of equitably and efficiently assigning resources in VANET based on VUEs' requirements.

3.6 Hybrid Traffic Management Solutions

3.6.1 Integrated Learning and Control Systems

Dynamic and Intelligent Traffic Light Control System (DITLCS) that utilizes deep reinforcement learning and fuzzy inference is proposed in [172]. The main objective is to effectively regulate traffic lights in response to live traffic data. The system functions in three different modes (Fair, Priority, Emergency) and adapts the timing of traffic lights accordingly. The methodology incorporates fuzzy logic for selecting the appropriate mode and deep reinforcement learning for controlling traffic signal phases. The system's performance was evaluated through simulation on an urban map.

The challenge of human-machine interaction in automation systems typically arises in control tasks within clearly defined technical processes. This is discussed in [173] where authors argue that integrated air traffic management and control systems are not easily defined due to varying purposes and levels of traffic. This leads to conflicts between individuals from different organizations attempting to assert control using their automation equipment. To prevent dangerous competition, specific design requirements are necessary for the automation system. A study discussed in [174] highlights

the role of system architecture in verifying system functions and examines how automated traffic control systems are developed and implemented. The study investigates the current methods used in developed countries for implementing smart traffic lights on roads. A prototype of a traffic light control system was developed for use in training students studying transport systems and information technologies. The significance of using gaming technologies to engage students in learning and help them develop essential skills was demonstrated. This work also outlines the key steps in creating a serious game based on the prototype.

3.6.2 Cross-Disciplinary Optimization Approaches

An innovative congestion control method named Congestion Avoidance by Reinforcement Learning Algorithm (CARLA) is introduced in for Intelligent Transportation Systems (ITS) using Wireless Sensor Networks (WSN) and IoT [175]. CARLA enhances the performance of the Zigbee protocol by effectively routing and adjusting the data rate of nodes through intelligent means. It employs a multiple Bloom filter for topology control and utilizes learning automata for distributed data rate adjustment, focusing on reducing congestion probability, improving packet delivery ratio, and minimizing energy consumption and end-to-end delay in ITS. In the realm of WSNs, which are crucial in the IoT landscape for various applications such as environmental monitoring, congestion challenges are investigated [176]. These challenges lead to increased packet loss, longer delays, and reduced throughput. To combat these challenges, [177] introduces a Fuzzy Logic-based Cross-Layered Optimization Model (FL-CLOM) for WSNs. FL-CLOM incorporates the signal-to-noise ratio of wireless channels into the Transmission Control Protocol (TCP), integrating fuzzy control with congestion control to dynamically manage queue sizes in busy nodes. Through simulations, FL-CLOM is shown to: (a) adapt to changes in queue size, (b) achieve rapid convergence, (c) reduce average delays, (d) minimize packet loss, and (e) enhance throughput. Moreover, [178] delves into the utilization of soft computing approaches, including ML, for intelligent congestion management. The study offers a comprehensive overview of various soft computing strategies and the challenges associated with congestion management. Additionally, an ensemble traffic classification model employing standard ML protocols is introduced and tested on a dataset of cell tower network congestion, demonstrating satisfactory performance in terms of f1-score, accuracy, precision, and recall.

3.6.3 Fusion of Predictive Analytics and Decision Support

The creation of the Fusion-based Intelligent Traffic Congestion Control System (FITCCS) for vehicular networks using

machine learning techniques is presented in [90]. The system leverages IoT-enabled devices for data collection and applies ML algorithms, such as Artificial Neural Networks (ANN) and Support Vector Machines (SVM), to predict traffic congestion and support [90]. By gathering traffic data, preprocessing it, and employing ML algorithms, the system aims to efficiently manage and control traffic flow in smart cities, demonstrating a practical application of ML in traffic management. The challenges of high computational demands, during the process of perception data fusion, are addressed by proposing vehicle–road cooperative network that allows for offloading computations during this process [163]. Through an architecture that enables task assignment and computation offloading, an integer nonlinear programming problem was devised within this framework. To address the dynamic nature of cooperative vehicle infrastructure systems, the Deep Deterministic Policy Gradient (DDPG) algorithm was introduced for perception fusion computing offloading (DDPG-PFCO). This method outperformed other comparison algorithms in evaluation experiments conducted on a real-world map. Research in [179] focused on developing data fusion-based traffic congestion control system in smart cities using deep learning models. A hybrid model consisting of convolution neural network (CNN) and LSTM architectures was utilized for region-based traffic flow predictions. CNN was employed for spatial data classification while LSTM was used for temporal data, with experimental results showing favorable outcomes compared to other baseline models. In summary, Table 3 revealed the adopted researches on the congestion control using AI models.

4 Survey of Actual Case Studies

In this section, there are several case studies that were collected from the literature to present the application of AI models in particular location around the world to reveal their feasibility.

A novel approach was proposed for addressing urban traffic congestion using reinforcement learning (RL), with a focus on Deep Q-Learning (DQL) [194]. As urbanization surges, traffic congestion increasingly impairs mobility, environment, and public health. The study introduces an RL-based traffic control framework capable of dynamically optimizing signal timings at intersections to improve traffic flow and reduce vehicle queue lengths. The study's primary focus is the design and evaluation of a DQL-driven traffic signal control system at a complex four-lane intersection. The framework is trained using traffic simulations and real-time intersection data. By modeling traffic as a Markov Decision Process (MDP), the researchers aimed to develop a policy that enables the RL agent to make optimal signal

control decisions based on environmental states such as vehicle count, velocity, and lane occupancy.

The proposed framework revolves around three main components: (i) definition of states, actions, and rewards for the RL agent, (ii) implementation of DQL with experience replay and target networks, and (iii) optimization via hyperparameter tuning. The action space includes possible signal phase changes across four directions: North–South, North–South Left, East–West, and East–West Left. The reward function is structured to minimize queue length and delay while maximizing traffic flow. Traffic data for model training was sourced from two XML datasets representing vehicle movements and route attributes. Through simulations, the model underwent training for 5400 steps and was tested across 30 episodes. The DQL agent successfully reduced average queue lengths by 49% and improved reward outcomes by 9%, showcasing its potential for real-world deployment. Various state transitions and their corresponding actions were mapped to quantify traffic conditions and control decisions.

A visual representation of states and decision outcomes demonstrated the agent's ability to anticipate optimal next states to reduce congestion. The study also benchmarked its system against existing RL and deep learning-based traffic control methods. Compared to prior works, the DQL model achieved comparable or superior performance with relatively simple and scalable architecture. Unique innovations include an adjusted action index to balance exploration and exploitation, and fine-tuning of hyperparameters such as learning rate, batch size, and reward discount factor ($\gamma = 0.75$). To validate the model's effectiveness, both queue reduction and reward gains were measured and visualized during the training and testing phases. The transition from high queue states to optimal flow states highlights the robustness and adaptability of the model. Further, the study provides actionable insights into model tuning, architecture design, and environment simulation for smart traffic systems. In conclusion, the study confirms the efficacy of reinforcement learning in dynamic urban traffic management. Finally, research future work aims to integrate real-time traffic feeds, reduce computational complexity, and extend the model to large-scale networks using distributed training techniques. The authors envision that such intelligent systems can significantly contribute to building sustainable and efficient urban mobility infrastructures. The modeling flowchart of the whole process is presented in Fig. 4.

A graph-theory-based framework was adopted for analyzing and managing expressway bottlenecks under catastrophic events, such as accidents and severe weather, with a focus on improving traffic safety and operational efficiency [195]. The methodology applied for Beijing–Hongkong–Macau expressway's Anyang–Xinxiang section in Henan Province, China, the framework introduces the concept of static and dynamic

Table 3 Congestion control related literature studies adopted using the application of AI models

Article references	Sensor type	Algorithm for congestion estimation	Finding	Quantitative performance summary
[102]	High-level sensor models	Deep Q-networks	Effective multi-intersection signal control	QT-CDQN reduced average queue length, waiting time, and improved speed across four and six-intersection scenarios, achieving best AQL (15.6 veh), AS (6.2 m/s), and AWT (29.7 s) under high traffic, outperforming CDRL, MADQN, and Distributed QL
[98]	Not specified	Deep reinforcement learning	Proposes a boosted genetic algorithm combined with machine learning (BGA-ML) for optimizing traffic signal control under incidents, reducing total travel time by 43% compared to traditional GA methods, with significantly improved computational efficiency (11 min vs. 8 h)	The proposed S-A DRL model reduced system delay by 15–20% and improved throughput by 17% over fixed timing in both saturated and oversaturated scenarios, outperforming traditional DQN with more stable convergence using dynamic discounting and action-input architecture
[180]	High-level sensor models	Reinforcement learning models	Enhances driving performance against perception errors	OU-based sensor models reduced collision rates up to 80% in scenarios with perception errors. Policies trained using these models showed enhanced robustness and fewer failures (3% vs. 84%) under noisy conditions, outperforming Gaussian-noise and ground-truth-only training setups
[92]	Not specified	RNN	Proposes an intelligent intersection scheduling system using BP neural networks and EDF-based CAN bus communication, predicting vehicle passage times with < 10% error and achieving > 95% successful data transmission, significantly improving real-time scheduling efficiency compared to traditional methods	BP neural network predicted vehicle crossing time with < 10% error. EDF-enhanced CAN bus achieved > 95% transmission success under 50–110% load, outperforming SPDS in high-load conditions, and improving scheduling efficiency and communication reliability in intelligent traffic systems
[88]	IoT-enabled devices	SVM	Improved traffic management	The TCC-SVM model achieved 98.7% training and 97.9% validation accuracy with miss rates of 1.3% and 2.1%, respectively—outperforming prior models (e.g., 76–96% accuracy). It used IoT sensor data and SVM to dynamically control smart city traffic congestion
[181]	Vehicle-to-vehicle, vehicle-to-road	Predictive analytics, ML	Enhances traffic efficiency with optimized signal control and routing	The PIT model achieved Pareto optimality in 73.33% of cases, reducing total driving time by up to 56.15% and delay time by 77.08% compared to fixed TSC + VRG. It integrates predictive signal control and vehicle routing via bidirectional feedback

Table 3 (continued)

Article references	Sensor type	Algorithm for congestion estimation	Finding	Quantitative performance summary
[182]	Not specified	Deep autoencoder, Z-score method	Detects unsafe driving behaviors, enhancing traffic safety and congestion management	The proposed autoencoder detected 81 anomalies (7.86%) with 96.31% accuracy (no early stopping), and 147 anomalies (14.26%) with 95.25% accuracy (with early stopping), using NGSIM data. It outperformed traditional outlier methods by combining multiple driving indicators
[96]	Cameras	Image and video processing	Proposes a Raspberry Pi-based smart traffic control method using image/video processing, dynamically adjusting traffic lights based on vehicle density and counts. Achieves ~35% improvement in congestion reduction compared to traditional fixed-time scheduling methods	The system allocated green light times dynamically (e.g., 20–54s) using vehicle density (overlap %) and vehicle count. Direction D increased from 20 to 50s when accounting for 11 vehicles, enhancing efficiency via Raspberry Pi and OpenCV without central servers
[136]	Image processing, IoT	Image processing for vehicle detection	Reduces congestion caused by illegal parking	the study did not report specific quantitative metrics
[137]	Vehicle identification data	Beacon frequency adjustment	Enhances message transmission efficiency in VANETs	TDCCA achieved 92.3% packet delivery, 2543 kbps throughput, and 39.5 ms delay, outperforming Newreno, Vegas, and Compound by 7.7–17.6% in delivery and up to 75% in throughput, while maintaining < 1 ms average latency for safety-related CAM transmission
[127]	Wi-Fi Direct-enabled smartphones	Dynamic route-selection algorithms	Proposes an intelligent vehicular ad-hoc network (iNET) for smart cities using Wi-Fi Direct-based V2V communication, significantly reducing overall travel time by up to 33.3%, without requiring roadside infrastructure or internet connectivity, compared to traditional fixed-route selection methods	The iNET framework reduced average travel time by up to 33.3% using MDSP over DSP algorithms, with only 2–3% battery usage, 230 B/s network load, and < 3% memory on Android devices—enabling low-cost, real-time congestion-aware routing without Internet or RSUs
[111]	IoT technology	ACO and fuzzy logic	Optimizing routes and managing traffic flow, improving congestion control in urban areas	The proposed ACO-based IoT model reduced average travel and waiting times compared to Dijkstra, Kruskal, and Prim algorithms. Although exact numbers weren't provided, performance graphs confirmed improved route efficiency and congestion handling in real-world Vellore district simulation >
[84]	Not specified	Boosted Genetic Algorithm with ML	Reducing travel time in urban networks during non-recurrent traffic incidents	The genetic algorithm-based signal optimization model reduced peak period average delay by 10.73%, travel time by 5.33%, and variance by 23.72%; during non-peak periods, delay dropped 4.47%, travel time 0.79%, and variance by 40.36% in Nanjing CBD

Table 3 (continued)

Article references	Sensor type	Algorithm for congestion estimation	Finding	Quantitative performance summary
[110]	IoT sensors	ACO	Efficiently managing vehicular traffic, optimizing traffic flow, reducing congestion	the study did not report specific quantitative metrics
[183]	Road measurements	Dynamic tolling optimization	Managing highway traffic congestion through dynamic tolling, enhancing urban mobility	the study did not report specific quantitative metrics
[138]	Wireless Sensor Network, GPS	Cooperative game-theoretic approach	Enhancing traffic flow management at multiple intersections, reducing vehicle waiting times	The cooperative game-theoretic approach reduced vehicle waiting time by 15–25% and increased throughput across simulated intersections compared to Green Wave and non-cooperative methods. Performance was evaluated under both uniform and non-uniform traffic volumes using NetLogo simulation
[100]	Surveillance cameras	Distributed multi-agent Q-learning	Proposes distributed multi-agent Q-learning for intelligent traffic control, significantly reducing vehicle and pedestrian queue lengths (~16.7%) and waiting times compared to fixed-time and sensor-based methods. It also achieves notable improvements in emissions (~12%) and fuel consumption (~15%)	Proposed Q-learning model reduced total queue length and waiting time for both vehicles and pedestrians by over 16.7%, and outperformed fixed-time, sensor-based, and MARL methods in emissions, fuel use, and noise under varying pedestrian rates on Sunnyvale network
[104]	Not specified	Deep Q-learning, reinforcement learning	Proposes GuidedLight, a pretrained reinforcement learning approach achieving superior adaptability and sustainability in traffic signal control. It reduces average travel time (~647s vs. ~774s baseline) and computational emissions (~sevenfold CO ₂ reduction) compared to conventional and machine-learning alternatives	Pretrained GuidedLight achieved lowest avg. travel time (647.45 s) on NY196, outperforming PressLight (706.36 s) and others. It reduced training CO ₂ emissions by 7×, showing superior adaptability to non-uniform topologies via length-invariant state design and analytic exploration
[184]	Not specified	Multi-agent systems, traffic light control	Reducing congestion, prioritizing emergency vehicles, improving journey times	The multi-agent model reduced emergency vehicle time loss by 52.6% and travel time by 37.2% (Scenario-A), and reduced gridlock-related time loss by 50.6% and journey time by 45.2% (Scenario-B), significantly improving intersection throughput using JADE-SUMO simulations
[185]	IoT cameras, RFID	AI and ML	AI and IoT-based system for urban traffic management, reducing congestion, enhancing safety	the study did not report specific quantitative performance metrics

Table 3 (continued)

Article references	Sensor type	Algorithm for congestion estimation	Finding	Quantitative performance summary
[103]	Cameras, Loop Sensors	Q-learning for Traffic Control	Improving traffic signal balance and throughput at intersections, enhancing traffic flow	The proposed Q-learning model reduced queue length by 25% vs. E-TS and 63% vs. C-TS; reduced standard deviation by 50% and 75%; and waiting time by 15% and 40%, respectively—demonstrating effective intersection delay reduction across multiple scenarios
[186]	RFID sensors	ABPNN	Enhances traffic flow with congestion prediction	MSR2C-ABPNN achieved 98.15% training and 97.56% validation accuracy (Time Series), with RMSE as low as 1.416×10^{-3} . It outperformed Tamimi & Zahoor (2010) and Pushpi & Dilip Kumar (2018) in regression ($R^2 = 0.98$ vs. 0.78–0.91)
[187]	–	Reinforcement learning	Improves autonomous driving maneuvers	The trained policy achieved 99.5% goal-reaching in crossroads scenarios, with reward-sharing improving speed (2.58 m/s vs. 1.9) and reducing collisions. Shared-reward agents outperformed baseline in high-density settings, showing improved cooperation and smoothness using PPO in decentralized multi-agent simulations
[112]	Traffic data	Switching dynamics	Optimizes urban traffic flow	The study did not report specific quantitative performance metrics
[188]	Traffic sensors	Adaptive algorithms	Reduces urban intersection congestion	The proposed multi-agent system reduced vehicle waiting time by 30.4% compared to default models in a simulated nine-intersection urban grid using JADE. Metrics included total waiting time, moving time, and distance traveled for 1000–6000 vehicles
[132]	IoT devices	Area-based management	Controls traffic flow dynamically	Compared to Webster, the proposed scheme reduced congestion by 25%, increased average speed by 15.78%, and shortened resolution time by 200 s. Regional coordination based on social traffic features significantly improved flow balance in SUMO urban simulations
[189]	RFID, cameras	Signal adjustment	Prioritizes emergency vehicles	The study did not report specific quantitative performance metrics
[190]	Simulations	Traffic control strategies	Manages traffic in underground loops	

Table 3 (continued)

Article references	Sensor type	Algorithm for congestion estimation	Finding	Quantitative performance summary
[133]	Vehicle data	Signal control	Improves vehicle servicing	The proposed ITMS reduced average waiting time by up to 92% (e.g., 370s → 18s), and increased serviced vehicles by $2.23 \times$ vs. fixed-time signals under high-density traffic in simulations—approaching optimal algorithm performance using an adaptive, vehicle-prioritized IoV strategy
[126]	Magnetometer sensors	Light synchronization	Manages urban traffic flow	ATLCS reduced average travel time by up to 39% and improved network-wide efficiency by 17% vs. fixed-time TLCS. Travel Time Index increased from 42.12% to 68.81%, demonstrating significant congestion mitigation in peak-hour arterial road scenarios
[28]	–	Graph neural networks	Robust traffic forecasting	VGRAN achieved lowest RMSE (2.32), MAE (1.53), and MAPE (3.7%) on METR; and RMSE (2.88), MAE (1.76), MAPE (4.2%) on PEMS, outperforming DCRNN, GraphWave, OGCNN, APTN, and other state-of-the-art traffic prediction models
[108]	Camera photos	Deep Q-learning	Learns and predicts traffic patterns	*PLMP achieved lowest average waiting time (0.225 s) and travel deviation (9.17 s) on real data (Hangzhou), outperforming PLM and periodic baselines (waiting: 6.87–12.13 s; deviation: 26.79–38.20 s), demonstrating efficient congestion control using structural and temporal pattern memory
[191]	Not specified	Deep learning velocity-based model predictive control	Effectively reduces urban congestion with predictive control	The proposed DL-VMPC controller reduced congestion by 19.8% vs. max-pressure and 35.5% vs. IDQN. It matched LQR's effectiveness but with smoother signals, achieving high stability and compliance with control constraints in real-world traffic simulations
[85]	Inductive loop detectors	kNN-based Shewhart and exponential smoothing	Enhances traffic management by analyzing deviations in traffic state	Nonparametric kNN-ES achieved top detection performance (AUC 0.985, TPR > 97%) across abrupt, intermittent, and gradual congestion. PWSL-KF estimation reached high accuracy (R^2 up to 0.9937, RMSE $\approx$ 0.45), outperforming conventional Shewhart/ES charts in precision and recall

Table 3 (continued)

Article references	Sensor type	Algorithm for congestion estimation	Finding	Quantitative performance summary
[141]	Not specified	Fuzzy self-adaptive PID control	Improves ramp metering to manage congestion	FSAPID reduced total travel time by 41–43% (stable demand) and 38–42% (fluctuating), outperforming PI RM. Though ramp delays slightly increased, FSAPID ensured faster response, congestion prevention, and robustness across bottleneck distances (up to 1.6 km)
[150]	Not specified	Fuzzy logic-based control	Manages traffic flow and signal timing for congestion reduction	The study did not report specific quantitative performance metrics
[192]	Not specified	Fuzzy logic-based control	Efficient emergency vehicle management for traffic congestion control	ITCC reduced emergency vehicle travel time and average vehicle wait time by up to 45% under high congestion. CARA routing improved packet delivery (PDR $\approx$ 95%), reduced end-to-end delay, and increased node energy efficiency vs. ENS_OR and GeRaF
[193]	Not specified	Mamdani fuzzy inference system	Enhances intelligent transportation systems for traffic management	Using MATLAB simulations, MFIS estimated congestion levels from 5 inputs (speed, signals, etc.). High congestion ($\approx$ 80%) occurred at low speed and dense signals; low congestion (<10%) was observed with high speed and wide road capacity. No time/delay metrics reported
[166]	Not specified	Cooperative game-theoretic approach	Utilizes cooperative strategies for improved traffic signal control	The Nash Bargaining (NB) controller cut delay by up to 67%, queue length by 77%, and emissions by 18%, outperforming fixed-time, actuated, and adaptive controllers in both isolated and arterial networks under varied traffic demands
[172]	Not specified	Deep reinforcement learning	Adapts traffic lights to dynamic traffic conditions for better flow	DTLCS reduced average waiting time and queue length, increased throughput (up to 68%), and boosted average speed. CO ₂ emissions dropped by up to 14.21% vs. state-of-the-art methods, with strong gains for emergency and priority vehicles across real-map SUMO simulations
[175]	Wireless sensor networks, IoT	CARLA, Zigbee protocol	Optimizes routing in WSN-IoT managed ITS for congestion control	CARLA improved PDR by 4.64%, reduced energy use by 11.44%, and cut end-to-end delay by 25.26% vs. Zigbee, CAR-MOSOA, and DCCP. Its learning automata-based data rate control enabled scalable, low-latency routing in dynamic ITS scenarios

Table 3 (continued)

Article references	Sensor type	Algorithm for congestion estimation	Finding	Quantitative performance summary
[90]	IoT-enabled devices	ANN and SVM	Provides effective traffic congestion control through data analysis and ML	FITCCS-VN achieved 95% accuracy and 5% miss rate in congestion prediction, outperforming prior ANN/SVM methods. ANN showed 94% validation accuracy, SVM 90.3%. Fusion improved decision reliability across dense urban traffic scenarios using real-world vehicular network datasets

bottlenecks and proposes methods for their identification using the Ford-Fulkerson labeling algorithm for single origin–destination (OD) scenarios and a super origin–destination transformation for multi-OD networks. The goal is to predict the location and capacity of bottlenecks under varying conditions, using variables like lane number, segment capacity, safety speed, and event-specific reduction factors. The methodology incorporates mathematical modeling and simulation tools such as MATLAB and LINGO to support traffic decision-making. Data was sourced from real-world traffic flow observations and simulations of three scenarios: normal flow, heavy traffic accidents, and fog-induced shut-downs. Key research questions addressed include how to detect and calculate bottlenecks dynamically, and how to optimize traffic flow and access control during disasters. The findings reveal that bottlenecks shift significantly in disaster scenarios, drastically reducing capacity, and introduce the concept of the “bottleneck chain corridor” to better capture upstream congestion risks. Validation is demonstrated through alignment between theoretical bottlenecks and real congestion during events. The study concludes that dynamic bottleneck analysis and chain corridor identification are critical for effective traffic diversion and network access control, offering a structured and practical approach for managing expressway networks during high-impact disruptions. The modeling process was formulated and presented in Fig. 5.

In another study, the use of high-resolution satellite imagery and deep learning to generate city-scale traffic data was explored, focusing on Barcelona, Spain [196] (the modeling process presented in Fig. 6). The objective was to address the challenge of fragmented and resource-intensive ground-based traffic data collection by leveraging WorldView-2 and WorldView-3 satellite imagery in combination with deep learning object detection (YOLOv3 and YOLOv8) models to identify vehicles across the city. Covering 100 km² of urban area, satellite images were manually annotated to distinguish parked, static, and moving vehicles, with over 19,000 vehicles labeled for training and validation. The best-performing model (YOLOv3 with Barcelona-specific anchor boxes) achieved a precision of 0.69 and recall of 0.79, with an F1 score of 0.74, effectively identifying vehicles on major roads but facing challenges with narrow streets, shadows, and vegetation cover. The framework evaluated multiple parameters—image resolution, color band composition, and anchor box configurations—to optimize detection accuracy. The study also compared YOLOv3 to YOLOv8, with YOLOv3 outperforming the newer version, especially in recall. Findings indicate that using customized anchor boxes and smaller image tiles improves accuracy and that classifying vehicles into multiple movement types reduces model reliability. This marks the first time deep learning has been used to detect vehicles across an entire city using satellite imagery, highlighting its global scalability—especially

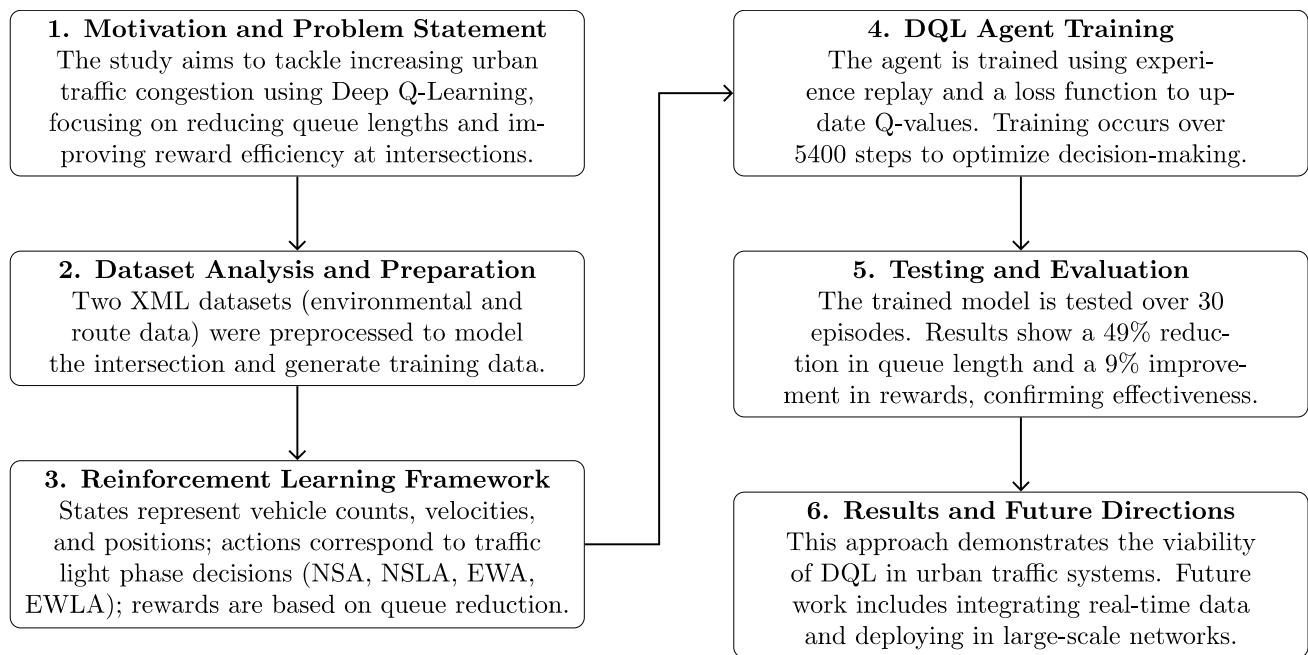


Fig. 4 The modeling flowchart of the proposed an integrated reinforcement learning (RL) with Deep Q-Learning (DQL) for addressing urban traffic congestion using

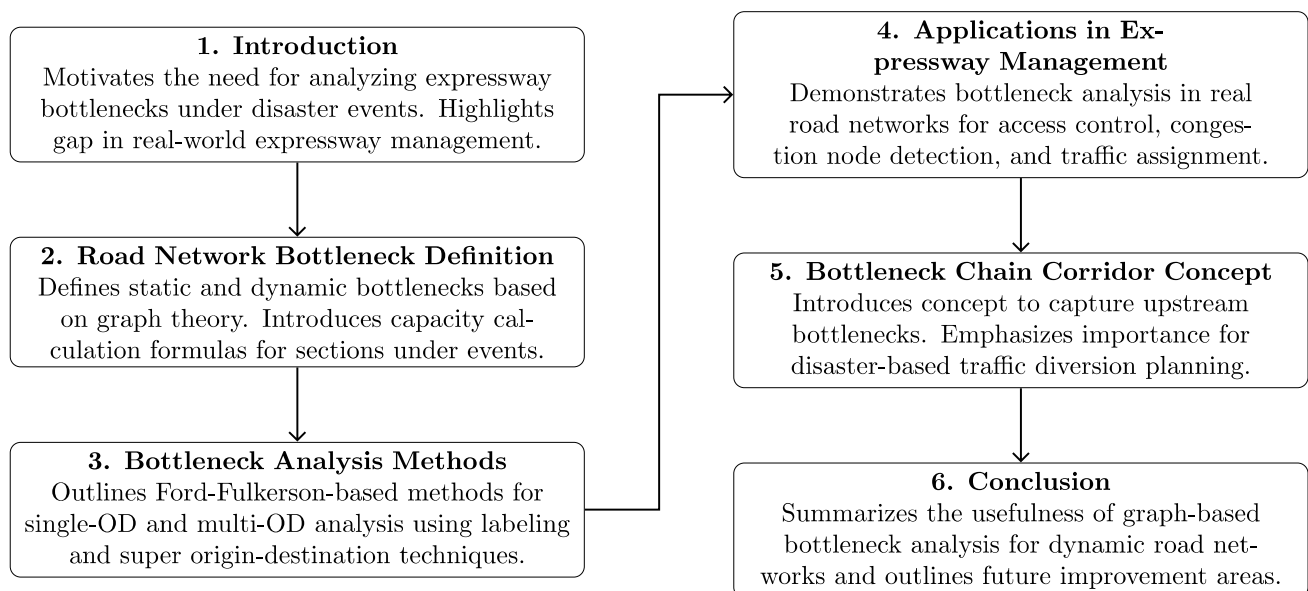


Fig. 5 The developed graph-theory-based framework for managing expressway bottlenecks under catastrophic events

for cities lacking conventional traffic data—despite limitations like partial occlusion and pixel constraints. The study concludes that satellite imagery, paired with well-trained DL models, offers a scalable, standardized, and cost-effective alternative to conventional urban traffic monitoring, with high relevance for transportation planning, air quality modeling, and policymaking in data-scarce regions.

A hybrid deep learning model that combines CNNs with a Boosted LSTM Ensemble (BLSTME) to effectively predict traffic congestion in smart cities [197]. This model is specifically designed to tackle the challenges posed by the dynamic behavior of vehicles, which can negatively impact the performance of traditional deep learning models. The proposed model combines the strengths of spatial

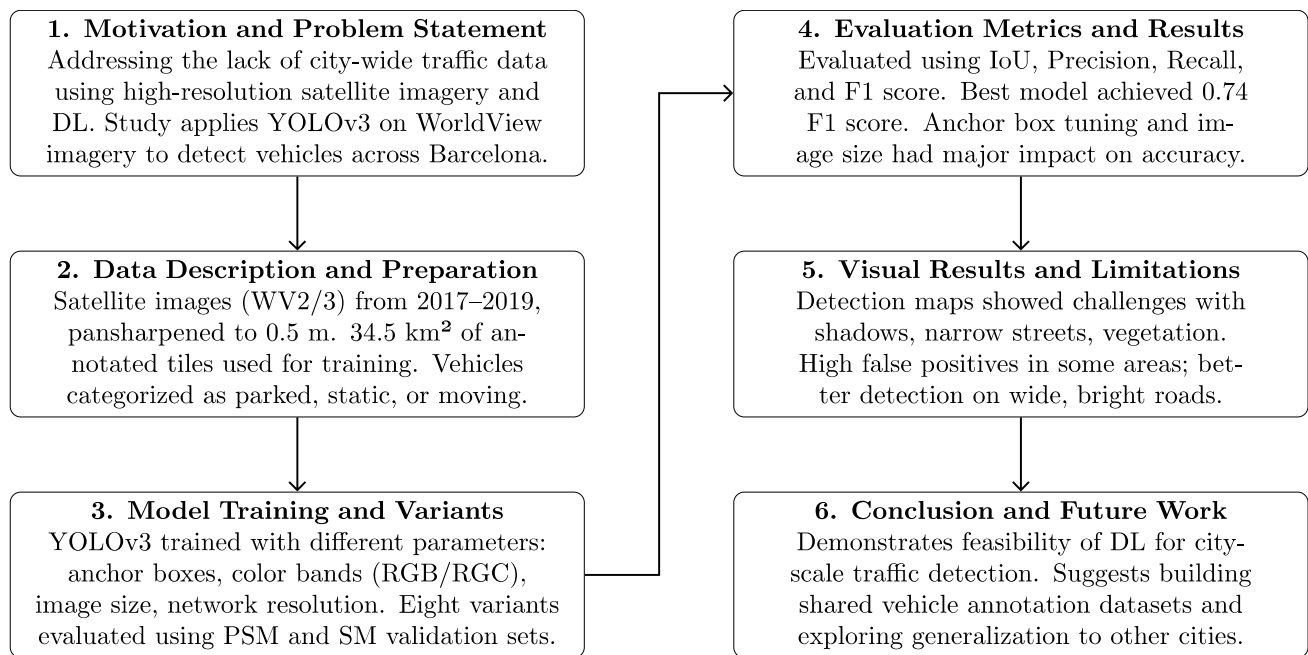


Fig. 6 The modeling framework of high-resolution satellite imagery and deep learning to generate city-scale traffic data of Barcelona, Spain

and temporal modeling. The CNN component extracts spatial features from traffic images, such as vehicle density and lane occupancy, using multiple convolutional, pooling, dropout, and batch normalization layers. The output of the CNN is a feature vector that is then processed by the BLSTME module. This module consists of an ensemble of LSTMs, each of which captures temporal patterns in the traffic flow data. To enhance prediction accuracy and reduce overfitting, the AdaBoost algorithm is applied to iteratively improve the ensemble by focusing on instances that were previously misclassified. This results in a more robust and generalizable predictor.

For data generation and evaluation, the study uses a simulation environment combining SUMO and OMNeT++ which model real-world traffic patterns in Seoul, South Korea. Traffic images are generated and classified into three categories based on pixel-based vehicle density: low congestion, medium congestion, and high congestion. Preprocessing steps include grayscale conversion, edge detection using the Prewitt kernel, frame differencing, thresholding, and dilation to isolate vehicle contours and enhance image clarity. The performance of this approach was evaluated using accuracy, precision, and recall metrics. The proposed BLSTME-CNN model achieves 98% accuracy, 96% precision, and 94% recall, outperforming existing models such as Autoencoders, ConvLSTM, and PredNet by approximately 10–12%. However, the integration of CNN and BLSTME increases the computational complexity of the model, which may

require significant processing power and time, especially for real-time applications.

Another investigation, a spatiotemporal analysis of traffic congestion in Casablanca, Morocco, using real-time traffic data collected via the Waze API [198] (Fig. 7). Traditional data collection methods for traffic analysis, such as road sensors or surveys, are often expensive and inefficient, particularly in developing countries. To overcome this, the study adopts an innovative data-driven strategy that leverages real-time traffic data collected via the Waze API. The dataset includes over 73,000 data points derived from hourly observations of 440 routes across 22 urban communes over one week. Using this data, the Travel Time Index (TTI) is computed for each trajectory, representing the ratio of travel time during peak hours to free-flow conditions. This macro-level congestion indicator allows for a spatiotemporal analysis of traffic patterns across the city. To categorize the communes based on their congestion levels, the authors employed three clustering algorithms: K-Means, Hierarchical Agglomerative Clustering (HAC), and DBSCAN. The clustering process identifies four congestion levels (low, moderate, congested, and highly congested) across the city. The study then explored how different land use variables to influence congestion. Infrastructure and spatial data were collected from OpenStreetMap and included features like residential and industrial zones, parks, commercial buildings, and highways. To determine the most influential variables, the authors used both Random Forests and Conditional Inference Trees to identify residential density, park area, commercial density,

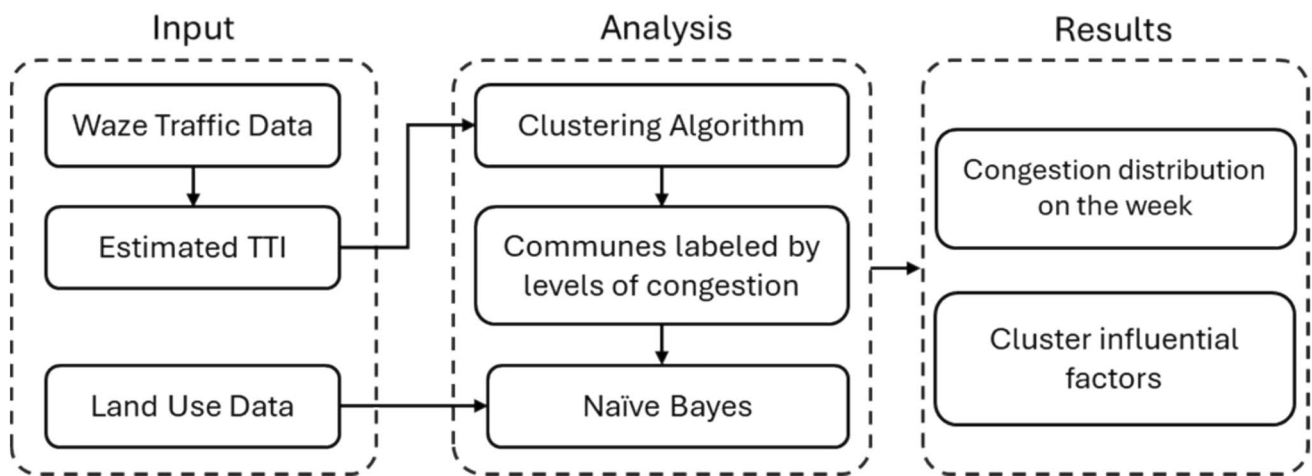


Fig. 7 The modeling procedure of a spatiotemporal analysis of traffic congestion in Casablanca, Morocco, using real-time traffic data collected via the Waze API

tram availability, and parking infrastructure as key contributors. The Naïve Bayes classifier is then employed to model the relationship between these variables and congestion levels. The findings of this study reveal that high residential coverage (above 73%), presence of large commercial zones, and limited parking are strongly correlated with higher congestion. Interestingly, while tram systems are generally expected to reduce congestion, in Casablanca, their layout that often takes space from existing road lanes has inadvertently worsened traffic in some areas. This is attributed to the city's insufficient road capacity and limited behavioral shift toward public transport. These results highlight a unique characteristic of developing cities, where increased density often correlates with congestion. The study recommended that effective congestion management in Casablanca must involve a coordinated approach integrating transportation and urban planning. Recommendations also include encouraging public transport use through improved service quality, expanding and managing parking infrastructure, considering congestion pricing, and revisiting tramway design to minimize its interference with vehicular flow. While this study offers valuable insights for policymakers, the authors acknowledge limitations such as the short time span (one week) and the absence of commuter behavioral data. They propose expanding future research to include extending the analysis to other MENA cities to validate and generalize their findings.

A deep learning-based traffic congestion was proposed for the detection system tailored for Addis Ababa, Ethiopia [199]. The authors aim to create a low-cost, efficient model that detects real-time traffic congestion from surveillance video cameras. This approach used YOLO models for congestion detection, which are one-stage object detectors known for their balance of speed and accuracy.

Unlike two-stage detectors like Faster-RCNN, one-stage detectors perform object detection in a single pass, making them suitable for real-time applications like traffic surveillance. YOLOv3, which uses the Darknet53 backbone, and YOLOv5, which is more modern and lightweight, were evaluated in this study. To develop the model, video footage was collected from the Addis Ababa Police Commission and online repositories. A total of 4,234 frames were extracted from traffic videos and manually labeled into congested (2,183) and uncongested (2,051) categories. The data was then split into training (90%), validation (10%), and testing sets. The images were annotated using bounding boxes-based models. The performance of YOLOv3 and YOLOv5 models was measured using precision, recall, and mean average precision (mAP) on both validation and test datasets. YOLOv3 achieved better accuracy, with 41.6 FPS, 68.6% mAP, and 41% precision, while YOLOv5, although faster (at 61.6 FPS), underperformed in accuracy, achieving only 32.1% precision and 62.3% mAP. These results indicate that YOLOv3 is more reliable for accurate congestion detection, whereas YOLOv5 is more suitable where speed is prioritized. The findings of this research demonstrate that traffic congestion in Addis Ababa can be accurately detected using YOLO-based deep learning models applied to surveillance video. YOLOv3, in particular, offers a good balance between speed and accuracy, making it suitable for integration into real-time traffic management systems.

AI-based solution proposed for the challenges of predicting traffic congestion when data is incomplete or missing [200]. The authors proposed a cost-effective methodology that leverages data from neighboring road segments and incorporates weather conditions to improve the accuracy of congestion prediction. The study selected the Tsimiski Street in Thessaloniki, Greece, as an ideal environment

to test the model in a complex urban setting with limited public transport infrastructure. The study used open-source datasets including real-time traffic data from the CERTH IMET Open API, weather data from VisualCrossing, and road metadata from OpenStreetMap and Emisia Wiseride. Data preprocessing involved normalizing and encoding variables, handling missing values, and engineering features such as road length, speed limits, and temporal factors like time of day and shop opening hours. To predict congestion for road segments with sparse data, the authors proposed using traffic data from adjacent segments. Four cases were evaluated, each combining the target segment with one or more neighboring segments. Case 2, which includes target segment with two preceding and one succeeding segment, provided the best results. Different ML models were evaluated including RF, Support Vector Classifier (SVC), KNN, LightGBM, CatBoost, and MLP. Initially, the model used a three-class label (low, medium, high congestion), but was later simplified into a binary classification (yes/no congestion) to address class imbalance and improve performance. Among the models, MLP achieved the best performance, and the accuracy improved from 73.1% (multi-class) to 77.04% (binary). When weather features like humidity, temperature, and precipitation were included, accuracy further increased to 79.81%. This finding reinforces the importance of weather conditions in traffic prediction. The proposed methodology was further validated using a second dataset

collected in 2022, which confirmed the model's robustness (Fig. 8). Overall, this research demonstrates that high-performance traffic congestion prediction is achievable even in data-scarce environments without relying on expensive sensors or cameras.

The study was conducted in Shanghai, China, using a sociotechnical approach to detect the social media activity-based accident and congestion hotspots for traffic accidents and congested areas [201]. Shanghai, as one of the most populous and rapidly urbanizing cities in China, offers a compelling environment for this type of analysis. The objective was to enhance the detection systems for monitoring frame traffic events from the public perception integrations element which many traditional sensor-based systems tend to bypass. The research lasted for more than eight months during which over 14 million Chinese micro-blogs on the popular Social Media platform Sina Weibo were analyzed in relation to public conversations around road traffic accidents. The study covered a broad spatial extent, including both the historic urban core and newer developments like Pudong New Area. Accident- and congestion-related Weibos were also collected from Lexing Shanghai, the city's official traffic Weibo account. From June to November 2012, 108 accident-relevant and 3,197 congestion-relevant posts were collected, manually verified, and incorporated into the analysis. The framework developed in this study uses natural language processing (NLP) and deep learning to identify posts

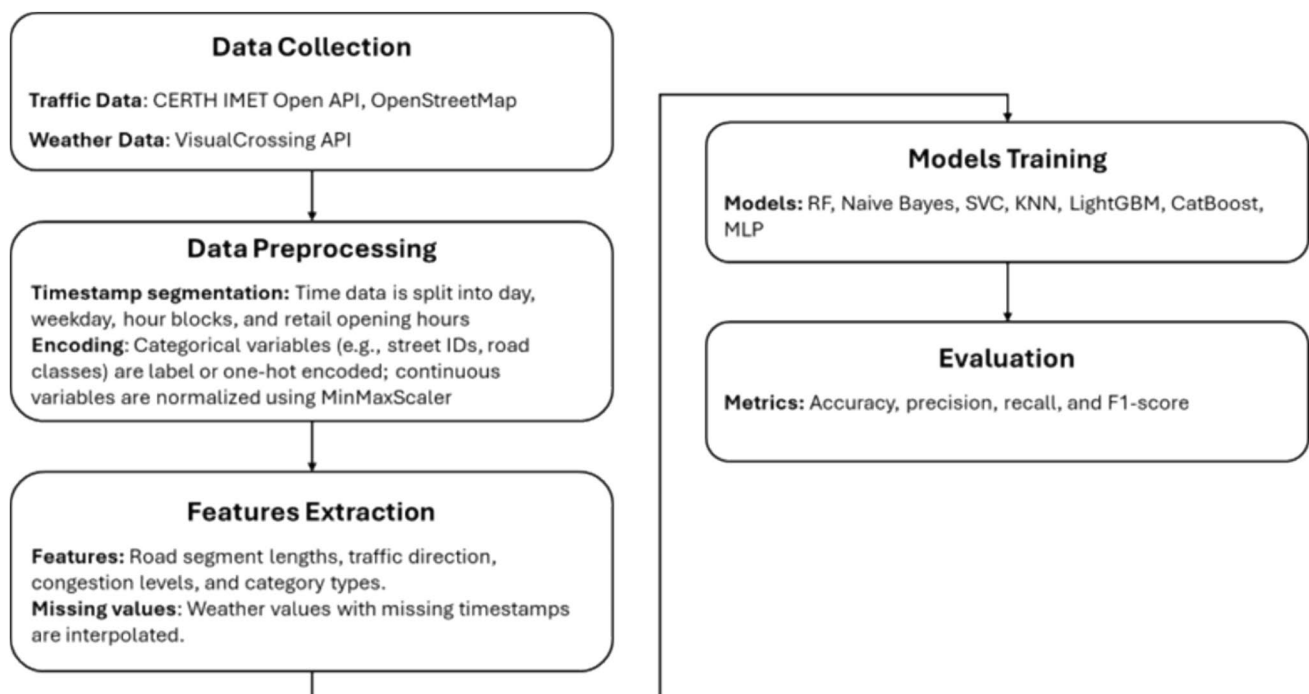


Fig. 8 Simulation process of an ideal environmental model in a complex urban setting with limited public transport infrastructure of Tsimiski Street in Thessaloniki, Greece

related to traffic accidents and congestion. Figure 9 depicts the flowchart of the proposed framework. To geolocate these events, the location information embedded in the text of each Weibo post was extracted and geocoded. A modified Kernel Density Estimation method (KDE*) was then applied. Unlike conventional KDE, this version integrates sentiment analysis to weigh the emotional intensity of the posts—thus emphasizing areas that not only experience frequent traffic events but also evoke strong public concern.

The analysis found that social media traffic congestion hotspots aligned with official traffic records, especially concentrating around the urban core and Northwest of Pudong. These zones were primarily location-driven and demonstrated chronic congestion behavior. Accident-prone zones, on the other hand, were more event-driven, aligning with where severe incidents were frequently talked about—areas that often fall outside high-congestion areas. Descriptive analysis was both spatial and temporal and every zone with a defined sentiment score and post volume count was verified. This also enabled researchers to focus on the Top Areas of Concern: 4 out of 15 accident zones and 6 out of 15 congested zones emerged as primary targets for enhanced traffic management. These highlighted zones are especially useful for traffic authorities to target optimal resource spend and infrastructure development. The proposed framework has a number of benefits: low resource requirements, scalability,

and greater alignment with public interest. Furthermore, many social network-posted events that are not captured within official records bolster the value of this approach. Still, there are drawbacks, most notably the lowered efficiency in regions with minimal social media use.

In another study, a practical roadmap for transportation agencies aiming to deploy data-driven traffic prediction systems [202]. The framework and induced insights can inform real-time traffic management strategies that contribute to improved roadway efficiency and reduced congestion. The study presented a ML framework designed for short-term traffic flow prediction, applied to a section of Interstate 64 (I-64) in St. Louis, Missouri. The proposed framework comprises two key tasks: to decide on suitable predictor variables that influence traffic flow and to develop predictive models using ML techniques. The flowchart of the short-term traffic flow prediction framework is illustrated in Fig. 10. The study was designed to assess how different ML-based models perform in predicting traffic conditions over short time intervals—in particular with considering 5, 10, and 15 min. The study focused on the use of traffic parameters including traffic volume (flow), vehicle speed, roadway occupancy, and the time of day. The research investigated whether including spatial data (e.g., upstream traffic data) would improve the accuracy of traffic predictions. The ML employed techniques include Deep Neural Networks

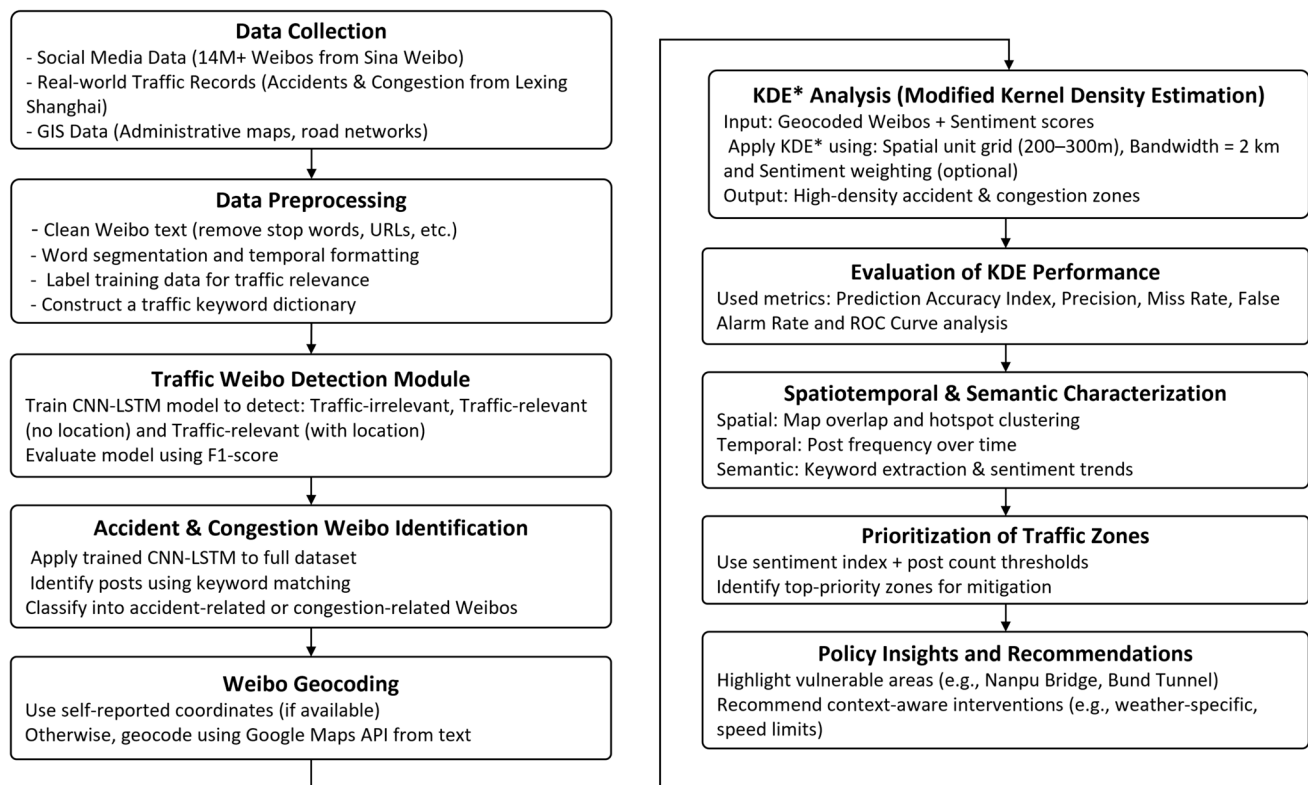


Fig. 9 Framework for identifying and profiling accident- and congestion-prone areas in Shanghai using social media data

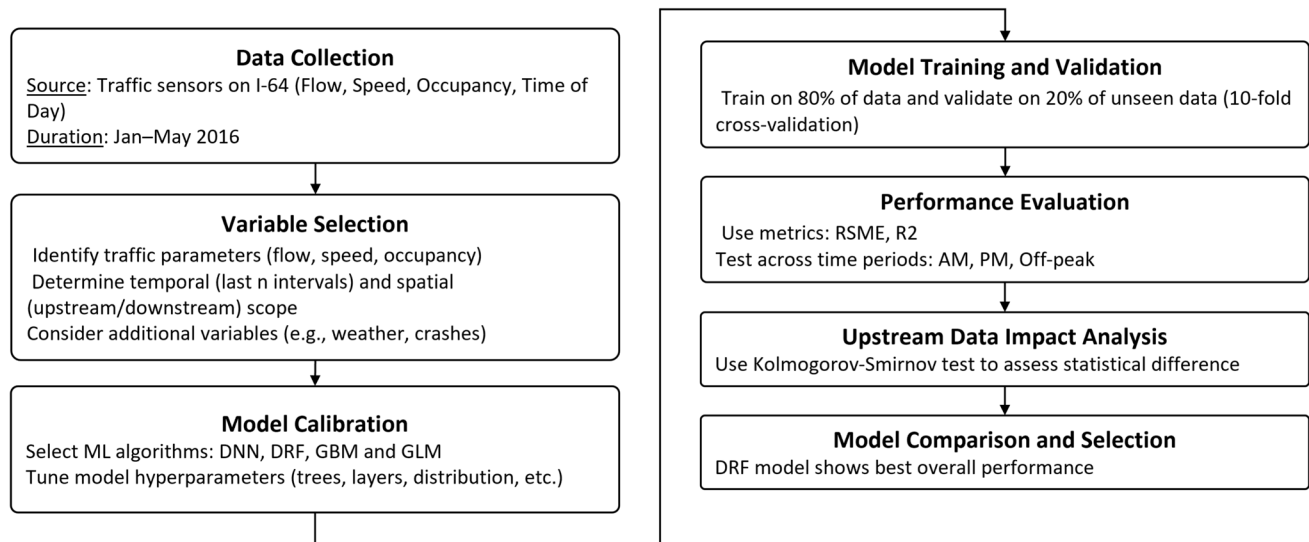


Fig. 10 Flowchart of the short-term traffic flow prediction framework: Interstate 64 in Missouri case study

(DNN), Distributed Random Forest (DRF), Gradient Boosting Machine (GBM), and Generalized Linear Model (GLM). The models were trained and validated using traffic data collected by remote microwave sensors from January to May 2016. The sensors captured 5-min aggregated data across all lanes for two adjacent segments of I-64—one downstream and one upstream. The downstream segment served as the target area for prediction and included three travel lanes, while the upstream segment had four lanes. Practically, the study was designed with the aim of answering two questions. The first one is about the influential traffic parameters and the associated spatial–temporal characteristics that may affect the prediction accuracy. The second question was about investigating which ML technique would provide the best performance with considering the generalization and reliability. The results obtained showed that the four models have comparable performance. Nevertheless, the DRF model slightly outperformed other models in learning complex patterns from historical data. Another observation from the experiments is that the inclusion of upstream data did not result in substantial improvement of the results. Two sample Kolmogorov–Smirnov tests were used to confirm the results. The results also confirmed that the quality and importance of the input variables are no less important than the capability of the method. In addition, it emphasized that the selection and combination of the predictors can contribute significantly for reducing prediction error. However, simply adding more spatial data, such as upstream flow, may not always enhance model performance.

This research put forth an integrated approach for estimating traffic congestion within the context of Casablanca city, Morocco [203]. The reason for choosing the city is due to its intensively populated region with its expanding

infrastructure, the evolving dynamics of traffic flows. Casablanca suffers from severe traffic congestion in critical areas as well as on the ring roads. In 2014, the number of registered vehicles in the city exceeds 1.4 million with complicated transportation links to other national routes. The study utilized a dataset from the Waze navigation platform that comprised 9,847 recorded accident pixels. It provided a holistic view of traffic behavior as the considered area covered both central and peripheral districts of Casablanca. The insights from Casablanca serve as a model for similar metropolitan regions across Africa and beyond. As shown in Fig. 11, an exploratory data analysis phase was conducted to induce critical patterns in traffic behavior that informed model development. Following model training and validation, a predicted traffic congestion map was generated. To forecast traffic jams using real-time data, the study employed four ML techniques including RF, XGBoost, K-NN, and ANN. One objective of the research is to determine which ML model could most accurately predict congested versus non-congested traffic conditions. The developed ML-based models were trained to identify spatial and temporal traffic patterns. Waze's crowd-sourced data was used for the purpose of training and evaluating the models. The Waze's dataset comprised traffic flow rates, incident reports, and congestion levels. In total, 72.7% of the instances represented congestion. It is worth noting, congestion was found to be most intense in city centers and along the southwestern bypass route. Among the developed models, Random Forest outperformed the other model by achieving 96% accuracy and an AUC of 0.997. The RF's high predictive ability due to its ensemble-based nature allowing affective handling of complex, nonlinear interactions among variables. The dataset's imbalance impact negatively the performance of the

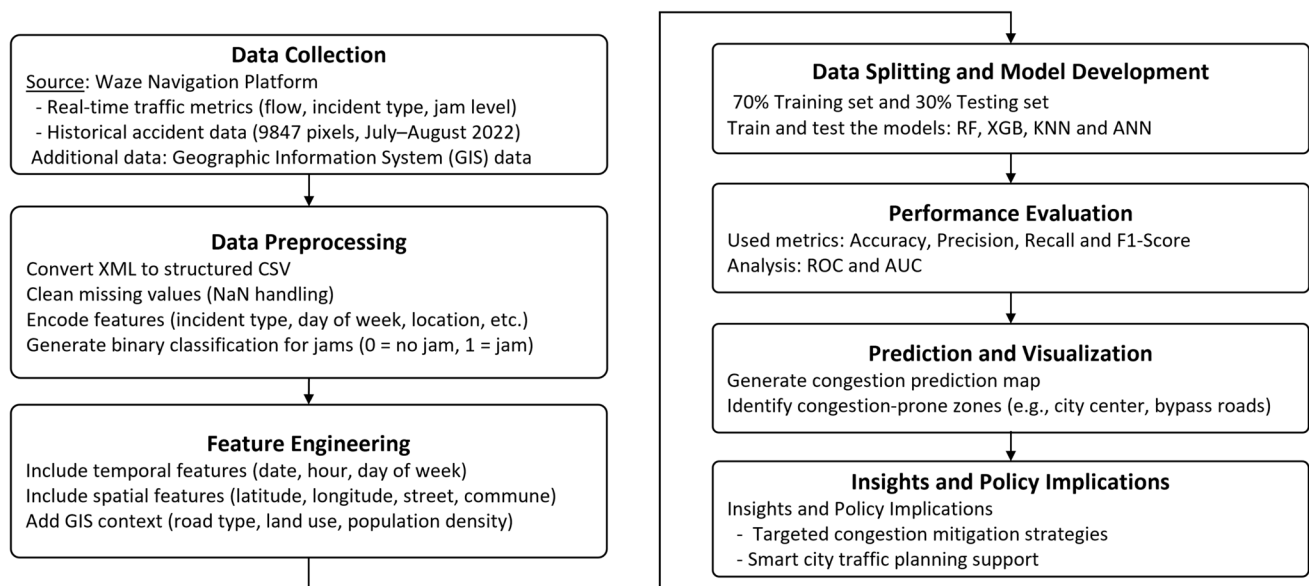


Fig. 11 Flowchart: methodology framework for traffic jam prediction in Casablanca, Morocco

other models, in particular ANN model that achieved low accuracy of 64% and an AUC of 0.5. The results showed the values of the ML-based models in traffic congestion prediction. It supports the integration of AI into smart city initiatives and traffic management systems, helping to reduce congestion, lower emissions, and improve urban livability. However, the study acknowledged that the use of single data source (Waze) is a key limitation as not all drivers use the investigated platform. The coverage may skew toward primary roads. In addition, the imbalanced dataset posed challenges for some models, in particular the ANN-based model. Injection of other external factors (e.g., weather, road conditions, and construction zones) would enhance model accuracy.

5 Big Data and AI Applications in Traffic Estimation and Control

AI and Big Data integration improves Intelligent Transportation Systems (ITS) development by optimizing traffic planning, forecasting, and management, while also lowering transportation costs through public transportation, ride-sharing, and smart parking. Big data platforms enable Intelligent Transportation Systems (ITS) to enhance decision-making and provide reliable transportation solutions to users. Utilizing these platforms enables ITS accurately real-time traffic flow, congestion patterns, and other vital data. Also, the big data provides real-time monitoring, predictive maintenance, and optimization, which lead to efficient and sustainable transportation ITS advancements are heavily reliant on AI algorithms such as deep neural networks, reinforcement

learning, and machine learning [205]. The advent of big data analytics and AI models has revolutionized ITS, enhancing sustainability and efficiency while paving the way for future transportation networks that are smarter and more interconnected.

Abirami et al. [206] executed an in-depth study and comprehensive investigation of the complementary integration of Big Data and AI in Intelligent Transportation Systems (ITS). The study clarifies the fundamental concepts of mentioned technologies, emphasizing their significant ability in handling existing transportation challenges. The study also addresses topics like data collection and computational issues. The systematic literature review primarily focuses on prediction, recognition, optimization, and autonomous vehicles, with most articles in prediction (37%), followed by recognition (25%), autonomous driving (20%), and optimization (18%). A total of 163 articles up to 2023 were retained for full-text examination. The study divided the paper into four main sections: background on big data and AI algorithms, various cloud-based and IoT-driven big data analytics platforms, the benefits, appropriateness, and limitations of AI algorithms in ITS, and a comprehensive overview of upcoming ITS research trends. The following are summaries of the major findings:

Big data frameworks in ITS: Sensors in Intelligent Transportation Systems collect real-time traffic data, processed by AI algorithms, for smart applications in smart cities. The data collection layer comprises sensors that collect traffic and congestion-related information in the transportation system, either in the form of video, image, signal, or numeric [207]. The collected data is converted into data analytics for processing to be used in fields like route planning, traffic

monitoring, accident detection, etc. It aids intelligent transportation systems by analyzing collected data for knowledge extraction. It utilizes distributed computing frameworks like Apache Spark and Hadoop. Lastly, the integration and decision layer are focused on extracting beneficial information from the processed data and displaying it in an approach that enables making decisions. The layer is where people interact for traffic prediction and assistance.

Big data acquired by sensors: the ITS highly deploys conventional on-road and vehicle-mounted sensors to gather data on traffic flow, vehicle speed, passenger routes, GPS coordinates, and travel timings. Some of regularly sensors used are: (i) Traffic Flow Sensors including Inductive Loop Detectors to identify vehicles in motion and Magnetic Sensors to detect moving objects to offer information on speed and the use of vehicles, (ii) Environmental Sensors including Weather Stations to gather data on humidity, temperature, rainfall, and wind speed all of which can impact the safety and efficiency of transportation and (ii) Air Quality Sensors to monitor pollutants using air quality sensors to evaluate environmental effects of transportation, (iii) Vehicle Detection Sensors including Radar Sensors are designed to track moving targets to provide information on the existence of a vehicle, its speed, and its movement patterns and other sensors like Infrared Sensors, Global Positioning System (GPS) sensors, and RFID and smart cards in Intelligent Parking System (IPS). The camera sensors to collect image and video data in traffic surveillance like Closed-Circuit Television (CCTV) Cameras and Automatic Number Plate Recognition (ANPR) Cameras. Additionally, the data generated by autonomous vehicles, recognized as an Intelligent Transportation System (ITS) technology, is utilized to gather and relay information such as location, destination, speed, and user preferences. This capability significantly improves their functionality, safety, and the overall transportation network. Communication data exchanges between vehicles (V2V) and infrastructure (V2I) involve alerts, safety warnings, traffic data, and coordination with traffic control systems [208].

Artificial intelligence algorithms: AI algorithms, including Evolutionary algorithms (EAs), Machine Learning (ML), Fuzzy Logic, and deep learning, effectively model and analyze large datasets, enhancing real-time analysis, data-driven decision-making, and automation in ITS. EAs are search-based optimization algorithms that use natural and genetic principles to optimize traffic signal timing and coordination in transportation networks, aiding effective management and route planning [209]. Fuzzy Logic is a reasoning technique that uses distorted and uncertain inputs to generate logic outputs, comparable to human thinking [210]. It serves as a methodology for managing vague and uncertain information within the transportation sector, facilitating effective traffic management and decision-making in Intelligent Transportation Systems (ITS). Its applications include

adaptive cruise control systems, assessment of route alternatives, and identification of unusual patterns. Nonetheless, challenges such as computational complexity, reliance on data-driven models, and the absence of standardized design and tuning approaches persist. ML is a widely used Big Data modeling technique that trains models to solve problems based on previous data instances without extensive programming [211]. Machine Learning (ML) can be categorized into three main types: (i) Supervised Algorithms, which utilize labeled data instances, including Support Vector Machines (SVM), Random Forest (RF), Decision Trees (DT), and Linear Regression (LR); (ii) Unsupervised Learning, which deals with unlabeled data instances, exemplified by the K-means algorithm; and (iii) Semi-Supervised Algorithms, which integrate both labeled and unlabeled data, combining elements of both supervised and unsupervised learning. Moreover, Deep learning algorithms, including Deep Neural Network, Convolutional Neural Network (CNN), and RNN, have significantly advanced ITS by extracting knowledge and patterns from large-scale data, enabling applications and improvements in various ITS applications, including hierarchical representations. In detail, CNN is an AI algorithm that uses multiple convolutional layers and filters to improve vehicle detection, license plate recognition, and traffic sign identification, enhancing ITS and protecting against cyberattacks. The Long-short term memory (LSTM) is a specific RNN used for analyzing time sequence data, consisting of a chain of cells that directly process inputs to extract patterns in the output sequences. It plays a vital role in forecasting pedestrian behavior and future movement patterns, thereby enhancing autonomous driving technologies and minimizing road accidents within ITS. Furthermore, in terms of Reinforcement Learning (RL), the algorithm interacts with the environment to learn a set of actions for a set of internal states and their associated actions using a specified set of rules and policies. Reinforcement learning is a machine learning technique that can analyze many kinds of data, including information about the current state of the transportation system, including traffic patterns and signal conditions [212].

Applications for big data in ITS: the study covered three categories (i) Prediction (ii) Recognition (iii) Optimization (iv) Autonomous driving (v) Big data platforms. More specifically, prediction in ITS pertains to forecasting traffic conditions and flow, as well as determining travel routes and coordinating traffic signal timings. Traffic data, including speed, flow, congestion, and travel time, is analyzed using AI for knowledge mining to predict future values at specific time instants. ML algorithms like SVM are utilized for forecasting traffic speed [213] and advanced learning algorithms such as CNN and DNN [214]. The RL algorithms are frequently employed to assess traffic conditions and establish traffic control signals [215]. The DL algorithms are utilized

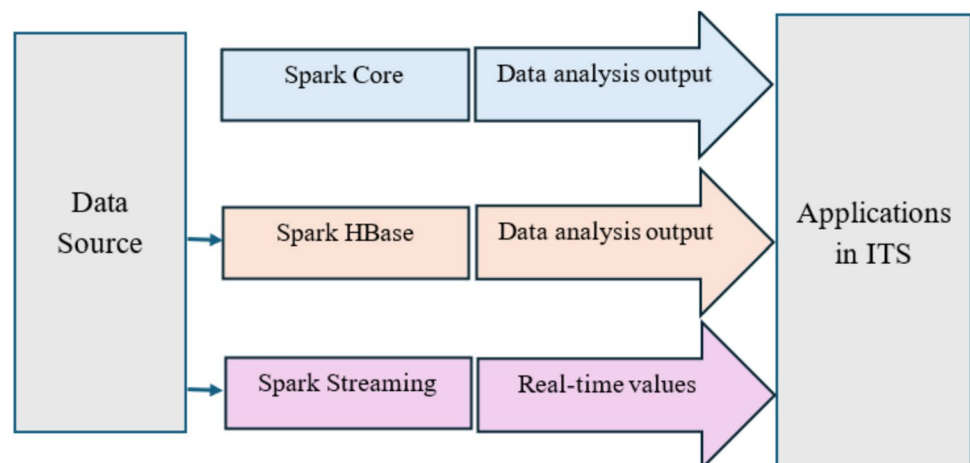
for forecasting traffic congestion. The DL algorithms are employed to predict traffic congestion. A type of RNN is employed to forecast traffic congestion through a traffic sensor prediction method [216]. Recognition plays a vital role in Intelligent Transportation Systems (ITS) for the identification of traffic signs, vehicle license plates, surveillance footage, pedestrian detection, and road hazards. Applications involving big data necessitate algorithms that can perform both object recognition and localization, which depend on feature extractors and classifiers. Machine learning classifiers have demonstrated effective results in the classification of traffic sign boards. A basic CNN with multi-stage non-linear transform layers and pooling layers is used for traffic sign identification [217]. The DL algorithms are also effective in vehicle detection tasks, which involve identifying specific objects within an image. This capability is crucial for detecting vehicle theft or the presence of other vehicles in proximity to an autonomous vehicle. Optimization is the process of enhancing the functionality or efficacy of a system or item by maximizing or decreasing specific parameters [218]. Operations planning, which involves utilizing available resources and determining travel routes, represents two critical optimization applications within ITS. Additionally, selecting the most efficient travel speed and minimizing waiting times are further optimization tasks in this domain. To address various optimization objectives for the most effective path segments in ITS, a fuzzy logic control system is employed, considering multiple traffic-related factors. A fuzzy bus holding system is suggested to enhance the waiting time for passengers in a public transportation network by utilizing GPS data. All the ITS applications referenced earlier are classified as part of autonomous driving. Autonomous driving, a subset of ITS applications, focuses on algorithms that reduce congestion, improve safety, and reduce delay time travel. The CNN [219] algorithm is utilized for range finding, aiming to minimize errors in autonomous vehicle highway driving. A mono camera mounted on a moving vehicle captures

images that are utilized for recognition tasks. The difficulties associated with implementing deep learning algorithms on these images for recognition purposes are addressed through contextual feature extraction and intraframe data fusion [220]. Advanced big data platforms like Hadoop and Apache Spark are used to process and analyze data from ITS sensors. Hadoop is universal and capable of large dataset storage, while Apache Spark is open-source and capable of machine learning-based applications. Spark's combination with Hadoop's shared storage features makes it the perfect platform for ITS applications that need large data analytics, also referred to as a distributed storage platform. The core of Spark regulates applications process by utilizing a dataset Hadoop's HBase database, so facilitating effective data management. For example, Godzilla proposed to perform traffic data analytics by handling growing big data under growing number of users under different workloads [221]. The major big data frameworks utilized in ITS encompass data streaming and data injection. A flexible Extract, Transform, and Load (ETL) architecture has been established for implementing dynamic toll charging on highways, leveraging big data platforms such as Spark and MongoDB, while Apache Sqoop facilitates the data injection process [222].

The development of big data models for ITS on big data platforms involves a systematic roadmap, starting with clear project objectives, data collection and preprocessing, and then integrating with big data platforms. This process optimizes distributed computing, deep learning, and real-time analytics, ensuring adaptability to evolving data patterns and low-latency processing (see Fig. 12).

Future research trends and challenges of AI algorithms in ITS: AI algorithms face various challenges when it comes to their implementation in ITS including (i) improving the algorithm's accuracy for processing big data (ii) ensure security when developing AI algorithms for smart city applications (iii) validate real time using AI in autonomous vehicles (iv) require a lightweight, scalable model for AI algorithms with

Fig. 12 Spark platform for intelligent transportation systems applications



faster inference (v) lack interpretability since the majority of AI systems, particularly deep learning models, are still function as black boxes [223], making reliability a concern.

Future research trends and challenges in the field of big data in ITS: The performance of Information and Communication Systems (ITS) has been significantly improved by big data, but further improvements are needed in areas such as data collection, storage, privacy, processing, and availability, requiring further research trends and challenges.

Akhtar and Moridpour [224] provide an overview of AI techniques used in predicting traffic congestion, highlighting the transition from conventional methods to more sophisticated machine learning and deep learning strategies. Forecasting traffic congestion requires the gathering of data, the creation of models, and subsequent validation. Each of these stages is essential for achieving precise outcomes. Traffic congestion has detrimental effects on the economy, environment, and public health. Accurate forecasting of congestion is essential for effective urban planning and enhancing the commuting experience. The application of AI, leveraging big data and GPS-equipped vehicles, is becoming increasingly important for the analysis of traffic trends. The data used in traffic congestion can be divided into main categories:

- *Stationary Sensor Data:* Stationary sensors collect traffic data; however, they can experience interruptions, which necessitates that authorities take this into account during their planning processes. A key benefit is that the location of vehicles is clearly defined. Stationary sensors collect spatiotemporal traffic information; however, they can experience interruptions, which should be considered by authorities during planning. The Performance Measurement System (PeMS) serves as the primary dataset for real-time traffic flow, sensor occupancy, and travel speed across major metropolitan areas in California. A significant number of studies utilize the I-5 highway dataset, which is updated every five minutes in San Diego, California [225].
- *Probe Vehicle Data:* Probe data, encompassing the complete road network, is extensively utilized in research, especially in studies focusing on a broad area. For instance, GPS data gathered every second from approximately 20,000 taxis in Beijing, China, comprises details such as taxi identification numbers, vehicle latitude and longitude coordinates, timestamps of the samples, and the number of passengers. The frequency of data updates in this dataset ranges from 10 s to 5 min, depending on the quality of the GPS device [225].

6 Literature Review Assessment

In this section, an assessment for the whole reviewed research adopted on the conducted methodologies of AI models for congestion estimation and control. This survey provides a systematic overview of the developments in traffic congestion estimation and control within intelligent transportation systems. The discussion includes application of various tools and techniques including machine learning, deep learning, reinforcement learning, sensor data fusion, evolutionary algorithms, and fuzzy logic systems for estimating and controlling traffic congestion.

The surveyed research highlighted a wide range of methods employed for traffic congestion estimation, including traditional statistical methods, ML techniques, evolutionary algorithms, deep learning, and hybrid models. The studies utilized various types of data, including sensor data (e.g., loop detectors, video cameras), GPS data from vehicles, and traffic flow data from intelligent transportation systems. In addition, some methods utilize external factors such as weather conditions, day of the week, and time of day. The integration of diverse data sources, such as real-time traffic data, historical data, and environmental factors, enhances the accuracy and robustness of congestion estimation models.

Several studies in the literature emphasized real-time traffic congestion prediction and the development of models capable of making short-term and long-term forecasts. Techniques such as RNNs, LSTM networks, and GRUs were employed to capture temporal dependencies in traffic data. These models aim to provide timely and accurate predictions to facilitate proactive traffic management and congestion mitigation. Image-based traffic congestion detection was investigated through advanced deep learning techniques (CNNs). Such evolution from traditional methods to sophisticated deep learning models reflects the ongoing advancement in leveraging computational power and data availability for improved traffic congestion estimation.

The lack of large, annotated data has motivated researchers to employ unsupervised learning techniques with the large unannotated publicly available data for congestion estimation. However, only classic unsupervised learning techniques, such as clustering and dimensionality reduction, were used for estimating road congestion. Additionally, these techniques were used to preprocess data for other models. More advanced end-to-end semi-supervised and unsupervised learning techniques should be explored for traffic congestion estimation. These techniques can boost performance given the huge amount of unannotated data publicly available.

Several filtering algorithms were utilized to handle data from different sources for real-time prediction of traffic

problems. However, the Kalman filter and its variants are the most commonly used in the literature for traffic congestion estimation.

The data-driven approaches, particularly those involving ML algorithms, have been reported to be effective in controlling road traffic congestion. Various ML algorithms such as kNN and SVM etc. have been incorporated into real-time congestion control systems which are more responsive and can dynamically adapt to the changing traffic conditions as opposed to the off-line preprogrammed methods. Deep learning algorithms i.e., neural networks have been utilized to work seamlessly with smart city technologies such as the IoT and various other detectors and sensors including video cameras. Since video cameras are installed in various countries for safety concerns and for detecting traffic signal violations, methods based on convolutional neural networks such as YOLO model provide an opportunity to utilize these video cameras for traffic congestion control. Reinforcement learning framework has been extensively used to optimize traffic signal timings in response to the changing traffic conditions to minimize traffic congestion. To enhance the capability of ML and AI techniques, hybrid approaches have been proposed which combine ML and AI techniques with other methods such as analytical approaches, game theory, and optimization algorithms. As their name suggests, data-driven approaches are highly data-dependent and inaccurate or insufficient data can lead to suboptimal performance. ML models are often termed black-box models due to their limited interpretability. In addition, scalability is also a concern when implementing real-time approaches that require data input from certain types of sensors or detectors.

Optimization methods have long been employed to alleviate traffic congestion. Reducing congestion by using optimization techniques happens at three scales i.e., at the areawide level, along corridors/arterials, and at isolated signalized intersections. Traditionally, the optimization of traffic signal settings at isolated intersections was performed offline and such signals were often termed as pretimed signals. More recently, several studies have proposed traffic signal optimization approaches that are more advanced than pretimed signals such as actuated and adaptive traffic signals, which gather traffic information through detectors and sensors and adjust the signal settings accordingly to reduce traffic congestion at signalized intersections. Since reducing traffic congestion at an intersection by optimizing its signal timing may affect the performance of nearby intersections, the arterial signal coordination problem has gained sufficient attention in literature. The arterial signal coordination problem often involves maximizing the bandwidth or minimizing the vehicular delay. Additionally, the literature contains various optimization approaches proposed to reduce congestion at an area-wide level. Various commercial and open-source simulation software have been utilized

to evaluate these signal timing approaches including PTV VISSIM, PARAMICS, and SUMO etc. Regardless of the scale of the signal timing problem, heuristic algorithms such as genetic algorithm, ant colony algorithm, and particle swarm algorithms are among the commonly used algorithms for optimizing signal settings. The objective function often consists of minimizing vehicular delays, maximizing vehicular efficiency, or minimizing the number of stops. In addition, bandwidth maximization and total travel time minimization are often the objective functions at the corridor level and network level, respectively. To prioritize high occupancy vehicles, objective functions have been proposed that minimize person-delay instead of vehicular delay. The signal timing optimization problem is sometimes formulated as a multi-objective optimization problem, aiming to maximize vehicular efficiency while simultaneously minimizing vehicular emissions. However, limited attention has been paid to the emerging mobility options such as micromobility vehicles while optimizing traffic signal settings.

While data-driven and optimization techniques primarily emphasize the methodological aspect, various approaches leverage recent technological advancements, including advancements in sensors and detectors, to alleviate road traffic congestion. Sensors and detectors are important components of congestion mitigation techniques. Data-driven and optimization approaches for alleviating traffic congestion often rely on the data input from various sensors and detectors deployed at either intersections or along road segments. These may include inductive loop detectors, WIFI devices, Bluetooth, LiDAR, GPS, RADAR, RFID, magnetic sensors, cameras etc. The data obtained from different sensors and detectors is sometimes fused together to provide a holistic picture of the traffic conditions to reduce traffic congestion. Moreover, the literature extensively investigates how IoT can alleviate traffic congestion by using interconnected devices and sensors to gather and share real-time data for improved traffic flow and decision-making. Finally, literature presents integrated approaches combining IoT, reliant on sensor and detector data, with data-driven approaches, all aimed at effectively reducing road traffic congestion.

Fuzzy-based control traffic refers to a method of controlling traffic flow using fuzzy logic systems. Traditional methods of traffic control typically rely on fixed algorithms and predetermined rules. However, fuzzy logic systems allow for more flexibility and adaptation in response to changing traffic conditions. By integrating fuzzy logic systems into traffic control management, traffic engineers can create more efficient and responsive traffic control systems that can better alleviate congestion, reduce travel times, and improve overall safety on the roads. In essence, fuzzy-based control of traffic can help to optimize traffic flow by adapting to dynamic conditions and making nuanced decisions based on real-time data. Fuzzy PID, fuzzy inference and fuzzy

priority systems have emerged as effective tools for controlling traffic in urban areas. These systems use fuzzy logic to model the complex and uncertain nature of traffic conditions, allowing for more adaptive and responsive control strategies. Lack of precision: Fuzzy inference and fuzzy priority systems rely on linguistic variables and membership functions to make decisions, which can lead to ambiguity and imprecision in the control of traffic flow. This lack of precision can result in suboptimal decision-making and potentially lead to traffic congestion or accidents. Having said that, fuzzy-based systems for traffic control can be complex and require a significant number of computational resources. This can lead to delays in response times and potentially exacerbating traffic congestion.

The Nash bargaining game-theoretic framework is another valuable tool for analyzing and optimizing traffic control systems. In this framework, the interaction between multiple decision-makers, such as drivers and traffic controllers, is modelled as a game where each player aims to maximize their own utility. Traffic control systems face the challenge of managing limited resources, such as road capacity and signal timing, to ensure efficient and safe traffic flow. This requires balancing the competing interests of various stakeholders, including minimizing travel time for drivers, reducing congestion and emissions, and maximizing revenue for transportation agencies. Overall, the Nash bargaining game-theoretic framework offers a comprehensive approach to traffic control that considers the complex interactions and trade-offs inherent in managing urban transportation systems. By applying this framework, planners and policymakers can make more informed decisions that improve efficiency, sustainability, and safety of traffic flow. One major drawback of using Nash bargaining game-theoretic frameworks to control traffic is the complexity and unpredictability of human behavior. Traffic flow is influenced by a variety of factors, including individual preferences, emotions, and decision-making processes. These variables can be difficult to accurately model and predict within the framework of Nash bargaining games, leading to potential inefficiencies in traffic management. Additionally, Nash bargaining games rely on the assumption that all players are rational and self-interested, seeking to maximize their own utility. Drivers may not always act in a rational manner, such as engaging in aggressive or risky behavior on the road. This can lead to breakdowns in the bargaining process and result in suboptimal traffic flow patterns.

Efficient and sustainable traffic management solutions are also addressed through the implementation of integrated learning and control systems. These systems combine advanced technology and ML algorithms to monitor and manage traffic flow in real-time. By analyzing data from various sources, such as traffic cameras, sensors, and GPS devices, these systems can make informed decisions to

improve traffic efficiency and reduce congestion. Integrated learning and control systems use real-time data to adjust traffic signal timings, coordinate traffic flow at intersections, and identify alternative routes for drivers to avoid congested areas. These systems can also predict traffic patterns and adjust their operations accordingly to prevent gridlock and ensure a smooth flow of vehicles throughout the city. Despite their benefits, integrated learning and control systems are complex involve numerous interconnected components such as sensors, cameras, algorithms, and control systems. This complexity can make the system difficult to understand and troubleshoot when issues arise. In addition, implementation can be expensive, requiring significant investment in hardware, software, and maintenance. This cost can be a barrier for smaller cities or municipalities looking to implement such a system.

From a congestion estimation perspective, AI models, particularly supervised learning algorithms such as support vector machines, decision trees, and random forests, have been frequently used to estimate congestion levels based on historical and real-time traffic parameters. These models have several advantages in terms of their interpretability and relatively low computational costs. The applications of deep learning models such as artificial neural networks, convolutional neural networks (CNNs), RNNs, and LSTM networks have also been explored for traffic congestion estimation. These models are capable of capturing both spatial and temporal dependencies. For instance, CNN-based frameworks have been utilized effectively with image and video data from roadside cameras, while RNN and LSTM models have shown high performance in predicting short-term traffic flow using time-series sensor data.

Unsupervised learning models such as clustering algorithms (e.g., DBSCAN), dimensionality reduction techniques (e.g., variational autoencoders), and fuzzy-based approaches have addressed the limitations of labeled data by extracting latent traffic patterns and estimating congestion in a data-driven yet interpretable manner. These approaches have proven useful for anomaly detection, unsupervised traffic state classification, and real-time monitoring in resource-constrained environments.

Multi-sensor fusion-based systems have been used to reduce the uncertainty of single-sensor data capturing devices. Data from different sensors were fused and fed into traffic management systems. Sensor data fusion has played a major role in improving traffic state estimation. Accuracy, robustness, and coverage have been enhanced by multi-sensor data fusion techniques that integrate loop detectors, GPS, radar, video feeds, and UAV observations. Kalman filters, particle filters, and Bayesian fusion models were particularly effective in integrating heterogeneous data sources, reducing noise, and dealing with missing values. Studies combining Kalman filtering with machine

learning or k-nearest neighbor models demonstrated increased forecasting accuracy over single-source techniques. Such hybrid approaches allowed for greater adaptability to varying traffic regimes and urban topologies.

From a congestion control perspective, AI-based strategies have played a vital role. Traditional AI models such as logistic regression, k-nearest neighbors, and support vector machines have been implemented to identify and control congestion at road segments. These models have been used for route guidance, signal timing adjustment, and traveler information systems. Deep learning approaches have extended this by leveraging traffic images, videos, real-time speed and flow data, and even social media feeds to develop predictive control models. CNNs and YOLO-based systems were employed for object detection and dynamic signal control, showing efficacy in both isolated and coordinated intersection scenarios.

From a congestion control perspective, data-driven and machine learning-based strategies have played a vital role. Road traffic congestion has been identified and managed using conventional machine learning models like logistic regression, k-nearest neighbors, and support vector machines. Traveler information systems, signal timing adjustments, and route guidance have all made use of these models. This has been expanded by deep learning techniques, which use real-time speed and flow data, videos, traffic photos, and even social media feeds to create predictive control models. For object detection and dynamic signal control, CNNs and YOLO-based systems were used, and they demonstrated effectiveness in both isolated and coordinated intersection scenarios.

Reinforcement learning has emerged as a powerful technique for adaptive traffic signal control. Intersections can learn optimal phase settings based on environmental feedback using deep Q-learning and multi-agent Q-learning frameworks, which improves throughput and decreases delay. Recent research has examined joint-action and decentralized multi-agent systems in which agents (intersections) learn both individual and collective policies, showing notable gains in performance in dense networks. Models that incorporate Q-value sharing among agents or use dynamic discounting in their Bellman updates have shown promising results in reducing overall system delay and enhancing network stability.

Fuzzy logic and neuro-fuzzy inference systems provided flexible, rule-based control strategies particularly effective in uncertain or imprecise environments. These systems have been deployed for ramp metering, congestion classification, and incident response. Neuro-fuzzy models were often used in tandem with sensor networks or IoT data streams to deliver adaptive control at signalized intersections or transport hubs.

The survey also identifies common features and differences among the discussed methods. Deep learning models offer strong prediction performance but often require large datasets and high computational resources. Fusion-based methods enhance accuracy and robustness, particularly under noisy or missing data scenarios. Fuzzy logic systems allow decision-making under uncertainty and are well-suited for expressway and incident-related applications. Reinforcement learning provides adaptive control capabilities, particularly effective at signalized intersections and under dynamic traffic conditions.

In the literature, case studies have been conducted in a wide range of settings and locations around the world. These diverse applications validate the generalizability of the reviewed models. Comparative analyses have shown that hybrid models, which combine statistical, machine learning, or deep learning components, often outperform solo approaches in terms of accuracy and responsiveness. The integration of diverse data sources, such as real-time traffic data, historical data, and environmental factors, enhances the accuracy and robustness of congestion estimation models.

Despite these advances, challenges still exist. Many models are not transferable across regions due to data and infrastructure heterogeneity. Despite their higher predictive accuracy, deep learning models require significant training costs. Furthermore, the integration of estimating and control components remains fragmented in most existing frameworks. Moving forward, there is a strong need for unified frameworks that combine congestion sensing, estimation, and control. Reinforcement learning methods with cooperative policies offer a promising direction, especially when coupled with edge computing for scalability. Privacy-aware sensor fusion and uncertainty-aware control systems are also needed to support deployment in smart cities. Addressing these areas will further enable real-time, adaptive, and context-sensitive traffic management systems that can operate effectively under the evolving demands of urban mobility. In conclusion, this review outlines a detailed taxonomy of congestion estimation and control models, provides cross-comparisons, and highlights the strengths of each methodological category. By bridging sensing, estimation, and control, the paper offers a comprehensive foundation for future work aiming to develop intelligent, data-driven, and context-aware traffic management systems.

7 Future Directions

Based on the reported literature review and assessment, several recommendations are proposed for future research directions:

- Utilizing semi-supervised techniques that leverage the limited amount of labeled data available with the extensive unlabeled data can enhance the performance of traffic congestion estimation systems.
- Few-shot learning techniques have not been explored for this problem. The advancements in few-shot learning techniques can assist in addressing the data scarcity issue in traffic congestion management.
- Large image models, such as CLIP, have demonstrated significant improvements in various computer vision downstream tasks. Evaluating the performance of these models on traffic data using zero-shot learning is a promising research direction. Additionally, fine-tuning these models on traffic data can help leverage the knowledge from these large models for the traffic congestion estimation problem.
- Focusing on integrating emerging technologies such as IoT sensors, 5G connectivity, and edge computing. Such technologies can provide real-time, high-resolution traffic data and enable decentralized processing, which can significantly enhance the responsiveness and accuracy of congestion prediction models.
- Investigating adaptive and personalized congestion prediction models that account for individual driver behavior and preferences. By incorporating user-specific data and context-aware information, such models can provide tailored traffic predictions and route suggestions.
- Currently, data-driven approaches typically consist of ML and AI models which are mostly considered black-box models and lack transparency in their decision-making process. Utilizing explainable ML techniques can help better understand the contribution of different factors in increasing road traffic congestion. Identifying the key factors contributing to congestion would assist in alleviating it.
- The integration of the integration of ML algorithms into frameworks for traffic control. ML algorithms can be integrated with fuzzy logic or game theory. By incorporating ML and predictive analytics, traffic signals could dynamically adjust their timing based on vehicles real-time data. This could lead to even more efficient traffic management and reduced congestion on roads.
- Many proposed frameworks have been evaluated through simulation studies using synthetic datasets. Although it is a challenge, evaluating the proposed congestion control mechanisms using real datasets would provide a more accurate evaluation.
- The majority of the literature focuses on optimizing signal settings at signalized intersections, primarily four-legged ones, to reduce congestion. Future research should aim to minimize congestion at alternative intersections, including signalized roundabouts, unsignalized intersections, and unconventional configurations.
- While some existing congestion mitigation strategies focus on reducing delays for both vehicles and pedestrians simultaneously, there is a need for more comprehensive approaches that account for diverse mobility modes. With the increasing adoption of micromobility options like electric bicycles, bike sharing, and e-scooters, it is vital to alleviate congestion for all users of the road. This may promote the use of alternative transportation modes and decrease car usage, ultimately reducing road traffic congestion.
- Rather than using AI as a reactive measure to address road traffic congestion, implementing traffic demand management strategies could proactively reduce travel demand from the outset. AI can assist in developing and implementing demand management policies. For instance, if certain vehicles are restricted from roads on specific days as a demand management strategy, image recognition techniques can identify violators, thereby helping to reduce congestion.
- Future research should consider the financial feasibility of implementing ML techniques for mitigating road traffic congestion on a larger scale. Some approaches rely heavily on extensive data inputs gathered from various sensors and detectors. While these methods may be effective at an intersection or within a small network, scaling them to a city-wide application could raise significant financial concerns. Consequently, a thorough financial evaluation of the city-level implementation of AI-based methods is essential. This evaluation should account for the costs associated with the deployment and maintenance of the necessary infrastructure and the potential economic benefits derived from reduced congestion.
- Sensors and detectors, particularly video cameras, track human activity and record their pictures and videos often raise privacy concerns. Advanced machine learning techniques such as federated learning can be explored to address privacy concerns.

Acknowledgements The authors would like to thank the reviewers and editors for their comprehensive and constructive comments for improving the manuscript. In addition, Zaher Mundher Yaseen would like to thank the Civil and Environmental Engineering Department, King Fahd University of Petroleum & Minerals, Saudi Arabia.

Authors Contributions Younus Hasan Taher: Conceptualization, reviewing, data collection, writing up, analysis, investigation. Jit Singh Mandeep: Conceptualization, reviewing, data collection, writing up, analysis, investigation. Haydar Abdulameer Marhoon: Conceptualization, reviewing, data collection, writing up, analysis, investigation. Hamdi A. Al-Jamimi: Conceptualization, reviewing, data collection, writing up, analysis, investigation. Hamzah Luqman: Conceptualization, reviewing, data collection, writing up, analysis, investigation. Farag Azzedin: Conceptualization, reviewing, data collection, writing up, analysis, investigation. Muhammad Abdullah: Conceptualization, reviewing, data collection, writing up, analysis, investigation. Hassan

M. Al-Ahmadi: Conceptualization, reviewing, data collection, writing up, analysis, investigation. Zaher Mundher Yaseen: Conceptualization, reviewing, data collection, writing up, analysis, investigation, supervision, project leader.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of Interest The authors declare no competing interests.

References

- Gohar A, Nencioni G (2021) The role of 5G technologies in a smart city: the case for intelligent transportation system. *Sustainability* 13:51–88
- Muthuramalingam S, Kumar ABR, Gayathri N et al (2019) IoT based intelligent transportation system (IoT-ITS) for global perspective: a case study
- Ghaleb M, Azzedin F (2023) Trust-aware fog-based IoT environments: artificial reasoning approach. *Appl Sci* 13:3665. <https://doi.org/10.3390/app13063665>
- SN S (2024) A decade of research in fog computing: Relevance, challenges, and future directions. *Softw Pract Exp* 54:3–23
- Cataldo-Díaz C, Escobar JW LR (2024) Mathematical models for the electric vehicle routing problem with time windows considering different aspects of the charging process. *Oper Res* 24
- Çil ZA, Öztö H, KD (2024) Mathematical models for disassembly line balancing and pickup-delivery vehicle routing problem. *J Transp Logist* 8:180–190
- Guerrero-Ibáñez J, Zeadally S, Contreras-Castillo J (2018) Sensor technologies for intelligent transportation systems. *Sensors* 18:1212. <https://doi.org/10.3390/s18041212>
- Chen Y, Chen C, Wu Q et al (2021) Spatial-temporal traffic congestion identification and correlation extraction using floating car data. *J Intell Transp Syst* 25:263–280
- Jia Y, Wu J, Xu M (2017) Traffic flow prediction with rainfall impact using a deep learning method. *J Adv Transp* 2017:1–10. <https://doi.org/10.1155/2017/6575947>
- Nama M, Nath A, Bechra N et al (2021) Machine learning-based traffic scheduling techniques for intelligent transportation system: opportunities and challenges. *Int J Commun Syst* 34
- Akhtar M, Moridpour S (2021) A review of traffic congestion prediction using artificial intelligence. *J Adv Transp*
- Kumar N, Raubal M (2021) Applications of deep learning in congestion detection, prediction and alleviation: a survey. *Transp Res Part C Emerg Technol* 133:103–432
- Kaffash S, Nguyen AT, Zhu J (2021) Big data algorithms and applications in intelligent transportation system: a review and bibliometric analysis. *Int J Prod Econ* 231:107868
- Yue W, Li C, Mao G et al (2021) Evolution of road traffic congestion control: a survey from perspective of sensing, communication, and computation. *China Commun* 18:151–177. <https://doi.org/10.23919/jcc.2021.12.010>
- N BK-H, Cho J, Yi H (2021) Patial-temporal graph neural network for traffic forecasting: an overview and open research issues. *Appl Intell*: 1–12
- Xin H, Liu S, Li T (2020) A multilogarithm linear regression traffic congestion prediction model based on R language New York City's five administrative districts (mainly in urban main roads and highways). In: *Proceedings—2020 international conference on big data and artificial intelligence and software engineering, ICBASE 2020*
- Tang J, Chen X, Hu Z et al (2019) Traffic flow prediction based on combination of support vector machine and data denoising schemes. *Physica A* 534:120642. <https://doi.org/10.1016/j.physa.2019.03.007>
- Gokasar I, Timurogullari A (2021) Real-time prediction of traffic density with deep learning using computer vision and traffic event information. In: *2021 international conference on innovations in intelligent systems and applications, INISTA 2021—proceedings*
- Sujatha R, Nithya RA, Subhpradha S, Srinithibharathi S (2018) Decision tree classification for traffic congestion detection using data mining. *Int J Eng Techn* 4:166–173
- Cheng L, Chen X, De Vos J et al (2019) Applying a random forest method approach to model travel mode choice behavior. *Travel Behav Soc*: 14. <https://doi.org/10.1016/j.tbs.2018.09.002>
- Wang G, Kim J (2016) The prediction of traffic congestion and incident on urban road networks using Naive Bayes classifier. In: *ATRF 2016—Australasian transport research forum 2016, proceedings*
- Cai L, Yu Y, Zhang S et al (2020) A sample-rebalanced outlier-rejected k-Nearest neighbor regression model for short-term traffic flow forecasting. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2020.2970250>
- Agafonov A, Yumaganov A (2018) Short-term traffic flow forecasting using a distributed spatial-temporal k nearest neighbors model. In: *Proceedings—21st IEEE international conference on computational science and engineering, CSE 2018*
- Lusiandro MA, Nasution SM, Setianingsih C (2020) Implementation of the advanced traffic management system using k-nearest neighbor algorithm. In: *2020 international conference on information technology systems and innovation, ICITSI 2020—proceedings*
- Chen Y, Li S, Pan Y, Zhang J (2022) Urban expressway congestion forewarning based on slope change of traffic flow fundamental diagram. *J Transp Eng A Syst* 148:4022030
- Narendra Babu C, Sure P, Bhuma CM (2021) Sparse Bayesian learning assisted approaches for road network traffic state estimation. *IEEE Trans Intell Transp Syst*. <https://doi.org/10.1109/TITS.2020.2971031>
- Kuang L, Yan H, Zhu Y et al (2019) Predicting duration of traffic accidents based on cost-sensitive Bayesian network and weighted K-nearest neighbor. *J Intell Transp Syst* 23:161–174. <https://doi.org/10.1080/15472450.2018.1536978>
- Zhou F, Yang Q, Zhong T et al (2021) Variational graph neural networks for road traffic prediction in intelligent transportation systems. *IEEE Trans Industr Inform* 17:2802–2812. <https://doi.org/10.1109/tii.2020.3009280>
- Afrin T, Yodo N (2021) A probabilistic estimation of traffic congestion using Bayesian network. *Measurement (Lond)*. <https://doi.org/10.1016/j.measurement.2021.109051>
- Guo J, Kong Y, Li Z et al (2019) A model and genetic algorithm for area-wide intersection signal optimization under user equilibrium traffic. *Math Comput Simul* 155:92–104. <https://doi.org/10.1016/j.matcom.2017.12.003>
- Hashmiejad SH-A, Hasheminejad SMH (2017) Traffic accident severity prediction using a novel multi-objective genetic algorithm. *Int J Crashworthiness* 22:425–440. <https://doi.org/10.1080/13588265.2016.1275431>
- Yao Y, Hong Y, Wu D et al (2018) Estimating the effects of “community opening” policy on alleviating traffic congestion in large Chinese cities by integrating ant colony optimization and complex network analyses. *Comput Environ Urban Syst* 70:163–174. <https://doi.org/10.1016/j.compenurbysys.2018.03.005>

33. Zhu D, Du H, Sun Y, Cao N (2018) Research on path planning model based on short-term traffic flow prediction in intelligent transportation system. *Sensors (Basel)* 18:4275. <https://doi.org/10.3390/s18124275>
34. Mondal MA, Rehena Z (2019) Intelligent traffic congestion classification system using artificial neural network. In: The web conference 2019—companion of the world wide web conference, WWW 2019
35. Fahs W, Chbib F, Rammal A et al (2023) Traffic congestion prediction based on multivariate modelling and neural networks regressions. *Procedia Comput Sci*
36. Shamsashtiany R, Ameri M (2018) Road accidents prediction with multilayer perceptron MLP modelling case study: roads of Qazvin, Zanjan and Hamadan. *JCEMA*
37. Gu Y, Lu W, Xu X et al (2020) An improved Bayesian combination model for short-term traffic prediction with deep learning. *IEEE Trans Intell Transp Syst*. <https://doi.org/10.1109/TITS.2019.2939290>
38. Wang J, Deng W, Guo Y (2014) New Bayesian combination method for short-term traffic flow forecasting. *Transp Res Part C Emerg Technol* 43:79–94. <https://doi.org/10.1016/j.trc.2014.02.005>
39. Guo Z, Li B, Ljfar H (2021) Urban traffic modeling and pattern detection using online map vendors and self-organizing maps. 10:715–728
40. Gu Y, Wang Y, Dong S (2020) Public traffic congestion estimation using an artificial neural network. *ISPRS Int J Geoinf* 9:152. <https://doi.org/10.3390/ijgi9030152>
41. Nazarov AA, Moiseev AN (2014) Distributed system of processing of data of physical experiments. *Russ Phys J* 57:984–990. <https://doi.org/10.1007/s11182-014-0334-3>
42. Guo K, Hu Y, Qian Z et al (2020) Optimized graph convolution recurrent neural network for traffic prediction. *IEEE Trans Intell Transp Syst* 22:1138–1149
43. Gao Y, Li J, Xu Z et al (2021) A novel image-based convolutional neural network approach for traffic congestion estimation. *Expert Syst Appl*. <https://doi.org/10.1016/j.eswa.2021.115037>
44. Chakraborty P, Adu-Gyamfi YO, Poddar S et al (2018) Traffic congestion detection from camera images using deep convolution neural networks. *Transp Res Rec*. <https://doi.org/10.1177/0361198118777631>
45. Li L, Lin H, Wan J et al (2020) MF-TCPV: a machine learning and fuzzy comprehensive evaluation-based framework for traffic congestion prediction and visualization. *IEEE Access* 8:227113–227125. <https://doi.org/10.1109/access.2020.3043582>
46. Ganapathy J, Sureshkumar T, Raghavendra Prasad M, Dhamini C (2022) Auto-Encoder LSTM for learning dependency of traffic flow by sequencing spatial-temporal traffic flow rate: a speed up technique for routing vehicles between origin and destination. In: 2022 international conference on innovative trends in information technology, ICITIIT 2022
47. Jeong M-H, Lee T-Y, Jeon S-B, Youm M (2021) Highway speed prediction using gated recurrent unit neural networks. *Appl Sci* 11:3059. <https://doi.org/10.3390/app11073059>
48. Zhang F, Wu T-Y, Wang Y et al (2020) Application of quantum genetic optimization of LVQ neural network in smart city traffic network prediction. *IEEE Access* 8:104555–104564. <https://doi.org/10.1109/access.2020.2999608>
49. Luo X, Niu L, Zhang S (2018) An algorithm for traffic flow prediction based on improved SARIMA and GA. *KSCE J Civ Eng* 22:4107–4115. <https://doi.org/10.1007/s12205-018-0429-4>
50. Zheng H, Lin F, Feng X, Chen Y (2021) A hybrid deep learning model with attention-based Conv-LSTM networks for short-term traffic flow prediction. *IEEE Trans Intell Transp Syst* 22:6910–6920. <https://doi.org/10.1109/tits.2020.2997352>
51. Yang H, Du L, Zhang G, Ma T (2023) A traffic flow dependency and dynamics based deep learning aided approach for network-wide traffic speed propagation prediction. *Transp Res Part B Methodol* 167:99–117. <https://doi.org/10.1016/j.trb.2022.11.009>
52. Oyeyemi Olayode I, Du B, Kwanda Tartibu L, Justice Alex F (2023) Traffic flow modelling of long and short trucks using a hybrid artificial neural network optimized by particle swarm optimization. *Int J Transp Sci Technol*. <https://doi.org/10.1016/j.ijst.2023.04.004>
53. Pan YA, Guo J, Chen Y et al (2024) A fundamental diagram based hybrid framework for traffic flow estimation and prediction by combining a Markovian model with deep learning. *Expert Syst Appl* 238:122219
54. Li A, Xu Z, Li W et al (2025) Urban signalized intersection traffic state prediction: a spatial-temporal graph model integrating the cell transmission model and transformer. *Appl Sci* 15:2377
55. Yong Benjamin TZ, Alam S, Ma CY (2021) A machine learning approach for the prediction of top of descent. In: 2021 IEEE/AIAA 40th digital avionics systems conference (DASC)
56. Shamitha C, Radhika SKM et al (2022) An intelligent IoT enabled traffic queue handling system using machine learning algorithm. In: 2022 international conference on innovative computing, intelligent communication and smart electrical systems (ICES). *IEEE*, pp 1–9
57. Reynolds HJD, Reynolds TG, Hansman RJ (2005) Human factors implications of continuous descent approach procedures for noise abatement in air traffic control. In: Proceedings of the 6th USA/Europe air traffic management research and development seminar, ATM 2005
58. Priambodo B, Ahmad A, Kadir RA (2021) Predicting traffic flow propagation based on congestion at neighbouring roads using hidden Markov model. *IEEE Access* 9:85933–85946. <https://doi.org/10.1109/access.2021.3075911>
59. Boquet G, Morell A, Serrano J, Vicario JL (2020) A variational autoencoder solution for road traffic forecasting systems: Missing data imputation, dimension reduction, model selection and anomaly detection. *Transp Res Part C Emerg Technol* 115:102622. <https://doi.org/10.1016/j.trc.2020.102622>
60. Ullah I, Youn J, Han Y-H (2021) Multisensor data fusion based on modified belief entropy in Dempster-Shafer theory for smart environment. *IEEE Access* 9:37813–37822. <https://doi.org/10.1109/access.2021.3063242>
61. Amini M, Hatwagner MF, Mikulaj GCS, Koczy LT (2021) An intelligent traffic congestion detection approach based on fuzzy inference system. 2021 IEEE 15th international symposium on applied computational intelligence and informatics (SACI)
62. Septyanto AW, Suryono S, Rosyida I (2018) Fuzzy rule-based system for monitoring traffic congestion using technology radio frequency identification. In: 2018 third international conference on informatics and computing (ICIC)
63. Toan TD, Wong YD, Lam SH, Meng M (2022) Developing a fuzzy-based decision-making procedure for traffic control in expressway congestion management. *Physica A* 604:127899. <https://doi.org/10.1016/j.physa.2022.127899>
64. Gollapalli M, Atta-ur-Rahman, Musleh D et al (2022) A Neuro-fuzzy approach to road traffic congestion prediction. *Comput Mater Continua* 73:295–310. <https://doi.org/10.32604/cmc.2022.027925>
65. Alawad H, An M, Kaewunruen S (2020) Utilizing an adaptive neuro-fuzzy inference system (ANFIS) for overcrowding level risk assessment in railway stations. *Appl Sci* 10:5156. <https://doi.org/10.3390/app10155156>
66. Olayode IO, Tartibu LK, Alex FJ (2023) Comparative study analysis of ANFIS and ANFIS-GA models on flow of vehicles at road intersections. *Appl Sci (Switzerland)* 13. <https://doi.org/10.3390/app13020744>

67. Sathyan A, Ma J, Cohen K (2021) Decentralized cooperative driving automation: a reinforcement learning framework using genetic fuzzy systems. *Transportmetrica B Transp Dyn* 9:775–797. <https://doi.org/10.1080/21680566.2021.1951394>
68. Trinh Dinh T (2022) A fuzzy-based methodology for anticipating trend of incident traffic congestion on expressways. *Transp Commun Sci J* 73:381–396. <https://doi.org/10.47869/tcsj.73.4.4>
69. Bachmann C, Abdulhai B, Roorda MJ, Moshiri B (2013) A comparative assessment of multi-sensor data fusion techniques for freeway traffic speed estimation using microsimulation modeling. *Transp Res Part C Emerg Technol* 26:33–48. <https://doi.org/10.1016/j.trc.2012.07.003>
70. Suseendran G, Akila D, Vijaykumar H et al (2021) Multi-sensor information fusion for efficient smart transport vehicle tracking and positioning based on deep learning technique. *J Supercomput* 78:6121–6146. <https://doi.org/10.1007/s11227-021-04115-6>
71. Liu J, Liu Z, Zhang H et al (2021) Multi-sensor information fusion for IoT in automated guided vehicle in smart city. *Soft comput* 25:12017–12029. <https://doi.org/10.1007/s00500-021-05696-3>
72. Jin G, Cui Y, Zeng L et al (2020) Urban ride-hailing demand prediction with multiple spatio-temporal information fusion network. *Transp Res Part C Emerg Technol* 117:102665. <https://doi.org/10.1016/j.trc.2020.102665>
73. Yahia CN, Scott SE, Boyles SD, Claudel CG (2021) Unmanned aerial vehicle path planning for traffic estimation and detection of non-recurrent congestion. *Transp Lett* 14:849–862. <https://doi.org/10.1080/19427867.2021.1951524>
74. Trinh X-S, Ngody D, Keyvan-Ekbatani M, Robertson B (2021) Incremental unscented Kalman filter for real-time traffic estimation on motorways using multi-source data. *Transportmetrica A Transp Sci* 18:1127–1153. <https://doi.org/10.1080/23249935.2021.1931548>
75. Liang S, Ma M, He S, Zhang H (2019) Short-term passenger flow prediction in urban public transport: Kalman filtering combined K-nearest neighbor approach. *IEEE Access* 7:120937–120949. <https://doi.org/10.1109/access.2019.2937114>
76. Wang Y, Work DB (2021) Estimation for heterogeneous traffic using enhanced particle filters. *Transportmetrica A Transp Sci* 18:568–593. <https://doi.org/10.1080/23249935.2021.1881186>
77. Hawes M, Amer HM, Mihaylova L (2017) Traffic state estimation via a particle filter over a reduced measurement space. In: 2017 20th international conference on information fusion (fusion)
78. Mil S, Piantanakulchai M (2018) Modified Bayesian data fusion model for travel time estimation considering spurious data and traffic conditions. *Appl Soft Comput* 72:65–78. <https://doi.org/10.1016/j.asoc.2018.06.046>
79. Gitahi J, Hahn M, Storz M et al (2020) Multi-sensor traffic data fusion for congestion detection and tracking. The international archives of the photogrammetry, remote sensing and spatial information sciences XLIII-B1-2:173–180. <https://doi.org/10.5194/isprs-archives-xliii-b1-2020-173-2020>
80. Cheng J, Li G, Chen X (2019) Developing a travel time estimation method of freeway based on floating car using random forests. *J Adv Transp*. <https://doi.org/10.1155/2019/8582761>
81. Toan TD, Wong YD (2021) Fuzzy logic-based methodology for quantification of traffic congestion. *Physica A* 570:125784. <https://doi.org/10.1016/j.physa.2021.125784>
82. Olayode OI, Tartibu LK, Okwu MO (2021) Application of adaptive neuro-fuzzy inference system model on traffic flow of vehicles at a signalized road intersections. Volume 9: Engineering Education
83. Liu Y, Wu H (2017) Prediction of road traffic congestion based on random forest. In: 2017 10th international symposium on computational intelligence and design (ISCID)
84. Mao T, Mihaita A-S, Chen F, Vu HL (2022) Boosted genetic algorithm using machine learning for traffic control optimization. *IEEE Trans Intell Transp Syst* 23:7112–7141. <https://doi.org/10.1109/TITS.2021.3066958>
85. Harrou F, Zeroual A, Sun Y (2020) Traffic congestion monitoring using an improved kNN strategy. *Measurement* 156:107534. <https://doi.org/10.1016/j.measurement.2020.107534>
86. Anjaneyulu M, Kubendiran M (2022) Short-term traffic congestion prediction using hybrid deep learning technique. *Sustainability* 15:74. <https://doi.org/10.3390/su15010074>
87. Seo SB, Singh D (2018) Smart town traffic management system using LoRa and machine learning mechanism. *IEEE Technol Policy Ethics* 3:1–4. <https://doi.org/10.1109/NTPE.2018.9778109>
88. Ata A, Khan MA, Abbas S et al (2021) Adaptive IoT empowered smart road traffic congestion control system using supervised machine learning algorithm. *Comput J* 64:1672–1679. <https://doi.org/10.1093/comjnl/bxx129>
89. Chhatpar P, Doolani N, Shahani S, Priya RL (2018) Machine learning solutions to vehicular traffic congestion. In: 2018 international conference on smart city and emerging technology (ICSCET). IEEE, pp 1–4
90. Saleem M, Abbas S, Ghazal TM et al (2022) Smart cities: fusion-based intelligent traffic congestion control system for vehicular networks using machine learning techniques. *Egyptian Inform J* 23:417–426. <https://doi.org/10.1016/j.eij.2022.03.003>
91. Mohanty S, Pozdnukhov A, Cassidy M (2020) Region-wide congestion prediction and control using deep learning. *Transp Res Part C Emerg Technol* 116:102624. <https://doi.org/10.1016/j.trc.2020.102624>
92. Lv Z, Lou R, Singh AK (2021) AI empowered communication systems for intelligent transportation systems. *IEEE Trans Intell Transp Syst* 22:4579–4587. <https://doi.org/10.1109/TITS.2020.3017183>
93. Abdullah SM, Periyasamy M, Kamaludeen NA et al (2023) Optimizing traffic flow in smart cities: soft GRU-based recurrent neural networks for enhanced congestion prediction using deep learning. *Sustainability* 15:5949. <https://doi.org/10.3390/su15075949>
94. Rangari AP, Chouthmol AR, Kadadas C, et al (2022) Deep learning based smart traffic light system using image processing with YOLO v7. In: 2022 4th international conference on circuits, control, communication and computing (I4C). IEEE, pp 129–132
95. Tiwari J, Deshmukh A, Godepure G et al (2020) Real time traffic management using machine learning. In: 2020 international conference on emerging trends in information technology and engineering (ic-ETITE). IEEE, pp 1–5
96. Razavi M, Hamidkhani M, Sadeghi R (2019) Smart traffic light scheduling in smart city using image and video processing. In: 2019 3rd international conference on internet of things and applications (IoT). IEEE, pp 1–4
97. Chaurasia A, Gautam A, Rajkumar R, Chander AS (2023) Road traffic optimization using image processing and clustering algorithms. *Adv Eng Softw* 181:103460. <https://doi.org/10.1016/j.advengsoft.2023.103460>
98. Wan C, Hwang M (2018) Value-based deep reinforcement learning for adaptive isolated intersection signal control. *IET Intel Transport Syst* 12:1005–1010. <https://doi.org/10.1049/iet-its.2018.5170>
99. Vlachogiannis DM, Wei H, Moura S, Macfarlane J (2024) HumanLight: incentivizing ridesharing via human-centric deep reinforcement learning in traffic signal control. *Transp Res Part C Emerg Technol* 162:104593. <https://doi.org/10.1016/j.trc.2024.104593>
100. Liu Y, Liu L, Chen W-P (2017) Intelligent traffic light control using distributed multi-agent Q learning. In: 2017 IEEE 20th

- international conference on intelligent transportation systems (ITSC). IEEE, pp 1–8
101. Labres JVB, Bazzan ALC, Abdoos M (2021) Improving traffic signal control with joint-action reinforcement learning. In: 2021 IEEE symposium series on computational intelligence (SSCI). IEEE, pp 1–8
102. Ge H, Song Y, Wu C et al (2019) Cooperative deep Q-learning with Q-value transfer for multi-intersection signal control. IEEE Access 7:40797–40809. <https://doi.org/10.1109/ACCESS.2019.2907618>
103. Joo H, Ahmed SH, Lim Y (2020) Traffic signal control for smart cities using reinforcement learning. Comput Commun 154:324–330. <https://doi.org/10.1016/j.comcom.2020.03.005>
104. Korecki M (2022) Adaptability and sustainability of machine learning approaches to traffic signal control. Sci Rep 12:16681. <https://doi.org/10.1038/s41598-022-21125-3>
105. Zhang G, Chang F, Jin J et al (2024) Multi-objective deep reinforcement learning approach for adaptive traffic signal control system with concurrent optimization of safety, efficiency, and decarbonization at intersections. Accid Anal Prev 199:107451. <https://doi.org/10.1016/j.aap.2023.107451>
106. Abdoos M (2021) A cooperative multiagent system for traffic signal control using game theory and reinforcement learning. IEEE Intell Transp Syst Mag 13:6–16. <https://doi.org/10.1109/ITS.2020.2990189>
107. Korecki M, Helbing D (2022) Analytically guided reinforcement learning for green it and fluent traffic. IEEE Access 10:96348–96358. <https://doi.org/10.1109/ACCESS.2022.3204057>
108. Han S, Gustafson J, Chung S-J (2023) Congestion control of vehicle traffic networks by learning structural and temporal patterns. In: Proceedings of the 5th annual learning for dynamics and control conference, pp 903–914
109. Fujdiak R, Masek P, Mlynek P et al (2016) Advanced optimization method for improving the urban traffic management. In: 2016 18th conference of open innovations association and seminar on information security and protection of information technology (FRUCT-ISPIT). IEEE, pp 48–53
110. VC SS (2020) A multi-agent ant colony optimization algorithm for effective vehicular traffic management. In: Tan Y, Shi Y, Tuba M (eds) Advances in swarm intelligence. Springer International Publishing, pp 640–647
111. Kumar PM, Devi GU, Manogaran G et al (2018) Ant colony optimization algorithm with Internet of vehicles for intelligent traffic control system. Comput Netw 144:154–162. <https://doi.org/10.1016/j.comnet.2018.07.001>
112. Rastgoftar H, Liu X, Jeannin J-B (2023) A concurrent switching model for traffic congestion control. IFAC-PapersOnLine 56:637–642. <https://doi.org/10.1016/j.ifacol.2023.12.096>
113. Cakici Z, Murat YS (2019) A differential evolution algorithm-based traffic control model for signalized intersections. Adv Civ Eng 2019:1–16. <https://doi.org/10.1155/2019/7360939>
114. Rida N, Ouadoud M, Hasbi A (2020) Ant colony optimization for real time traffic lights control on a single intersection-international association of online engineering
115. Liao S, Wu Y, Ma K, Niu Y (2024) Ant colony optimization with look-ahead mechanism for dynamic traffic signal control of IoV systems. IEEE Internet Things J 11:366–377. <https://doi.org/10.1109/JIOT.2023.3286799>
116. Korkmaz E, Akgüngör AP (2021) Optimizing of phase plan, sequence and signal timing based on flower pollination algorithm for signalized intersections. Soft comput 25:4243–4259. <https://doi.org/10.1007/s00500-020-05438-x>
117. Christofa E, Papamichail I, Skabardonis A (2013) Person-based traffic responsive signal control optimization. IEEE Trans Intell Transp Syst 14:1278–1289. <https://doi.org/10.1109/TITS.2013.2259623>
118. He R, Zhao H, Ye Z, Yang L (2024) Time optimization of urban arterial coordinated control based on multidimensional neighborhood artificial bee colony algorithm. Transp Plan Technol 47:113–145. <https://doi.org/10.1080/03081060.2023.2246959>
119. Wang X, Jerome Z, Wang Z et al (2024) Traffic light optimization with low penetration rate vehicle trajectory data. Nat Commun 15:1306. <https://doi.org/10.1038/s41467-024-45427-4>
120. Jing BB, Xu JM, Yan XW (2018) Arterial signal coordination control model for double-cycle signal control. J Transp Syst Eng Inf Technol 18:1–73
121. Jiao P, Wang H, Sun T (2014) Real-time arterial coordination control based on dynamic intersection turning fractions estimation using genetic algorithm. Math Probl Eng 2014:1–10. <https://doi.org/10.1155/2014/430497>
122. Christofa E, Ampountolas K, Skabardonis A (2016) Arterial traffic signal optimization: a person-based approach. Transp Res Part C Emerg Technol 66:27–47. <https://doi.org/10.1016/j.trc.2015.11.009>
123. Fu T, Wang L, Garg S et al (2024) Adaptive signal light timing for regional traffic optimization based on graph convolutional network empowered traffic forecasting. Inf Fusion 103:102072. <https://doi.org/10.1016/j.inffus.2023.102072>
124. Zhang X, Fan X, Yu S et al (2023) Multi-objective optimization method for signalized intersections in intelligent traffic network. Sensors 23:6303. <https://doi.org/10.3390/s23146303>
125. Zhou Y, Jia S, Guan W, Liu S (2014) Evaluation on traffic congestion mitigation in Beijing with variable message signs. J Transp Syst Eng Inf Technol 14:201–206. [https://doi.org/10.1016/S1570-6672\(13\)60146-4](https://doi.org/10.1016/S1570-6672(13)60146-4)
126. Aleko DR, Djahel S (2020) An efficient adaptive traffic light control system for urban road traffic congestion reduction in smart cities. Information 11:119. <https://doi.org/10.3390/info11020119>
127. Zaheer T, Malik AW, Rahman AU et al (2019) A vehicular network-based intelligent transport system for smart cities. Int J Distrib Sens Netw 15:155014771988884–155014771988884. <https://doi.org/10.1177/1550147719888845>
128. Wen W (2010) An intelligent traffic management expert system with RFID technology. Expert Syst Appl 37:3024–3035. <https://doi.org/10.1016/j.eswa.2009.09.030>
129. Yang B, Zhang H, Du M et al (2024) Urban traffic congestion alleviation system based on millimeter wave radar and improved probabilistic neural network. IET Radar Sonar Navig 18:327–343. <https://doi.org/10.1049/rsn2.12443>
130. Maripini H, Vanajakshi L, Chilukuri BR (2022) Optimal signal control design for isolated intersections using sample travel-time data. J Adv Transp 2022:1–16. <https://doi.org/10.1155/2022/7310250>
131. Djahel S, Jabeur N, Barrett R, Murphy J (2015) Toward V2I communication technology-based solution for reducing road traffic congestion in smart cities. In: 2015 international symposium on networks, computers and communications (ISNCC). IEEE, pp 1–6
132. Gu K, Hu J, Jia W (2023) Adaptive area-based traffic congestion control and management scheme based on fog computing. IEEE Trans Intell Transp Syst 24:1359–1373. <https://doi.org/10.1109/TITS.2022.3183687>
133. Elsaagheer Mohamed SA, AlShalfan KA (2021) Intelligent traffic management system based on the internet of vehicles (IoV). J Adv Transp 2021:1–23. <https://doi.org/10.1155/2021/4037533>
134. Sahil SSK (2021) Smart vehicular traffic management: an edge cloud centric IoT based framework. Internet of Things 14:100140. <https://doi.org/10.1016/j.iot.2019.100140>
135. Alsaawy Y, Alkhodre A, Abi Sen A et al (2022) A comprehensive and effective framework for traffic congestion problem based on the integration of IoT and data analytics. Appl Sci 12:2043. <https://doi.org/10.3390/app12042043>

136. Gomathi V, Sai Sumanth G, Charansai G, Charan G (2023) Traffic congestion control by monitor illegal parking and enabling intimidation. *J Phys Conf Ser* 2484:12047. <https://doi.org/10.1088/1742-6596/2484/1/012047>
137. Subramaniam M, Rambabu C, Chandrasekaran G, Kumar NS (2022) A traffic density-based congestion control method for VANETs. *Wirel Commun Mob Comput* 2022:1–14. <https://doi.org/10.1155/2022/7551535>
138. Nam Bui K-H, Jung JJ (2018) Cooperative game-theoretic approach to traffic flow optimization for multiple intersections. *Comput Electr Eng* 71:1012–1024. <https://doi.org/10.1016/j.compeleceng.2017.10.016>
139. Matsui Y, Yoshihiro T (2024) A flow-based centralized route guidance system for traffic congestion mitigation. In: *IEEE transactions on intelligent transportation systems*, 1–14. <https://doi.org/10.1109/TITS.2024.3375267>
140. Rezaei M, Noori H, Mohammadkhani Razlighi M, Nickray M (2021) ReFOCUS+: multi-layers real-time intelligent route guidance system with congestion detection and avoidance. *IEEE Trans Intell Transp Syst* 22:50–63. <https://doi.org/10.1109/TITS.2019.2952524>
141. Zhao L, Li Z, Ke Z, Li M (2020) Fuzzy self-adaptive proportional–integral–derivative control strategy for ramp metering at distance downstream bottlenecks. *IET Intel Transport Syst* 14:250–256. <https://doi.org/10.1049/iet-its.2019.0618>
142. Yang X, Chen J, Yan M et al (2021) Regional boundary control of traffic network based on MFD and FR-PID. *J Adv Transp* 2021:9730813
143. Jiang T, Liang X (2009) Fuzzy self-adaptive PID controller for freeway ramp metering. In: *2009 international conference on measuring technology and mechatronics automation*. IEEE, pp 570–573
144. Lihan S, Jie M, Baoqing Y (2020) Fuzzy PID design of vehicle attitude control systems. In: *2020 Chinese control and decision conference (CCDC)*. IEEE, pp 1826–1830
145. Jin L, Zhang R, Tang B, Guo H (2020) A fuzzy-PID scheme for low speed control of a vehicle while going on a downhill road. *Energies (Basel)* 13:2795
146. SY H, JF D, Zhou J, YH C (2022) Adaptive fuzzy PID control strategy for vehicle active suspension based on road evaluation. *Electronics (Basel)* 11:921
147. E. U (2018) A brief review on fuzzy logic used in vehicle dynamics control. *J Innov Sci Eng (JISE)* 2:1–7
148. Li R, Deng S, Hu Y (2022) Autonomous vehicle modeling and velocity control based on decomposed fuzzy PID. *Int J Fuzzy Syst* 24:2354–2362
149. Dawood YS, Mahmood AK, Ibrahim MA (2018) Comparison of PID, GA and fuzzy logic controllers for cruise control system. *Int J Com Dig Sys* 7
150. Atta A, Abbas S, Khan MA et al (2020) An adaptive approach: smart traffic congestion control system. *J King Saud Univ Comput Inf Sci*. <https://doi.org/10.1016/j.jksuci.2018.10.011>
151. VM MA, JR PC, Hurtado-Beltrán A, Trumpold J (2022) Efficient intersection management based on an adaptive fuzzy-logic traffic signal. *Appl Sci* 12:6024
152. Wu W, Huang L, Du R (2019) Simultaneous optimization of vehicle arrival time and signal timings within a connected vehicle environment. *Sensors* 20:191
153. Savithramma RM, Sumathi R (2020) Road traffic signal control and management system: a survey. In: *2020 3rd international conference on intelligent sustainable systems (ICISS)*. IEEE, pp 104–110
154. Chowdhury A (2020) Enhancing service quality and reliability in intelligent traffic system
155. Ikidid A, A EF, Sadgal M (2021) A fuzzy logic supported multi-agent system for urban traffic and priority link control. *JUCS J Univ Comput Sci* 27
156. Selase GS (2019) Enhanced adaptive neuro-fuzzy inference system for the control of vehicle traffic at road intersections
157. Sadhukhan P, Banerjee S, Das PK (2021) Road traffic congestion monitoring in urban areas: a review. Challenges and solutions for sustainable smart city development, 199–211
158. Pasandideh F, AA R (2018) A fuzzy priority based congestion control scheme in wireless body area networks. *Int J Wirel Mobile Comput* 14:1–5
159. Jana S, BC G, Sarkar A et al (2024) Application of fuzzy AHP in priority based selection of financial indices: a perspective for investors. *ECONOMICS-Innov Econ Res J* 12:1–27
160. Zm M, Zs X, Zw F, Yang W (2023) Deriving priorities based on representable uninorms from fuzzy preference relations. *Fuzzy Sets Syst* 458:201–220
161. Meshram K, Satpute S, Darade S, Chaudhari S (2022) Dynamic traffic scheduling using emergency vehicle detection. In: *2022 international conference on futuristic technologies (INCOFT)*. IEEE, pp 1–4
162. Hegazy A, Soubra H, Aboulyousr Y (2023) ML driven priority-based constrained routing for pollution management in ITS. In: *2023 eleventh international conference on intelligent computing and information systems (ICICIS)*. IEEE, pp 290–297
163. Wu R, Li B, Hou P, Hou F (2024) Perception data fusion-based computation offloading in cooperative vehicle infrastructure systems. *J Supercomput* 80:17688–17710
164. Azzedin F, Suwad H, Alyafeai Z (2017) Countermeasure zero day attacks: asset-based approach. In: *2017 international conference on high performance computing & simulation (HPCS)*. IEEE, pp 854–857
165. Yece Q, Guoping S, Yufeng Z (2024) A fuzzy adaptive sliding-mode-based SoC estimation for lithium-ion batteries in electric vehicles. *Int J Dyn Control* 12:915–923
166. Abdelghaffar HM, Rakha HA (2021) Development and testing of a novel game theoretic de-centralized traffic signal controller. *IEEE Trans Intell Transp Syst*. <https://doi.org/10.1109/TITS.2019.2955918>
167. Azzedin F, Yahaya M (2016) Modeling BitTorrent choking algorithm using game theory. *Futur Gener Comput Syst* 55:255–265. <https://doi.org/10.1016/j.future.2015.02.007>
168. Ruan S, Ma Y, Yang N et al (2023) Multiobjective optimization of longitudinal dynamics and energy management for HEVs based on nash bargaining game. *Energy* 262:125422
169. Wang Y, Wang X, Shao C, Gong N (2020) Distributed energy trading for an integrated energy system and electric vehicle charging stations: a Nash bargaining game approach. *Renew Energy* 155:513–530
170. Dutta P, Boulanger A (2014) Game theoretic approach to offering participation incentives for electric vehicle-to-vehicle charge sharing. In: *2014 IEEE transportation electrification conference and expo (ITEC)*. IEEE, pp 1–5
171. Zhao X, Zhang X, Li Y (2019) A hierarchical resource allocation scheme based on Nash bargaining game in VANET. *Information* 10:196
172. Kumar N, Rahman SS, Dhakad N (2021) Fuzzy inference enabled deep reinforcement learning-based traffic light control for intelligent transportation system. *IEEE Trans Intell Transp Syst* 22:4919–4928. <https://doi.org/10.1109/tits.2020.2984033>
173. Bangen H-J (1993) Validation problems in air traffic control systems. Verification and validation of complex systems: human factors issues, 497–520
174. Tomar I, Sreedevi I, Pandey N (2022) State-of-art review of traffic light synchronization for intelligent vehicles: current status, challenges, and emerging trends. *Electronics (Basel)* 11:465

175. He Z, Chen L, Li F, Jin G (2023) Congestion avoidance in intelligent transport networks based on WSN-IoT through controlling data rate of Zigbee protocol by learning automata. *Electronics* (Basel) 12:2070. <https://doi.org/10.3390/electronics12092070>
176. Pandey D, Kushwaha V (2024) An exploratory study of optimization techniques for congestion control in wireless sensor networks. *Adhoc Sensor Wirel Netw* 58
177. Nithya S, Maithili K, Kumar TS et al (2024) A fuzzy logic and cross-layered optimization for effective congestion control in wireless sensor networks to improve efficiency and performance. In: *MATEC web of conferences*. EDP sciences, p 01145
178. Kumar A, Lobiyal DK (2024) Traffic classification using soft computing for congestion management in WSN. In: *2024 international conference on automation and computation (AUTOCOM)*. IEEE, pp 464–469
179. Khan S, Nazir S, García-Magariño I, Hussain A (2021) Deep learning-based urban big data fusion in smart cities: towards traffic monitoring and flow-preserving fusion. *Comput Electr Eng* 89:106906
180. Turlej W (2022) High-level sensor models for the reinforcement learning driving policy training. *Electronics* (Basel) 12:71. <https://doi.org/10.3390/electronics12010071>
181. Zhang L, Khalgui M, Li Z (2021) Predictive intelligent transportation: alleviating traffic congestion in the internet of vehicles. *Sensors* (Basel) 21:7330. <https://doi.org/10.3390/s21127330>
182. Abbas S, Malik MO, Javed AR, Hong S-P (2023) Naturalistic driving data-based anomalous driving behavior detection using hypertuned deep autoencoders. *Electronics* (Basel) 12:2072. <https://doi.org/10.3390/electronics12092072>
183. Ochoa DE, Poveda JI (2022) High-performance optimal incentive-seeking in transactive control for traffic congestion. *Eur J Control* 68:100696. <https://doi.org/10.1016/j.ejcon.2022.100696>
184. Hamadeh N, Karouni A, Farhat Z (2022) Intelligent transportation systems to mitigate road traffic congestion. *Intelligenza Artificiale* 15:91–104. <https://doi.org/10.3233/ia-200079>
185. El Hansali Y, Farrag S, Yasar A, Zavanis D (2022) Artificial intelligence-based smart traffic enforcement and management system in urban areas. *SSRN Electron J*. <https://doi.org/10.2139/ssrn.4227717>
186. Ata A, Khan MA, Abbas S et al (2019) Modelling smart road traffic congestion control system using machine learning techniques. *Neural Netw World* 29:99–110. <https://doi.org/10.14311/nnw.2019.29.008>
187. Orłowski M, Skruch P (2023) Multiagent maneuvering with the use of reinforcement learning. *Electronics* (Basel) 12:1894. <https://doi.org/10.3390/electronics12081894>
188. Wided A, Assia B, Fatima B (2023) Traffic management system and traffic light control in smart city to reduce traffic congestion. *Int J Autom Smart Technol* 13:2464. <https://doi.org/10.5875/ausmt.v13i1.2464>
189. Rajasekar T, Mohanraj P, Abishek RK, Haries M, Dhivya Dharshini PA (2023) Adaptive traffic congestion control approach with emergency vehicle protocol. In: *2023 8th international conference on communication and electronics systems (ICCES)*
190. Sun P, Yu Q, You K (2024) Intelligent traffic management strategy for traffic congestion in underground loop. *Tunn Undergr Space Technol* 143:105509. <https://doi.org/10.1016/j.tust.2023.105509>
191. Yin Z, Liu T, Wang C et al (2024) Reducing urban traffic congestion using deep learning and model predictive control. *IEEE Trans Neural Netw Learn Syst*: 1–12. <https://doi.org/10.1109/tnnls.2023.3264709>
192. Shelke M, Malhotra A, PN M (2019) Fuzzy priority based intelligent traffic congestion control and emergency vehicle management using congestion-aware routing algorithm. *J Ambient Intell Humaniz Comput*: 1–8
193. Iqbal K, Khan MA, Abbas S et al (2018) Intelligent transportation system (ITS) for smart-cities using Mamdani Fuzzy Inference System. *Int J Adv Comput Sci Appl*. <https://doi.org/10.14569/IJACSA.2018.090215>
194. Swapno SM, Nobel SM, Meena P et al (2024) A reinforcement learning approach for reducing traffic congestion using deep Q learning. *Sci Rep* 14:1–20
195. Qu Q, Chen F, Zhou X (2019) Road traffic bottleneck analysis for expressway for safety under disaster events using blockchain machine learning. *Saf Sci*. <https://doi.org/10.1016/j.ssci.2019.06.030>
196. Sheehan A, Beddows A, Green DC, Beevers S (2023) City scale traffic monitoring using worldview satellite imagery and deep learning: a case study of barcelona. *Remote Sens* (Basel). <https://doi.org/10.3390/rs15245709>
197. Kothai G, Poovammal E, Dhiman G et al (2021) A new hybrid deep learning algorithm for prediction of wide traffic congestion in smart cities. *Wirel Commun Mob Comput*. <https://doi.org/10.1155/2021/5583874>
198. Rouky N, Bousouf A, Benmoussa O, Fri M (2024) A spatiotemporal analysis of traffic congestion patterns using clustering algorithms: a case study of Casablanca. *Decision Analytics J*. <https://doi.org/10.1016/j.dajour.2024.100404>
199. Dursa BB, Tune KK (2020) Developing traffic congestion detection model using deep learning approach: a case study of Addis Ababa city road
200. Mystakidis A, Tjortjis C (2024) Traffic congestion prediction and missing data: a classification approach using weather information. *Int J Data Sci Anal*: 1–20
201. Chang H, Li L, Huang J et al (2022) Tracking traffic congestion and accidents using social media data: a case study of shanghai. *Accid Anal Prev*. <https://doi.org/10.1016/j.aap.2022.106618>
202. Mohammed O, Kianfar J (2018) A Machine learning approach to short-term traffic flow prediction: a case study of interstate 64 in Missouri. In: *2018 IEEE international smart cities conference, ISC2 2018*
203. Hammoui L, Farah S, Benayad M et al (2025) Leveraging machine learning to predict traffic jams: case study of Casablanca, Morocco. *J Urban Manage*
204. Verma SK, Richa Verma BKS, Sinha RS (2024) Management of intelligent transportation systems and advanced technology. In: *Intelligent transportation system and advanced technology. Energy, environment, and sustainability intelligent transportation system and advanced technology*
205. Prakash J, Murali L, Manikandan N et al (2024) A vehicular network based intelligent transport system for smart cities using machine learning algorithms. *Sci Rep*: 14. <https://doi.org/10.1038/s41598-023-50906-7>
206. Abirami S, Pethuraj M, Uthayakumar M, Chitra P (2024) A systematic survey on big data and artificial intelligence algorithms for intelligent transportation system. *Case Stud Transp Policy*: 17
207. Shengdong M, Zhengxian X, Yixiang T (2019) Intelligent traffic control system based on cloud computing and big data mining. *IEEE Trans Industr Inform* 15:6583–6592. <https://doi.org/10.1109/TII.2019.2929060>
208. Aldridge S, Stehel V (2021) Intelligent vehicular networks, deep learning-based sensing technologies, and big data-driven algorithmic decision-making in smart transportation systems. *Contemp Read Law Soc Justice* 13:107–120
209. Abbasi M, Rafiee M, Khosravi MR et al (2020) An efficient parallel genetic algorithm solution for vehicle routing problem in cloud implementation of the intelligent transportation systems. *J Cloud Comput* 9. <https://doi.org/10.1186/s13677-020-0157-4>

210. Khan A, Abbas S (2018) Intelligent transportation system for smart-cities using fuzzy logic. *Lahore Garrison Univ Res J Comput Sci Inf Technol (LGURJCSIT)* 2:64–79
211. Abdelrahman AE, Hassanein HS, Abu-Ali N (2022) Robust data-driven framework for driver behavior profiling using supervised machine learning. *IEEE Trans Intell Transp Syst* 23:3336–3350. <https://doi.org/10.1109/TITS.2020.3035700>
212. Zhang R, Ishikawa A, Wang W et al (2018) Using reinforcement learning with partial vehicle detection for intelligent traffic signal control
213. Yao B, Chen C, Cao Q et al (2016) Short-term traffic speed prediction for an urban corridor: short-term traffic speed prediction for an urban corridor. *Computer-Aided Civ Infrastr Eng* 32:154–169
214. Song C, Lee H, Kang C et al (2017) Traffic speed prediction under weekday using convolutional neural networks concepts. *IEEE Intelligent Vehicles Symp (IV)*:1293–1298
215. Gao J, Shen Y, Liu J et al (2017) Adaptive traffic signal control: deep reinforcement learning algorithm with experience replay and target network
216. Toque F, Come E, El Mahrssi M, Oukhellou L (2016) 2016 IEEE 19th international conference on intelligent transportation systems (ITSC): 1–4 Nov. 2016. IEEE
217. Sermanet P, Lecun Y. Traffic sign recognition with multi-scale convolutional networks
218. Sathananthavathi V, Indumathi G (2021) BAT optimization based retinal artery vein classification. *Soft comput* 25:2821–2835. <https://doi.org/10.1007/s00500-020-05339-z>
219. Parmar Y, Natarajan S, Sobha G (2019) Deeprange: deep-learning-based object detection and ranging in autonomous driving. *IET Intel Transport Syst* 13:1256–1264. <https://doi.org/10.1049/iet-its.2018.5144>
220. Yuan Y, Xiong Z, Wang Q (2017) An incremental framework for video-based traffic sign detection, tracking, and recognition. *IEEE Trans Intell Transp Syst* 18:1918–1929. <https://doi.org/10.1109/TITS.2016.2614548>
221. Shtern M, Mian R, Litoiu M et al (2015) Towards a multi-cluster analytical engine for transportation data. In: *Proceedings—2014 international conference on cloud and autonomic computing, ICCAC 2014*. Institute of Electrical and Electronics Engineers Inc., pp 249–257
222. Guerreiro G, Figueiras P, Silva R et al (2016) An architecture for big data processing on intelligent transportation systems. An application scenario on highway traffic flows. In: *2016 IEEE 8th international conference on intelligent systems, IS 2016—proceedings*. Institute of Electrical and Electronics Engineers Inc., pp 65–71
223. Doornenbal BM, Spisak BR, van der Laken PA (2022) Opening the black box: Uncovering the leader trait paradigm through machine learning. *Leadership Quarterly* 33. <https://doi.org/10.1016/j.leaqua.2021.101515>
224. Akhtar M, Moridpour S (2021) A review of traffic congestion prediction using artificial intelligence. *J Adv Transp* 2021
225. Kong X, Xu Z, Shen G et al (2016) Urban traffic congestion estimation and prediction based on floating car trajectory data. *Futur Gener Comput Syst* 61:97–107. <https://doi.org/10.1016/j.future.2015.11.013>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.